INTRODUCTION TO PSYCHOLOGY

Year 1 | First Semester | Prelims

1: WHAT IS PSYCHOLOGY?

PSYCHOLOGY AS A SCIENCE

- **PSYCHOLOGY** - scientific study of behavior and mental processes
- It seeks to describe, explain, predict, and control behavior and mental processes
  - Controlling in a way that psychologists try to understand the factors that influence behavior and apply this knowledge for public good
- **THEORIES** - set of hypothesized statements about relationships among events
  - Satisfactory if it allows us to predict behavior
  - If observations cannot be adequately explained by or predicted from a given theory, consider revising or replacing the theory

WHAT PSYCHOLOGISTS DO

- Engage in research, practice, and teaching
- **PURE RESEARCH** - undertaken out of curiosity
  - No immediate application to personal or social problems
- **APPLIED RESEARCH** - designed to find solutions to specific personal or social problems
- Practicing psychology means applying psychological knowledge to help individuals change their behavior
- Teaching psychology is sharing psychological knowledge

FIELDS OF PSYCHOLOGY

- **CLINICAL PSYCHOLOGISTS** - help people with psychological disorders and adjust to demands of life
- **COUNSELING PSYCHOLOGISTS** - help people with adjustment problems
- **SCHOOL PSYCHOLOGISTS** - assist students with problems that interfere with learning
- **EDUCATIONAL PSYCHOLOGISTS** - focus on course planning and instructional methods
- **DEVELOPMENTAL PSYCHOLOGISTS** - study changes throughout life span
- **PERSONALITY PSYCHOLOGISTS** - identify and measure human traits
- **SOCIAL PSYCHOLOGISTS** - study individuals' thoughts, feelings, and behavior in social situations.
- **ENVIRONMENTAL PSYCHOLOGISTS** - study the interaction between people and the environment.
- **EXPERIMENTAL PSYCHOLOGISTS** - specializes in basic processes like sensation, perception etc.
- **INDUSTRIAL / ORGANIZATIONAL PSYCHOLOGISTS** - focus on the relationship between people and work.
- **HEALTH PSYCHOLOGISTS** - study the effects of stress on health
- **FORENSIC PSYCHOLOGISTS** - apply psychology to the criminal justice system
- **SPORT PSYCHOLOGISTS** - help athlete with mental strategies to enhance performance

HISTORICAL FOUNDATIONS OF PSYCHOLOGY

ARISTOTLE

- Argued that science could rationally treat only information gathered by the senses
- Enumerated 5 senses of vision, hearing, smell, taste, and touch
- Explored nature of cause and effect
- Pointed out that people differ from other living things in their capacity for rational thought
- Outlined laws of associationism that have lain at the heart of learning theory
- Declared that people are motivated to seek pleasure and avoid pain

DEMOCRITUS

- Suggested that we could think of behavior in terms of a body and mind
- Pointed out that behavior is influenced by external stimulation

SOCRATES

- Suggested that we should rely on rational thought and **introspection** to gain self-knowledge
- **INTROSPECTION** - careful examination of one's own thoughts and emotions
- Also pointed out that people are social creatures who influence one another

GUSTAV THEODOR FECHNER

- Published his book "Elements of Psychophysics," which showed how physical events stimulate psychological sensations and perception

WILHELM WUNDT

- Established first psychological laboratory in **Leipzig, Germany**

STRUCTURALISM

- Found by **WILHELM WUNDT**
- Attempted to break conscious experience into objective sensations (sight or taste), and subjective feelings (emotional responses, mental images)
- Argues that mind consists of 3 basic elements — sensation, feelings, and images — that combine to form experience

FUNCTIONALISM

- Found by **WILLIAM JAMES**
- Focused on behavior as well as mind or consciousness
- Looked at how experiences helps us function more adaptively in our environments
- Was influenced by **Charles Darwin's Theory of Evolution** - argued that organisms with adaptive features survive and reproduce

BEHAVIORISM

- Found by **JOHN B. WATSON**, with **B.F. SKINNER** as a contributor
- Focuses on learning observable behavior
- Define psychology as scientific study of behavior only
- **B.F. Skinner** believed that organisms learn to behave in certain ways because they have been **reinforced** to do so

GESTALT PSYCHOLOGY

- Found by **MAX WERTHEIMER, KURT KOFFKA, WOLFGANG KOHLER**
- Focused on perception and how perception influences thinking and problem-solving
- Emphasizes the tendency to organize perceptions into wholes

and to integrate separate stimuli into meaningful patterns
- **Kohler's** findings in his research about chimpanzees suggest that we often manipulate the elements of problems until we group them in such a way that we can reach a goal

PSYCHOANALYSIS

- Both a theory and a method of psychotherapy
- Developed by **SIGMUND FREUD**
- As a theory, it proposes that much of our lives is governed by unconscious ideas and impulses that originate in childhood conflicts
- As a method of therapy, it aims to help patients gain insight into their conflicts and to find socially acceptable ways of expressing wishes and gratifying needs

CONTEMPORARY PERSPECTIVES IN PSYCHOLOGY

- Many contemporary psychologists in the behaviorist tradition look on themselves as **social cognitive theorists**, who see human learning as intentional rather than mechanical

THE BIOLOGICAL PERSPECTIVE

- Seek relationships between the brain, hormones, heredity, and evolution, and behavior and mental processes
- Basically examines relationship between biological processes and behavior
- **Evolutionary Psychologists** - suggest that much human social behavior has a hereditary basis

COGNITIVE PERSPECTIVE

- Focuses on mental processes to understand human nature
- Investigate the ways we perceive and mentally represent the world
- Study those things we refer to as the mind
- Rooted to **Socrates**' "Know thy self" and introspection

HUMANISTIC-EXISTENTIAL PERSPECTIVE

- Emphasizes role of subjective (personal) experience
- **HUMANISM** - stresses human capacity for self-fulfillment and the central roles of consciousness, self-awareness, and decision making
  - **Consciousness** - our sense of being in the world and the force that unifies our personalities
- **EXISTENTIALISM** - views people as free to choose and as being responsible for choosing ethical conduct
- Grounded in the work of **CARL ROGERS** and **ABRAHAM MASLOW**

PSYCHODYNAMIC PERSPECTIVE

- Based on **FREUD**'s Psychoanalysis
- Contemporary psychologists who follow Freud are likely to call themselves **Neoanalysts**, who focused less on unconscious processes and more on conscious choice and self-direction

PERSPECTIVES ON LEARNING

- Learning is an essential factor in describing, explaining, predicting, and controlling behavior
- Contemporary behaviorists emphasize environmental influences and learning of habits

through repetition and reinforcement
- **Social cognitive theorists** suggest that people can modify and create their environments
  - Noted that people engage in intentional learning by observing others

THE SOCIOCULTURAL PERSPECTIVE

- Addresses ways that people differ from one another
- Studies the influences of ethnicity, gender, culture, and socioeconomic status on behavior and mental processes

ETHNICITY

- Members of ethnic group share their cultural heritage, race, language, or history

GENDER

- Culturally defined concepts of masculinity and femininity
- **Mary Whiton Calkins** - introduced method of paired associates to study memory, discovered primacy and recency effects, and engaged in research into the role of frequency of repetition in the vividness of memories
- **Mary Salter Ainsworth** - revolutionized our understanding of attachment between parents and children by means of cross-cultural studies
- **Elizabeth Loftus** - show that memories are not snapshots of the past, instead consists of something old, something new, something borrowed, and something blue

HOW PSYCHOLOGISTS STUDY BEHAVIOR AND MENTAL PROCESSES

THE SCIENTIFIC METHOD

- Organized way of using experience and testing ideas to expand and refine knowledge
- Begins with **formulating a research question** - can be based on a daily experience, theory, or folklore
- Develop a **hypothesis** - a testable statement about behavior or mental processes
- Test the **hypothesis** - done through controlled methods like experiments
- Draw **conclusions** - based on observations and findings
- When observations do not bear out their hypotheses, they may modify the theories from which the hypotheses were derived
- **CORRELATION** - association or relationship among variables
- **SELECTION FACTOR** - source of bias that may occur in research findings when participants are allowed to choose for themselves a certain treatment in a scientific study

SAMPLES AND POPULATIONS

- **POPULATION** - group of interest
- **SAMPLE** - segment of a population
- Only **representative samples** allow us to generalize or extend our findings from research samples to target populations

PROBLEMS IN GENERALIZING FROM PSYCHOLOGICAL RESEARCH

- Nature of the research sample
- We must question whether findings of research with men can be generalized to women

RANDOM AND STRATIFIED SAMPLING

- **RANDOM SAMPLE** - each member of a population has an equal chance of being selected to participate

- **STRATIFIED SAMPLE** - Selected so that identified subgroups in the population represented proportionately in the sample
- **VOLUNTEER BIAS** - source of bias or error in research reflecting the prospect that people who offer to participate in research studies differ systematically from people who do not

METHODS OF RESEARCH

METHODS OF OBSERVATION

CASE STUDY

- Collect information about individuals and small groups
- Sometimes used to investigate rare occurrences
- Subject to inaccuracies

SURVEY

- Study many thousands of people at the same time
- Used to learn about behavior and mental processes that cannot be observed in natural setting or studied experimentally
- Collects data from a large sample through questionnaires or interviews

NATURALISTIC OBSERVATION

- Observing behavior of people in their natural habitat without interference

CORRELATION

- **CORRELATIONAL METHOD** - investigate whether observed behavior or a measured trait is related to another
- **CORRELATION COEFFICIENT** - number that varies from r = +1.00 - r = -1.00, expressing the strength and direction of the relationship
- **Positive correlation** - as one variable increases, the other also increases
- **Negative correlation** - as one variable increases, the other decreases
- Correlation may suggest, but does not prove, cause and effect

THE EXPERIMENTAL METHOD

- **EXPERIMENT** - group of participants obtains a treatment, and observed if treatment makes a difference in their behavior
- Allow psychologists to control experiences of participants to draw conclusions about cause and effect

INDEPENDENT AND DEPENDENT VARIABLE

- **INDEPENDENT VARIABLE** - manipulated by the experimenters so that its effects may be determined
- **DEPENDENT VARIABLE** - measured results or outcomes, assumed to be affected by the independent variable

EXPERIMENTAL AND CONTROL GROUPS

- **EXPERIMENTAL GROUPS** - received the treatment
- **CONTROL GROUPS** - do not obtain the treatment
- Effort is made to ensure that all other conditions are held constant for both groups

BLINDS AND DOUBLE BLINDS

- **PLACEBOS** - bogus treatment that has the appearance of being genuine
- **BLIND** - participants are unaware of the treatment they receive
- **DOUBLE-BLIND STUDIES** - study in which neither the subjects nor the observers to know who has received the treatment

ETHICS IN PSYCHOLOGICAL RESEARCH

ETHICS OF RESEARCH WITH HUMANS

- **INFORMED CONSENT** - participant's agreement to participate in research after receiving information about the purposes of the study and the nature of the treatments
- **CONFIDENTIALITY** - maintaining the privacy of participants' data
- **DEBRIEFING** - to explain the purposes and methods of a completed procedure to a participant
- Harming animals is only justified when there is no alternative and the benefits outweigh the harm

CRITICAL THINKING

- Way of evaluating the claims and comments of other people that involves skepticism and examination of evidence

PRINCIPLES OF CRITICAL THINKING

- Be skeptical
- Insist on evidence
- Examine definitions of terms
- Examine assumptions
- Be cautious in drawing conclusions.
- Be skeptical of anecdotes.
- Consider alternative interpretations.
- Do not oversimplify.
- Do not overgeneralize.
- Apply critical thinking to all areas of life.

2: RESEARCH METHODS IN PSYCHOLOGY

CRITICAL THINKING

- Intellectually disciplined process of actively and skillfully conceptualizing, applying, analyzing, synthesizing, and/or evaluating information gathered from observation, experience, reflection, reasoning, or communication as a guide to belief and action
- It's not just knowing the right answer, it digs deep to answering the question about why is it the right answer

THE SCIENTIFIC METHOD

- Organized way of using experience and testing ideas to expand and refine knowledge
- **RESEARCH QUESTIONS** - start of the method
- **HYPOTHESIS** - statement about behavior or mental processes that is testable through research
- **TEST THE HYPOTHESIS** - provide evidence
- **DRAW CONCLUSIONS**
- **THEORY CONSTRUCTION**
- **NEW RESEARCH QUESTIONS**

SAMPLES AND POPULATIONS

SAMPLE

- Segment of population that must be drawn so that it accurately represents the population
- **Generalizing** allows us to apply findings from research sample to target populations
- Problems:
  - **Nature of research** - gender, age, intelligence score
  - **Random sampling** - equal chance
  - **Stratified random** - identified subgroups
  - **Volunteer bias** - reason why they participate

METHODS OF OBSERVATION

- **CASE STUDY** - individuals and small groups
  - For rare occurrences
  - Inaccuracies: gaps and errors in memory, societal expectations
- **SURVEY** - behavior and mental processes that cannot be observed naturally or experimentally by questionnaires, interviews, public records
- **NATURALISTIC OBSERVATION** - natural habitats

CORRELATION

- Follows observation
- Whether observed behavior or a measured trait is related to or correlated with another by a mathematical method
- **CORRELATION COEFFICIENT** - number between 1.00 - -1.000 that expresses the strength and direction of the relationship between 2 variables
- **POSITIVE CORRELATION** - as one variable increases, the other also increases
- **NEGATIVE CORRELATION** - as one variable increases, the other decreases and vice versa
- Does not show cause and effect

EXPERIMENTAL RESEARCH

- **HYPOTHESIS** - testable prediction concerning possible relationship between two or more variables
  - Never a question, always a statement
  - Must always be operationalized, meaning it must include the independent variable, dependent variable. And how much the independent variable will differ from the dependent variable, population, and sample
- **INDEPENDENT VARIABLE** - manipulated by the researcher
  - consists of two or more treatment conditions to which participants are exposed
  - Assumed to have a direct effect on the dependent variable
- **DEPENDENT VARIABLE** - measured by the researcher
  - Observed for changes in order to assess the effect of the treatment
  - Assume to be directly affected by changes in the independent variable
- **EXTRANEOUS VARIABLE** - any variable that has the potential to influence the dependent variable in an experiment and therefore bias the result
  - May be associated with the characteristics of participants, the experimenter or the experimental design
- **CONFOUNDING VARIABLE** - does not have an unwanted effect on the dependent variable in an experiment
  - Results are biased and cannot be determined whether they are caused by the independent variable of the confounding variable

PLACEBO EFFECT

- A positive or therapeutic result due to a participant's expectations of the effect of an inactive substance or dummy treatment
- **Single blind procedure** - method of control for placebo effect
  - Participants are unaware of their research group allocation
  - Used to avoid contamination of results from bias and preconceptions of the participants

EXPERIMENTER EFFECT

- Where the expectations of preconceptions of the experimenter bias the results by either consciously or unconsciously revealing the desired outcome of the experiment, or through unconscious procedural or recording bias
- **DOUBLE BLIND PROCEDURE** - method of control for experimenter effect
  - Neither experimenter nor participants are aware of the allocation of participants to research groups
  - Used to avoid contamination of results from biases and preconceptions of the experimenter or participants

EXPERIMENTAL RESEARCH: COUNTER BALANCING

- Involves allocating participants to groups and presenting the independent variable in a different order
- All participants still undergo each condition of the independent variable, but in a different order

SAMPLING PROCEDURES

- **CONVENIENCE SAMPLING** - selecting participants who are easily obtainable
- **RANDOM SAMPLING** - allocates participants from the populations of interest in synch a way that each member of the population has an equal chance of being selected
- **STRATIFIED SAMPLING** - attempts to prevent biases by making the sample more representative of the problem
  - Involves identifying some of the relevant factors (strata) present in a population such as age, sex, etc. then selecting a separate sample from each group
- **RANDOM ALLOCATION** - experimental control which is used to place participants into groups
  - **Experimental group** - participants are exposed to the independent variable
  - **Control group** - participants are not exposed to the independent variable
    - Used as standard of comparison against the experimental group

TECHNIQUES OF QUALITATIVE AND QUANTITATIVE DATA COLLECTION

- **QUANTITATIVE DATA** - numerical, usually collected through experimental research
- **QUALITATIVE DATA** - descriptive data, usually collected through non-experimental methods
- **SUBJECTIVE DATA** - obtained by self-report measures in which subjects give verbal or written responses to a series of research questions
- **OBJECTIVE DATA** - data that has been gathered using systematic observation which is not influenced by any personal bias

STATISTICS

MEASURES OF CENTRAL TENDENCY

- Statistics that indicate information about the middle scores of a data set
- **MEAN** - average score
- **MEDIAN** - middle point in a set of data
- **MODE** - most frequently occurring score in a set of data
- **Measures of dispersion** include range and standard deviation

INFERENTIAL STATISTICS

- Formal data analysis that measures the likelihood of results obtained for a study occurring by chance
- **Measures of statistical significance** - indicate whether the results obtained in an experiment do not occur by chance and therefore may be attributed to other variables
- **P-value** represent the probability level which forms the basis for deciding whether chance factors are responsible for results obtained
- **0.05** significance level occurs when the probability of chance is 5x or fever times in a 100 repetitions of the research

CONCLUSION

- Drawn based on the results obtained
- Draws an interference as to whether the hypothesis has been supported or rejected
- **Generalisation** - application of the conclusion based on the results obtained to a wider population in other similar settings outside the study

VALIDITY

- Extent to which an experiment measures what it is supposed to measure
- **Internal validity** - whether results gained from a measure is truly due to the variable that is thought to be measuring
- **Construct validity** - involves deciding whether the instrument can be used to support the theory that is being tested
- **External validity** - examines whether the results gained from the sample used in the research can confidently be generalised to the population
- **Reliability** - how consistent a measuring instrument
- **Internal reliability** - refers to the extent to which all the items in a research instrument contribute equally to the final score

INTRODUCTION TO PSYCHOLOGY

Year 1 | First Semester | Midterms

3: BIOLOGY AND PSYCHOLOGY

THE NERVOUS SYSTEM

NEURONS

- **NEURONS** - specialized cells of the nervous system that conduct impulses
  - We are born with over 100 billion neurons that vary in length
- **GLIAL CELLS** - remove dead neurons and waste products from the nervous system, nourish and insulate neurons, form myelin, and play a role in neural transmission
- **CELL BODY** - contains the core or nucleus of the cell
  - Nucleus uses oxygen to carry out the work of the cell
- **DENDRITES** - short fibers that received impulses from other neurons
- **AXON** - long, thin part of the neuron that transmits impulses to other neurons
  - **AXON TERMINALS** - bulb-shaped structures at the end of the axon
- As child matures, axons of neurons become longer and dendrites and terminal proliferate
- Number of glial cells also increase
- Neurons carry messages in one direction: dendrites/cell body → axon → axon terminals.

MYELIN

- Fatty substance that encases and insulates axons from electrically charged atoms or ions
- Found in fluids that surround the nervous system
- Myelin sheath minimizes leakage of electrical current being carried along the axon
- **Myelination** - part of maturation

AFFERENT AND EFFERENT NEURONS

- **AFFERENT NEURONS** - sensory neurons that transmit messages from sensory receptors to spinal cord and brain
- **EFFERENT NEURONS** - motor neurons that send messages from the brain or spinal cord to muscles and glands

THE NEURAL IMPULSE

- **Neural impulses** are the electrochemical discharge of cell or neuron
- Messages that travel within neurons at 2-225 miles per hour
- Neurons and body fluid contain **ions** which are positively or negatively charged atoms
- When neuron is at resting state, there are a lot of Cl- within it, giving it an overall negative charge in relation to the outside
- **POLARIZATION** - Difference in electrical charge polarizes a neuron for firing by creating an internal negative charge in relation to the body fluid outside the cell membrane
- **RESTING POTENTIAL** - electrical potential across the neural membrane when it is not responding to other neurons
- **DEPOLARIZATION** - Reduction of resting potential of a cell membrane
- **ACTION POTENTIAL** - electrical impulse that provides the basis for the conduction of a neural impulse along an axon of a neuron
- **FIRING** - conduction of neural impulse along the length of an axon
- Neurons release neurotransmitters
- Neurons fire when incoming messages combine to reach a threshold

- **ALL-OR-NONE PRINCIPLE** - neuron fires an impulse of the same strength whenever its action potential is triggered
  - Either a neuron fires or it doesn't
- **REFRACTORY PERIOD** - phase following firing during which a neuron is less sensitive to messages from other neurons and will not fire

SYNAPSE

- Junction between axon terminals of one neuron and the dendrites or cell body of another neuron where neuron relays its messages to another neuron
- **SYNAPTIC CLEFT** - fluid-filled gap between neurons

NEUROTRANSMITTERS

- **SYNAPTIC VESICLES** - sacs in the axon terminals that contain neurotransmitters
- **NEUROTRANSMITTERS** - chemical substances involved in the transmission of neural impulses from one neuron to another
- **RECEPTOR SITE** - location on a dendrite of a receiving neuron tailored to receive a neurotransmitter
- Loose neurotransmitters are usually either broken down or reabsorbed by axon terminal in process called **reuptake**
- Some neurotransmitters act to **excite** other neurons, causing them to fire
- Other neurotransmitters **inhibit** receiving neurons
- Sum of excitatory and inhibitory determine whether a neuron will fire or not
- Neurotransmitters are involved in physical and psychological processes
  - Excesses or deficiencies have been linked to psychological disorders
- **ACETYLCHOLINE** - controls muscle contractions
  - Excitatory at synapses between nerves and muscles, but inhibitory at the heart and some other locations
  - **CURARE** (poison) and **BOTULISM** (disease) prevents AcH from binding to its receptor, causing paralysis
  - Found in **Hippocampus** that is involved in formation of memories
  - Deficiency can lead to memory formation impairment
- **DOPAMINE** - acts in the brain and affects the ability to perceive pleasure, voluntary movement, and learning and memory
  - Deficiencies are linked to Parkinson's disease
  - Excesses are linked to Schizophrenia
- **NOREPINEPHRINE** - produced largely by neurons in brain stem
  - Acts both as a neurotransmitter and as a hormone
  - Excitatory neurotransmitter that speed up heartbeat and other body processes, and is involved in general arousal, learning, memory, and eating
  - Excesses and deficiencies are linked to mood disorders
  - Cocaine and amphetamines prevent their reuptake, leading to persistent arousal
- **SEROTONIN** - involved in emotional arousal and sleep
  - Deficiencies linked to eating disorders, alcoholism, depression, aggression, and insomnia.
  - LSD (drug) decreases action of serotonin

- **GAMMA-AMINOBUTYRIC ACID** (GABA) - Inhibitory neurotransmitter that may help calm anxiety reactions
  - Deficiencies linked to depression
- **ENDORPHINS** - inhibitory neurotransmitters that reduce pain
  - Occur naturally in the brain and in the blood stream
  - Similar to narcotic morphine in effects

DIVISIONS OF THE NERVOUS SYSTEM

- Nervous system consists of the brian, spinal cord, and nerves
  - **NERVES** - bundle of axons from many neurons

PERIPHERAL NERVOUS SYSTEM

- Consists of sensory and motor neurons that transmit messages to and from the central nervous system
- **SOMATIC NERVOUS SYSTEM** - contains sensory and motor neurons
  - Transmits messages that control purposeful body movements
  - Connects the CNS with sensory receptors, skeletal muscles, and the surface of the body.
- **AUTONOMIC NERVOUS SYSTEM** - regulates glands and the muscles of internal organs
  - Controls activities that occur automatically
  - **Sympathetic Division:** most active during processes that involve spending body energy from stored reserves
    - Fight or flight response
  - **Parasympathetic nervous system**: most active during processes that restore energy
    - Rest and digest

CENTRAL NERVOUS SYSTEM

- Consists of the spinal cord and brain
- **SPINAL CORD** - column of nerves that transmits messages from sensory receptors to the brian and from the brian to muscles and glands
  - **Spinal reflex** - unlearned response to a stimulus that may involve two neurons
- Spinal cord and brain contain **gray matter,** which consists of non-myelinated neurons
  - Cross-section of spinal cord shows that gray matter is distributed in a butterfly pattern
- **White matter** is composed of bundles of longer, myelinated axons that carry messages to and from the brain

THE BRAIN

- Weighs about 3 pounds

UNDERSTANDING THE BRAIN

- Mind is a function of the brain
- Damage to left side of the brain is connected with loss of sensation or movement on the right side of the body

EXPERIMENTING WITH THE BRAIN

ELECTROENCEPHALOGRAPH

- Detects minute amounts of electrical activity (brain waves) that pass between electrodes
- Certain brain waves are associated with feelings of relaxation, with various stages of sleep, and with neurological problems

BRAIN-IMAGING TECHNIQUES

- **COMPUTERIZED AXIAL TOMOGRAPHY** (CAT or CT SCAN) -

Passes X-rays through the head and measures the structures that reflect the beams from various angles, generating a 3D image
    - Reveals deformities in shape and structure that are connected with blood clots, tumors, and other health problems
- **POSITRON EMISSION TOMOGRAPHY SCAN** (PET SCAN) - forms a computer-generated image of the activity of parts of the brain by tracing the amount of glucose used by these parts
    - tracer , a harmless amount of radioactive compound, is used
- **MAGNETIC RESPONSE IMAGING** (MRI) - use of powerful magnetic field and is exposed to radio waves that cause parts of the brain to emit signals, which are measured from multiple angles
    - Relies on subtle shifts in blood flow
    - **FUNCTIONAL MRI** - provides more rapid picture and enables researchers to observe the brain while it works by repeated scans
    - **prefrontal cortex** is considered to be the executive center, where decisions are made to keep information in working memory and to solve problems

A VOYAGE THROUGH THE BRAIN

- **HINDBRAIN**
    - **MEDULLA** - oblong area of the hindbrain involved in regulation of heartbeat, blood pressure, movement, and respiration
    - **PONS** - a bulge structure of the hindbrain that transmits information about body movement in respiration, attention, and sleep and arousal
    - **CEREBELLUM** - part of the hindbrain involved in maintaining balance and in controlling motor (muscle) behavior
    - **RETICULAR FORMATION** - lower part of the hindbrain vital in the functions of attention, sleep, and arousal
- **FOREBRAIN**
    - **THALAMUS** - located near the center of the brain and could be said to lie between the forebrain and the midbrain
        - consists of two joined egg shaped structures.
        - serves as a relay station for sensory information to the cortex and in the functions of sleep and attention
    - **HYPOTHALAMUS** - lies beneath the thalamus and above the pituitary gland
        - Vital in the regulation of body temperature, concentration of fluids, storage of nutrients, and motivation and emotion
        - Dubbed as the pleasure center
    - **LIMBIC SYSTEM** - forms a fringe along the inner edge of the cerebrum and is fully evolved only in mammals
        - Involved in memory and emotion and in the drives of hunger, sex, and aggression.
        - **AMYGDALA** - near the bottom of the limbic system and looks like two little almonds; apparently facilitates stereotypical aggressive responses
        - **HIPPOCAMPUS** - involved in memory function
    - **CEREBRUM** - responsible for thinking and language

- **CEREBRAL CORTEX** - the wrinkled surface area (gray matter) of the cerebrum
- **CORPUS CALLOSUM** - a thick fiber bundle that connects the hemispheres of the cortex

THE CEREBRAL CORTEX

- outer coating of the cerebrum
- involved in almost every bodily activity, including most sensations and responses
- It is also the part of the brain that frees people from the tyranny of genetic dictates and instinct
- It is the seat of thinking and language, and it enables humans to think deeply about the world outside and to make decisions

STRUCTURE OF CEREBRAL CORTEX

- Has two hemispheres (left and right), each divided into four lobes
- **Frontal lobe** - lies in front of the central fissure
- **Parietal lobe** - behind frontal lobe
- **Temporal lobe** - lies below the side, or lateral, fissure—across from the frontal and parietal lobes
- **Occipital lobe** - lies behind the temporal lobe and behind and below the parietal lobe
- **SOMATOSENSORY CORTEX** - behind the central fissure in the parietal lobe
  - Receives messages from skin senses all over the body
- Left hemisphere controls, acts on, and receives inputs from the right side of the body
- Right hemisphere controls, acts on, and receives inputs from the left side of the body
- **MOTOR CORTEX** - motor area of the cerebral cortex
  - Lies in the frontal lobe just across the central fissure from the somatosensory cortex
  - Neural impulses in the motor cortex cause parts of our body to move

THINKING, LANGUAGE, AND THE CEREBRAL CORTEX

- **ASSOCIATION AREAS** - areas not primarily involved in sensation or motor activity
  - Involved in learning, thought, memory, and language
- Association areas in the prefrontal region are the brain's executive center
  - Where we solve problems and make plans and decisions
  - retrieves visual, auditory, and other memories and manipulates them

LANGUAGE FUNCTIONS

- Left hemisphere for most people
- Two key language areas lie within the hemisphere of the cortex that contains language functions (usually the left hemisphere): **Broca's area** and **Wernicke's area**
  - Damage to either can cause **aphasia** - disruption of the ability to understand or produce language
- **Wernicke's area** - lies in the temporal lobe near the auditory cortex
  - It responds mainly to auditory information (sounds)
  - Essential to understanding relationships between words and their meanings
- **Broca's area** - located in the frontal lobe, near the section of the motor cortex that controls the muscles of the tongue, throat, and other areas of the face used when speaking

  - Broca's area processes the information and relays it to the motor cortex
  - The motor cortex sends the signals that cause muscles in your throat and mouth to contract.
  - Speech production
- **Wernicke's aphasia** - difficulty comprehending the meaning of spoken language
- **Broca's Aphasia** - language disorder characterized by slow, laborious speech
- **Angular gyrus** - between the visual cortex and Wernicke's area
  - Translates visual information, as in perceiving written words, into auditory information (sounds) and sends it on to Wernicke's area.

LEFT-BRAIN, RIGHT BRAIN

- According to this view, left-brained people would be primarily logical and intellectual
- Right-brained people would be intuitive, creative, and emotional
- Those of us who are fortunate enough to have our brains "in balance" would presumably have the best of it—the capacity for logic combined with emotional richness
- Hemispheres respond simultaneously

SPLIT-BRAIN EXPERIMENTS

- Severing the corpus callosum
- Purpose of the operation is to confine seizures to one hemisphere of the cerebral cortex rather than allowing a neural tempest to reverberate.
- Used to treat patients with **epilepsy** - temporary disturbances of brain functions that involve sudden neural discharges

THE ENDOCRINE SYSTEM

- **GLAND** - organ that secretes one or more chemical substances such as hormones, saliva, or milk
- Body has two types of glands: glands with ducts, and glands without ducts
- **Duct** - passageway that carries substances to specific locations
- **DUCTLESS GLANDS** - make up the endocrine system, and release hormones into the bloodstream
- **ENDOCRINE SYSTEM** - body's system of ductless glands that secrete hormones and release them directly into the bloodstream
- **HORMONES** - substance secreted by an endocrine gland that regulates various body functions

PITUITARY AND THE HYPOTHALAMUS

- **PITUITARY GLAND** - gland that secretes growth hormone, prolactin, antidiuretic hormone, and other hormones
  - **Growth hormone** - regulates the growth of muscles, bones, and glands
  - **Prolactin** - regulates maternal behavior in lower mammals such as rats and stimulates production of milk in women
  - **Vasopressin** - antidiuretic hormone; inhibits production of urine when the body's fluid levels are low
  - **Oxytocin** - stimulates labor in pregnant women and is connected with maternal behavior in some mammals
- Blood vessels between the hypothalamus and the pituitary

gland provide a direct route for these factors

PINEAL GLAND

- Secretes melatonin
- **Melatonin** - helps regulate the sleep-wake cycle and may affect the onset of puberty

THYROID GLAND

- Produces **thyroxin**, which affects the body's **metabolism**—the rate at which the body uses oxygen and produces energy
- **Hypothyroidism** - condition that results from too little thyroxin
- **Cretinism** - Thyroxin deficiency in children characterized by stunted growth and intellectual disability
- **Hyperthyroidism** - too much thyroxin characterized by excitability, insomnia, and weight loss

ADRENAL GLANDS

- Located above the kidneys, have an outer layer, or cortex, and an inner core, or medulla
- **Adrenal cortex** is regulated by the pituitary hormone ACTH (adrenocorticotrophic hormone)
  - Secretes **corticosteroids** - regulate the heart beat, increase resistance to stress, promote muscle development, and cause the liver to release stored sugar, making more energy available in emergencies, such as when you see another car veering toward your own
- **Medulla**:
  - **Epinephrine** - a.k.a adrenaline; manufactured exclusively by the adrenal glands
    - intensifies most emotions and is central to the experience of fear and anxiety
  - **Norepinephrine** - produced elsewhere in the body

THE TESTES AND OVARIES

- Testes produces **testosterone**, ovaries produce **estrogen** and **progesterone**
- **TESTOSTERONE** - causes the male's sex organs to develop
  - stokes the growth of muscle and bone and the development of primary and secondary sex characteristics
  - **Primary sex characteristics** - directly involved in reproduction and include the increased size of the penis and the sperm-producing ability of the testes
  - **Secondary sex characteristics** - differentiate males from females but are not directly involved in reproduction
- **ESTROGEN** - fosters female reproductive capacity and secondary sex characteristics such as accumulation of fatty tissue in the breasts and hips
- **PROGESTERONE** - stimulates growth of the female reproductive organs and prepares the uterus to maintain pregnancy

EVOLUTION AND HEREDITY

- **EVOLUTION** - concept of a struggle for existence
- **NATURAL SELECTION** - Species better adapted to their environment are more likely to survive and reproduce

- **MUTATIONS** - sudden variation in an inheritable characteristic
  - Leads to differences which affect the ability to adapt to change

EVOLUTIONARY PSYCHOLOGY

- Studies ways in which adaptation and natural selection are connected with mental processes and behavior
- **INSTINCTIVE** - stereotyped pattern of behavior that is triggered in a specific situation
  - nearly identical among the members of the species in which it appears
  - It tends to resist modification, even when it serves no purpose or results in punishment
  - language and even consciousness are considered instinctive to humans

HEREDITY, GENETICS, AND BEHAVIORAL GENETICS

- **HEREDITY** - defines one's nature, which is based on biological structures and processes
  - biological transmission of traits that have evolved from generation to generation
- **GENETICS** - subfield of biology that studies heredity
- **BEHAVIORAL GENETICS** - concerned with the genetic transmission of traits that give rise to patterns of behavior

GENES AND CHROMOSOMES

- **GENES** - most basic building blocks of heredity
  - Regulate the development of specific traits
  - Segments of chromosomes
- **Chromosomes** - made up of strings of genes
  - Large complex molecules of DNA
- **Human genome project** - learned that the sequencing of your DNA consists of about three billion DNA sequences spread throughout your chromosomes
- **POLYGENIC** - traits that are influenced by combinations of genes
- **GENOTYPE** - full genetic potential, as determined by the sequencing of the chemicals in your DNA
  - Provides what psychologists refer to as your **nature**
- **PHENOTYPE** - manner in which your genetic code manifests itself because of your experiences and environmental circumstances
  - Represents the interaction of your nature (heredity) and your nurture (environmental influences) in the origins of your behavior and mental processes
- **SEX CHROMOSOMES** - 23rd pair of chromosome that determine whether we are female or male
- **DOWN SYNDROME** - results from an extra chromosome on the 21st pair

KINSHIP STUDIES

- ways in which psychologists compare the presence of traits and behavior patterns in people who are biologically related or unrelated to help determine the role of genetic factors in their occur rence

TWIN STUDIES

- **ZYGOTE** - fertilized egg cell (ovum) that carries genetic messages from both parents
- **MONOZYGOTIC (MZ) TWINS** - twins that develop from a single

fertilized ovum that divides in two early in prenatal development
    - Share the same genetic code; also called identical twins
- **DIZYGOTIC (DZ) TWINS** - develop from two fertilized ova and who are thus as closely related as brothers and sisters in general
    - also called fraternal twins
- **TWIN STUDIES** show how strongly genetic factors influence physical features

ADOPTION STUDIES

- Comparing children who have been separated from their parents at an early age and reared in different environments.

CHAPTER 4: SENSATION AND PERCEPTION

SENSATION AND PERCEPTION DEFINED

- **SENSATION** - stimulation of sensory receptors and the transmission of sensory information to the central nervous system (spinal cord or brain)
  - It's an automatic process resulting from energy sources like light and sound, or chemicals in smell and taste.
- **PERCEPTION** - An active process where sensations are organized and interpreted to form an inner representation of the world
  - It's not mechanical and reflects experiences and expectations.

ABSOLUTE THRESHOLD

- weakest level of a stimulus necessary to produce a sensation
- Psychophysicists determine this by gradually increasing stimulus strength until detected 50% of the time
- **PITCH** - highness or lowness of a sound as determined by the frequency of the sound waves

SUBLIMINAL STIMULATION

- Sensory stimulation below the absolute threshold for conscious perception
- It may influence behavior even without conscious awareness.

DIFFERENCE THRESHOLD

- The minimum difference in magnitude between two stimuli required to perceive them as different (detected 50% of the time).
- **WEBER'S CONSTANT -** fraction by which a source of physical energy must be increased or decreased so that a difference in intensity will be perceived
  - Ex. threshold for perceiving differences in intensity of light is 2% (~1/60)
- **JUST NOTICEABLE DIFFERENCE** - minimal amount by which a source of energy must be increased or decreased so that a difference in intensity will be perceived

SIGNAL DETECTION THEORY

- Perception involves the interaction of physical, biological, and psychological factors, not just stimulus intensity. Factors like training, motivation, and alertness influence detection

FEATURE DETECTORS IN THE BRAIN

- Neurons in the sensory cortex that fire in response to specific features of sensory information (e.g., lines, edges, colors)

SENSORY ADAPTATION

- Becoming more sensitive to low-magnitude stimuli (**sensitization/positive adaptation**) and less sensitive to constant stimuli (**desensitization/negative adaptation**)

VISION

LIGHT

- **Visible light** is a small portion of the electromagnetic spectrum that stimulates the eye and produces visual sensations
- **HUE** - color of light, as determined by its wavelength
- Wavelengths range from approximately 400 nm (violet) to 700 nm (red).

THE EYE

- **CORNEA** - Transparent outer layer.
- **SCLERA** - white of the eye that is composed of hard protective tissues
- **IRIS** - Colored part of the eye; controls pupil size.
- **PUPIL** - Opening in the iris that allows light to enter.
- **LENS** - Focuses images on the retina
  - It adjusts its thickness to focus on near or far objects.
- **RETINA** - Contains photoreceptors that are sensitive to light (rods and cones)
  - Contains several layers of cells: rods and cones, bipolar cells, and ganglion cells
  - **BIPOLAR CELLS** - Neurons that conduct neural impulses from rods and cones to ganglion cells.
  - **GANGLION CELLS** - Neurons whose axons form the optic nerve
- **OPTIC NERVE** - Transmits sensory information from the eye to the brain

RODS AND CONES

- **RODS** - Sensitive to light intensity; black and white vision
- **CONES** - Transmit sensations of color
  - They are most densely packed in the fovea
- **FOVEA** - Central area of the retina with high visual acuity (sharpness).
- **BLIND SPOT** - Area on the retina where the optic nerve exits; insensitive to light
- **VISUAL ACUITY** - Sharpness of vision
  - **Nearsightedness** - the need to be close to an object to discriminate its details
    - Result when eyeballs are elongated, such that images of distant objects are focused in front of the retina
  - **Farsightedness** - the need to be far from an object to discriminate its details
    - Result when eyeballs are too short, such that images of nearby objects are focused behind the retina
- **PRESBYOPIA** - Condition of brittleness of the lens that makes it difficult to focus on nearby visual stimuli
  - typically occurs in middle adulthood

LIGHT ADAPTATION

- **DARK ADAPTATION** - Adjusting to lower lighting conditions by increasing the sensitivity of rods and cones

COLOR VISION

- **HUE** - color of light determined by wavelength of light
- **VALUE** - color's degree of brightness or darkness
- **SATURATION** - How intense a color appears

WARM AND COOL COLORS

- Psychologically, the colors on the **green-blue** part of the color wheel are considered to be cool in temperature

- Colors in the **yellow–orange-red** area are considered to be warm

COMPLEMENTARY COLORS

- Colors opposite each other on the color wheel that, when mixed, produce gray
- **Red-green** and **blue-yellow** are major complementary pairs
- **Light** is the source of all colors
- **Pigments** reflect and absorb different wavelengths of light selectively
- Mixture of lights is an **additive process**
- Mixture of pigments is **subtractive**

AFTERIMAGE

- lingering visual impression made by a stimulus that has been removed

THEORIES OF COLOR VISION

- **TRICHROMATIC THEORY** - Color vision is based on three types of cones: red, green, and blue
- **OPPONENT-PROCESS THEORY** - Color vision is based on three types of color receptors: red-green, blue-yellow, and brightness.
  - Afterimages support this theory

COLOR BLINDNESS

- **TRICHROMAT** - Normal color vision
- **MONOCHROMAT** - Sensitive only to lightness and darkness (total color blindness)
- **DICHROMAT** - Sensitive to black-white and either red-green or blue-yellow (partial color blindness)

VISUAL PERCEPTION

- Process by which we organize or make sense of the sensory impressions caused by the light that strikes our eyes
- **CLOSURE** - tendency to perceive a broken figure as being complete or whole

PERCEPTUAL ORGANIZATION

- Integrating sensory stimulation into meaningful wholes

FIGURE-GROUND PERCEPTION

- Distinguishing objects (figures) from their backgrounds

OTHER GESTALT PRINCIPLES FOR ORGANIZATION

- **PROXIMITY** - Grouping nearby objects together
- **SIMILARITY** - Grouping similar objects together
- **CONTINUITY** - Perceiving a series of points or lines as having unity
- **COMMON FATE** - Perceiving elements that move together as belonging together

TOP-DOWN VERSUS BOTTOM-UP PROCESSING

- **TOP-DOWN PROCESSING** - Using contextual information or knowledge to organize parts of a pattern
- **BOTTOM-UP PROCESSING** - Organizing parts of a pattern to form an image of the pattern

PERCEPTION OF MOTION

- Perception of movement is based on change of position relative to other objects
- **ILLUSIONS** - sensations that give rise to misperceptions

STROBOSCOPIC MOTION

- Illusion of movement created by rapidly showing a series of stationary images

DEPTH PERCEPTION

MONOCULAR CUES

- Depth cues perceived with one eye
- **PERSPECTIVE** - Parallel lines converge in the distance
- **RELATIVE SIZE** - Distant objects look smaller
- **CLEARNESS** - Nearby objects show more detail
- **OVERLAPPING** - Closer objects block farther ones
- **SHADOWING** - Shadows provide information about shape and position
- **TEXTURE GRADIENT** - Closer objects have rougher textures
- **MOTION PARALLAX** - Nearby objects appear to move more rapidly relative to our own motion

BINOCULAR CUES

- Depth cues involving both eyes
- **RETINAL DISPARITY** - Difference in the images cast on the retinas of the eyes
- **CONVERGENCE** - Inward movement of the eyes to focus on a near object

PERCEPTUAL CONSTANCIES

SIZE CONSTANCY

- Perceiving objects as the same size even when their retinal image size varies

COLOR CONSTANCY

- Perceiving objects as retaining their color, even when lighting conditions change

BRIGHTNESS CONSTANCY

- Perceiving objects as being equally bright, even though lighting conditions change

SHAPE CONSTANCY

- Perceiving objects as maintaining their shape, even when viewed from different angles

VISUAL ILLUSIONS

- Occur when perceptual constancies are misapplied
- **HERING-HELMHOLTZ** - horizontal lines are straight and parallel, but the radiating lines cause them to appear to be bent outward near the center
- **MULLER-LYER** - two lines are the same length, but the line on the right, with its reversed arrowheads, looks longer

HEARING

- **AUDITORY STIMULATION** - also known as sound
  - vibration of molecules in a medium such as air or water
- Sound travels through the medium like waves

PITCH AND LOUDNESS

- **PITCH** - Determined by frequency (Hz)
- **LOUDNESS** - Corresponds to amplitude (dB)
- **HERTZ** - Unit expressing the frequency of sound waves. One hertz equals one cycle per second
- **DECIBELS** - Unit expressing the loudness of a sound

THE EAR

- **OUTER EAR** - shaped to funnel sound waves to the eardrum
  - **EARDRUM** - thin membrane that vibrates in response to sound waves, and thereby transmits them to the middle and inner ears
- **MIDDLE EAR** - contains the eardrum and three small bones (malleus, incus, stapes), which also transmit sound by vibrating

  - functions as an amplifier, increasing the pressure of the air entering the ear
  - **STIRRUP** - attached to another vibrating membrane, the **oval window**
  - **OVAL WINDOW** - works in conjunction with the **round window**, which balances the pressure in the inner ear
  - **ROUND WINDOW** - pushes outward when the oval window pushes in, and is pulled inward when the oval window vibrates outward
- **INNER EAR**
  - **COCHLEA** - Bony tube containing the basilar membrane and organ of Corti
  - **BASILAR MEMBRANE** - Membrane within the cochlea that vibrates in response to sound
  - **ORGAN OF CORTI** - Receptor for hearing located on the basilar membrane; contains hair cells
  - **AUDITORY NERVE** - Transmits neural impulses from the organ of Corti to the brain

PERCEPTIONS OF LOUDNESS AND PITCH

- **PLACE THEORY** - Pitch is sensed according to the location on the basilar membrane that vibrate
- **FREQUENCY THEORY** - Pitch is reflected in the frequency of neural impulses generated
- **VOLLEY PRINCIPLE** - Groups of neurons take turns firing to transmit sensory information about pitches in the intermediate range

DEAFNESS

- **CONDUCTIVE DEAFNESS** - Damage to the middle ear structures
- **SENSORINEURAL DEAFNESS** - Damage to the inner ear structures or auditory nerve

OTHER SENSES

THE CHEMICAL SENSES: SMELL AND TASTE

SMELL

- Detects odors (molecules in the air) via receptor neurons in the olfactory membrane
- **FLAVOR** - complex quality of food based on odor, texture, temperature, and taste
- **OLFACTORY NERVE** - Transmits information concerning odors from olfactory receptors to the brain

TASTE

- Detects molecules via taste cells (receptor neurons) in taste buds on the tongue
- **TASTE CELLS** - Receptor cells that are sensitive to taste
- **TASTE BUDS** - sensory organs for taste
  - they contain taste cells and are located mostly on the tongue

THE SKINS SENSES

- Include touch, pressure, warmth, cold, and pain

TOUCH AND PRESSURE

- Sensory receptors fire when the skin is touched
- **ACTIVE TOUCHING** - continuously moving your hand along the surface of an object so that you continue to receive sensory input from the object

TEMPERATURE

- Receptors for warmth and cold respond to changes in skin temperature

PAIN

- Results from stimulation of nociceptors in the skin
- **PROSTAGLANDINS** - facilitate transmission of the pain message to the brain and heighten circulation to the injured area, causing the redness and swelling that we call inflammation
- Pain-relieving drugs such as aspirin and ibuprofen help by inhibiting production of prostaglandins
- **PHANTOM LIMB PAIN** - Perception of pain in amputated limbs
- **GATE THEORY OF PAIN** - Nervous system can only process a limited amount of stimulation at a time
- **ACUPUNCTURE** - Use of pins to diminish pain, possibly because of stimulation of release of endorphins and cortisol

KINESTHESIS AND THE VESTIBULAR SENSE

KINESTHESIS

- Sense of body position and movement

VESTIBULAR SENSE

- Sense of equilibrium and body position relative to gravity; located in the semicircular canals of the ears

EXTRASENSORY PERCEPTION (ESP)

- Perception of objects or events through means other than the known sensory organs
- Includes telepathy, clairvoyance, precognition, and psychokinesis
- **ESP** - also referred to as parapsychological or psi phenomena
- **PSI COMMUNICATION** - transfer of information through an irregular or unusual process—not through the usual senses
- Skepticism exists due to the lack of consistent, reliable evidence and the "file-drawer problem" (failure to report negative results)

INTRODUCTION TO PSYCHOLOGY

Year 1 | First Semester | Midterms

CHAPTER 5: LEARNING

LEARNING, EXPERIENCE AND CHANGE

FROM THE BEHAVIORIST POV…

- **learning** is a relatively permanent change in behavior that arises from practice or experience
- Plays down the roles of cognition and choice

COGNITIVE PSYCHOLOGISTS

- Define learning as the process by which organisms change the way they represent the environment because of experience
- Changes influence the organism's behavior, but do not fully determine it.
- Learning may be shown by changes in behavior, but learning itself is a mental process
- Suggest that people choose whether or not to imitate aggressive and other behaviors they observe, and that people are most likely to imitate behaviors that are consistent with their values

CLASSICAL CONDITIONING: LEARNING WHAT IS LINKED TO WHAT

CLASSICAL CONDITIONING

- Involves ways in which we learn to associate events with other events
- Involuntary, automatic learning
- Simple form of learning in which organisms come to anticipate or associate events with one another
- Simple form of learning in which a neutral stimulus comes to evoke the response usually evoked by another stimulus by being paired repeatedly with the other stimulus
- Organisms come to anticipate or associate events with one another

IVAN PAVLOV (1927)

- Conditioning and learning with dogs
- Discovered that reflexes can also be learned, or conditioned, by association.
- In his initial experiments, he trained dogs to salivate when he sounded a tone or a bell. He termed this trained salivary responses **conditional reflexes.**
  - The reflexes were conditional on the repeated pairing of a previously neutral stimulus and a stimulus that evoked the target response.
  - Today, these are generally referred to as **conditioned responses**

In Pavlov's experiment, the **neutral stimulus** was the clanging of the food tray, the **stimulus** was the food, and the **target response** was the salivation.

RFFLEXES

- Unlearned and evoked by certain stimuli
- A simple, unlearned response to a stimulus

STIMULUS

- Environmental condition that elicits a response

EXPLAINING CLASSICAL CONDITIONING

BEHAVIORISTS

- Explain outcome of classical conditioning in terms of the publicly observable conditions of learning.

- Conditioning is a simple form of learning in which one stimulus comes to evoke the response usually evoked by another stimulus because the stimuli are paired repeatedly.
- We can only outline the conditions under which targeted behaviors occur.

In Pavlov's demonstration, the dog learned to salivate in response to the tone because the tone had been paired with meat powder.

COGNITIVE PSYCHOLOGISTS

- View classical conditioning as the learning of relationships among events
- Relationships allow organisms to mentally represent their environments and make predictions.

In Pavlov's demonstration, the dog salivated in response to the tone because the tone became mentally connected with the meat.

- Cognitive focus is on the information learned by the organism.
- Organisms are seen as seekers of information that generate and test rules about relationships among events.

STIMULI AND RESPONSES IN CLASSICAL CONDITIONING

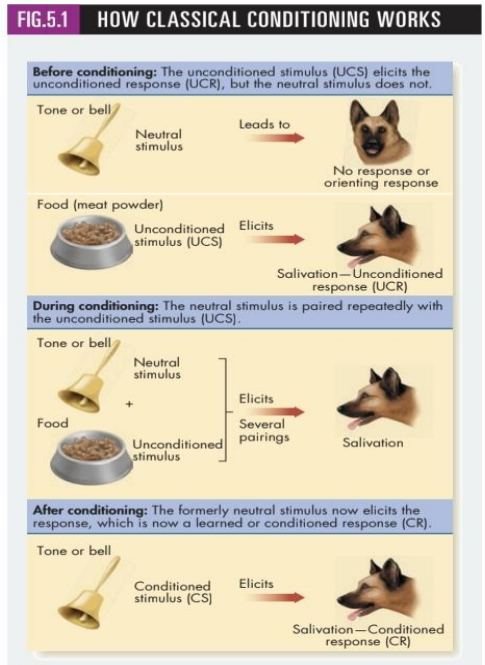
In Palov's experiment, the meat powder is an **unconditioned stimulus (UCS).**

Salivation in response to the meat powder is an unlearned or **unconditioned response (UCR).**

The tone was at first a meaningless or neutral stimulus. It might have caused the dog to look in the direction of the sound, an **orienting flex.** But the tone was not yet associated with food.

Then, through repeated association with the meat powder, the tone became a learned or **conditioned stimulus (CS)** for the salivation response.

Salivation in response to the tone is a learned or **conditioned response (CR).** Therefore, salivation can be either a CR or a UCR, depending on the method used to evoke the response.



Prior to conditioning, food elicits salivation. The tone, a neutral stimulus, elicits either no response or an orienting response. During conditioning, the tone is rung just before meat powder is placed on the dog's tongue. After several repetitions, the tone, now a CS, elicits salivation, the CR.

UNCONDITIONED STIMULUS (UCS)

- Stimulus that elicits a response from an organism prior to conditioning

UNCONDITIONED RESPONSE

- Unlearned response to an unconditioned stimulus

ORIENTING FLEX

- An unlearned response in which an organism attends to a stimulus

CONDITIONED STIMULUS (CS)

- Previously neutral stimulus that elicits a conditioned response because it has been paired repeatedly with a stimulus

CONDITIONED RESPONSE (CR)

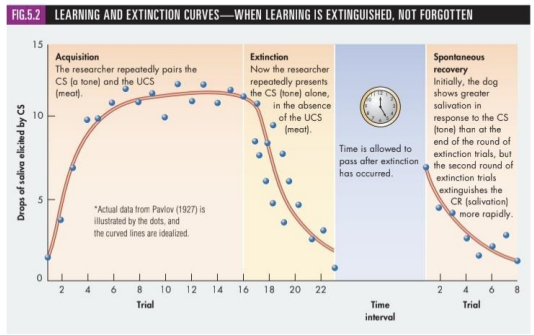
- a learned response to a conditioned stimulus

EXTINCTION AND SPONTANEOUS RECOVERY

- Extinction and spontaneous recovery are aspects of conditioning that help us adapt by updating our expectations or revising our thinking about (representations of) the changing environment

EXTINCTION

- Process by which stimuli lose their ability to evoke learned responses because the events that had followed the stimuli no longer occur
- CS loses the ability to elicit the CR because the CS is no longer associated with UCS
- Learner responses are said to be extinguished.
- From the cognitive perspective, it changes the child\s mental representation of its environment because the CS no longer allows the child to make the same prediction.
- Inhibits the response, but the response remains available for the future under the "right" conditions.
- In experiments, the extinction of CR, Pavlov found that repeated presentations of the CS without the UCS led to extinction of the CR.



- Figure 5.2 shows that after the initial conditioning, a dog's responsive salivation increased for a while, and then leveled off.
- Continued pairings of the stimuli led to increased salivation. After seven or eight trials, salivation leveled off at 11 to 12 drops

SPONTANEOUS RECOVERY

- Recurrence of an extinguished response as a function of the passage of time
- Evolutionary psychologists noted that it is adaptive
- As time passes and the seasons change, things sometimes follow circular paths and arrive where they were before.
- Helps organisms adapt to situations that recur from time to time

GENERALIZATION AND DISCRIMINATION

- Adaptation requires us to respond similarly (or generalize) to stimuli that are equivalent in function and to respond differently to (or discriminate between) stimuli that are not.

GENERALIZATION

- Tendency for a CR to be evoked by stimuli that are similar to the stimulus to which the response was conditioned

For example, Pavlov demonstrated generalization by getting his dog to salivate when it was shown a circle. Then later, the dog salivated in response to being shown closed geometric figures—even squares

- Organisms must also learn that many stimuli perceived as being similar are functionally different, and they must respond adaptively to each.

DISCRIMINATION

- In conditioning, the tendency for an organism to distinguish between a CS and similar stimuli that do not forecast a UCS
- Cognitive theorists, in disagreement, said that in Pavlov's experiment, the dog lost its ability to adjust its mental map of the environment as the ellipses grew more circular. Thus, it was frustrated

HIGHER-ORDER CONDITIONING

- A classical conditioning procedure in which a previously neutral stimulus comes to serve as a learned or CS after being paired repeatedly with a stimulus that has already become a learned or CS.

APPLICATIONS OF CLASSICAL CONDITIONING

- Some involve the conditioning of fear and the counterconditionning or extinction of fear.

TASTE AVERSION

- Intriguing examples of classical conditioning
- Adaptive because they motivate organisms organisms to avoid harmful foods
- Differ from other kinds of classical conditioning in a couple of ways:
  1. Only one association may be required
  2. Most kinds of classical conditioning require that the UCS and CS be close together in time. In taste aversion, UCS can occur hours after the CS

THE EVOLUTION OF TASTE AVERSION

- Challenges the view that organisms learn to associate any stimuli that are linked in time.
- In reality, not all stimuli are created equal
- Evolutionary perspective suggests that animals (and humans) are biologically prepared to develop aversions that are adaptive in their environments
- Those of us who acquire taste aversions quickly are less likely to eat poisonous food, more likely to survive, and more likely to contribute our genes to future generations.

"LITTLE ALBERT": CLASSICAL CONDITIONING OF EMOTIONAL RESPONSES

- In **1920, John B. Watson** and his future wife, Rosalie Rayner, published an article describing their demonstration that emotional reactions, such as fears, can be acquired through principles of classical conditioning.
  - Subject of demonstration was "Little Albert," a phlegmatic fellow at the age of 11 months
- Watson startled Little Albert by clanging steel bars behind his head when he played with the rat. After seven pairings, Albert showed fear of the rat even though clanging was suspended. Albert's fear was also generalized to objects similar in

appearance to the rat, such as a rabbit and the fur collar on his mother's coat.

- **Counterconditioning** - fear-reduction technique in which pleasant stimuli are associated with fear-evoking stimuli so that the fear-evoking stimuli lose their aversive qualities

PREPAREDNESS AND THE CONDITIONING OF FEAR

- Arne Öhman and Susan Mineka suggested that humans may be **biologically prepared** by evolutionary forces to rapidly develop fears of certain animals that could do them harm
- **Biological preparedness** - readiness to acquire a certain kind of CR due to the biological makeup of the organism

COUNTERCONDITIONING

- Organism learns to respond to a stimulus in a way that is incompatible with a response that was conditioned earlier.
- The reason behind is, if fears, as Watson had shown, could be conditioned by painful experiences like a clanging noise, perhaps fears could be counterconditioned by substituting pleasant experiences.
- In 1924, Watson's protégé **Mary Cover Jones** attempted to countercondition fear in a two-year-old boy named Peter.
    - Peter had an intense fear of rabbits. Jones had a rabbit gradually brought closer to Peter while he munched candy and cookies. Jones first placed the rabbit in a far corner of the room while Peter munched and crunched. Peter cast a wary eye, but he continued to consume the treats. Over a couple of months, the animal was brought closer until Peter simultaneously ate and touched the rabbit.
    - Jones theorized that the joy of eating was incompatible with fear and thus counterconditioned it.

FLOODING AND SYSTEMATIC DESENSITIZATION

FLOODING

- a behavioral fear-reduction technique based on principles of classical conditioning
- fear-evoking stimuli (CSs) are presented continuously in the absence of actual harm so that fear responses (CRs) are
- like counterconditioning, is a behavior therapy method for reducing fears
- based on the classical conditioning principle of extinction
- client is exposed to the fear-evoking stimulus until fear is extinguished

In extinction, the CS is presented repeatedly in the absence of the UCS until the CR is no longer evoked.

- Although flooding is usually effective, it is unpleasant. With that, behavior therapists frequently prefer to use **systematic desensitization**

SYSTEMATIC DESENSITIZATION

- a behavioral fear-reduction technique in which a hierarchy of fear-evoking stimuli is presented while the person remains relaxed
- Client is gradually exposed to fear-evoking stimuli under circumstances in which he or she remains relaxed
- Takes longer than flooding but is not as unpleasant

In any event, people can learn by means of simple association. In terms of the **evolutionary perspective**, organisms that can learn by several routes—including conditioning and conscious reflection—would stand a greater chance of survival than organisms whose learning is limited to conditioning.

OPERANT CONDITIONING: LEARNING WHAT DOES WHAT TO WHAT

- Through classical conditioning, we learn to associate stimuli. As a result, a simple, usually passive response made to one stimulus is then made in response to the other.
- However, classical conditioning is only one kind of learning that occurs in these situations.

OPERANT CONDITIONING

- organisms learn to do things, or not to do things, because of the consequences of their behavior.
- While classical conditioning focuses on how organisms form anticipations about their environments, operant conditioning focuses on what they do about them.

EDWARD L. THORNDIKE AND THE LAW OF EFFECT

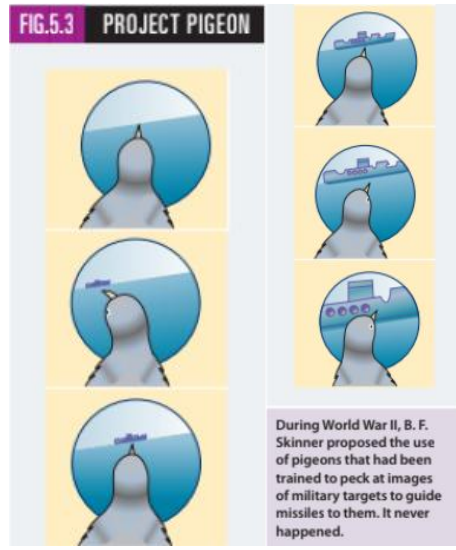
- Thorndike used stray cats as subjects in experiments on the effects of rewards and punishments on learning, 1890s in the Columbia University
- Thorndike placed a cat in a "puzzle box." If it pulled a dangling string, a latch would be released, allowing it to jump out and reach a bowl of food.
- Thorndike explained the cat's learning to pull the string in terms of his **law of effect**

LAW OF EFFECT

- Thorndike's view that pleasant events stamp in responses, and unpleasant events stamp them out
- A response would be, in Thorndike's term, **"stamped in"** in a particular situation by a reward. But punishments **"stamp out"** response
- Organisms would learn not to behave in ways that bring on punishment.

B.F. SKINNER AND REINFORCEMENT

- During World War II, Skinner proposed that pigeons be trained to guide missiles to their targets.



  - Once trained, the pigeons would be placed in missiles. Their pecking at similar targets displayed on a screen would correct the missile's flight path, resulting in a "hit" and a sacrificed pigeon.
- Skinner taught pigeons and other animals to engage in **operant behavior**, behavior that operates on, or manipulates, the environment.

OPERANT CONDITIONING

- behavior that operates on, or manipulates, the environment
- Organisms learn to engage in certain behavior because of the effects of that behavior.
- We learn to engage in operant behaviors, also known as **operants,** that result in presumably desirable outcomes

METHODS OF OPERANT CONDITIONING

- While other psychologists might focus on how organisms think or feel, **Skinner** focused on measurable things they do.
- Many psychologists have found these kinds of behavior inconsequential, especially when it comes to explaining and predicting human behavior. But Skinner's supporters point out that focusing on discrete behavior creates the potential for helpful changes.
- To study operant behavior, Skinner devised an animal cage (or "operant chamber") that has been dubbed the Skinner box, is ideal for laboratory experimentation because experimental conditions can be carefully introduced and removed, and their effects on laboratory animals can be observed.

TYPES OF REINFORCERS

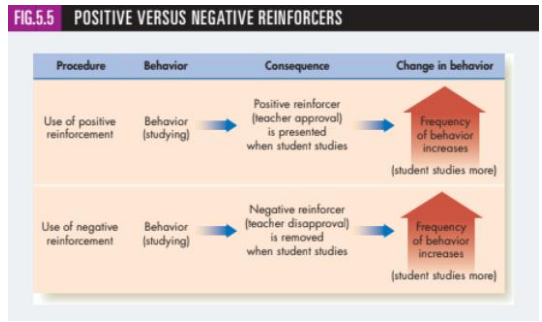
- Any stimulus that increases the probability that responses preceding it serves as a reinforcer

POSITIVE REINFORCERS

- Increase the probability that a behavior will occur when they are applied.

NEGATIVE REINFORCERS

- Increase the probability that a behavior will occur when the reinforcers are removed



IMMEDIATE VS. DELAYED REINFORCERS

- **Immediate reinforcers** are more effective than **delayed reinforcers**
  - short-term consequences of behavior often provide more of an incentive than the long-term consequences.

PRIMARY AND SECONDARY REINFORCERS

- **Primary reinforcers** are effective because of the organism's biological make-up
  - an unlearned reinforcer whose effectiveness is based on the biological makeup of the organism and not on learning
- **Secondary reinforcers** acquire their value through being associated with established reinforcers, that's why they are also termed **conditioned reinforcers**
  - a stimulus that gains reinforcement value through association with established reinforcers

EXTINCTION AND SPONTANEOUS RECOVERY IN OPERANT CONDITIONING

- In **Pavlov's experiment**, the meat powder was the event that followed and confirmed the appropriateness of salivation, while in **operant conditioning**, the ensuing events are reinforcers.
  - The extinction of learned responses results from the repeated performance of operant behavior without reinforcement.
- Reinforcers maintain operant behavior or strengthen habitual behavior in operant conditioning.

SPONTANEOUS RECOVERY

- Occurs in operant conditioning and classical conditioning
- Adaptive in operant conditioning and classical conditioning

REINFORCERS VS. REWARDS AND PUNISHMENTS

- **Reinforcers** are defined as stimuli that increase the frequency of behavior
  - They are known by their effects, whereas rewards and punishments are more known by how they feel
- A list of reinforcers is arrived at scientifically and empirically—that is, by observing what sorts of stimuli increase the frequency of the behavior.
  - Whereas reinforcers, even negative reinforcers, increase the frequency of the behavior they follow, punishments decrease it
- Punishment can rapidly suppress undesirable behavior and may be warranted in "emergencies"
- Psychologists distinguish between positive punishments and negative punishments, both being aversive events and has the capability to decrease the frequency of the behavior they follow

POSITIVE PUNISHMENT

- application of an aversive stimulus to decrease unwanted behavior

NEGATIVE PUNISHMENT

- removal of a pleasant stimulus

DISCRIMINATIVE STIMULI

DISCRIMINATIVE STIMULI

- Indicate whether behavior will be reinforced
- Behaviors that are not reinforced tend to be extinguished

DISCRIMINATIVE STIMULUS

- A stimulus that indicates that reinforcement is available

SCHEDULES OF REINFORCEMENT

- **Continuous reinforcement** a schedule of reinforcement in which every correct response is reinforced
- **Partial reinforcement** one of several reinforcement schedules in which not every correct response is reinforced
- Responses that have been maintained by partial reinforcement are more resistant to extinction than responses that have been maintained by continuous reinforcement
- From the cognitive perspective, we could suggest that organisms that have experienced partial reinforcement do not expect reinforcement every time they engage in a response. Therefore, they are more likely to persist in the absence of reinforcement.

INTERVAL SCHEDULES

- In a **fixed-interval schedule**, a fixed amount of time must elapse before the correct response will result in a reinforcer.

  - An organism's response rate falls off after each reinforcement and then picks up again as the time when reinforcement will occur approaches.
  - **Fixed-interval schedule** is a schedule in which a fixed amount of time must elapse between the previous and subsequent times that reinforcement is available
- Reinforcement is more unpredictable in **variable-interval schedule;** response rate is lower, but it is also steadier
  - a schedule in which a variable amount of time must elapse between the previous and subsequent times that reinforcement is available

RATIO SCHEDULES

- In a **fixed-ratio schedule,** reinforcement is provided after a fixed number of correct responses have been made.
  - It is a schedule in which reinforcement is provided after a fixed number of correct responses
- In a **variable-ratio schedule,** reinforcement is provided after a variable number of correct responses have been made
  - It is a schedule in which reinforcement is provided after a variable number of correct responses
- Fixed- and variable-ratio schedules both maintain a high response rate

APPLICATIONS OF OPERANT CONDITIONING

- We use operant conditioning everyday in our efforts to influence other people
- Operant conditioning also plays a role in attitude formation

BIOFEEDBACK TRAINING (BFT)

- Based on operant conditioning. It has enabled people and lower animals to learn to control autonomic responses to attain reinforcement
- In BFT, people receive reinforcement in the form of information.
- People use other instruments to learn to lower muscle tension, heart rates, and blood pressure

SHAPING

- Reinforces progressive steps toward the behavioral goal
- a procedure for teaching complex behaviors that at first reinforces approximations of the target behavior
- As training proceeds, we come to expect more before we are willing to provide reinforcement. We reinforce **successive approximations** of the goal.

SUCCESSIVE APPROXIMATIONS

- behaviors that are progressively closer to a target behavior

BEHAVIOR MODIFICATION

- Adults frequently reinforce undesirable behavior in children by paying attention to them, or punishing them, when they misbehave but ignoring them when they behave in desirable ways.

PROGRAMMED LEARNING

- B. F. Skinner developed an educational method called programmed learning that is based on operant conditioning.

- This method assumes that any complex task can be broken down into a number of small steps, which can be shaped individually and then combined in sequence to form the correct behavioral chain.
- It does not punish errors, instead, correct responses are reinforced, usually with immediate feedback.
- One can learn without making mistakes

COGNITIVE FACTORS IN LEARNING

- Classical and operant conditioning were originally conceived of as relatively simple forms of learning.
- In addition to **association** and **reinforcement,** cognitive psychologists use concepts, such as mental structures, schemas, templates, and information processing.
  - Cognitive psychologists see people as searching for information, weighing evidence, and making decisions.

LATENT LEARNING: FORMING COGNITIVE MAPS

- Many behaviorists argue that organisms acquire only responses, or operants, for which they are reinforced.
- E. C. Tolman, however, showed that rats also learn about their environment in the absence of reinforcement.
- He showed that rats must form **cognitive maps** of their surroundings
  - Tolman concluded that the rats had learned about the mazes by exploring them even when they were unrewarded for doing so, distinguishing **learning** and **performance**

LATENT LEARNING

- learning that is hidden or concealed

CONTINGENCY THEORY

- Behaviorists explain classical conditioning in terms of the pairing of stimuli.
- Cognitive psychologists explain classical conditioning in terms of the ways in which stimuli provide information that allows organisms to form or revise mental representations of their environment.
- **Robert Rescorla** conducted research in an effort to demonstrate which view is more accurate.

CONTINGENCY THEORY

- the view that learning occurs when stimuli provide information about the likelihood of the occurrence of other stimuli
- Suggests that learning occurs only when the conditioned stimulus (CS) provides information about the unconditioned stimulus (UCS).
- Rescorla concluded that learning occurs only when the CS provides information about the UCS

OBSERVATIONAL LEARNING

- It is the acquisition of knowledge and skills through the observation of others rather than by means of direct experience
- A person who engages in a response that is imitated is a **model**.
  - When we see modeled behavior being reinforced, we are said to be vicariously reinforced.
- Observational learning is not acquired through reinforcement.

MODEL

- an organism that engages in a response that is then imitated by another organism

MIRROR NEURONS

- Neurons that fire when an animal observes the behavior of another animal and that behavior stimulates imitative behavior in the first animal
- Allow us to anticipate other people's intentions when they reach for things.

VIOLENCE IN THE MEDIA AND AGGRESSION

- research strongly suggests that media violence is a risk factor for increasing emotional arousal, aggressive behavior, and violent thoughts
- It seems that violent video games create the greatest risk because they require audience participation
- Television is also a risk factor since it is one of our major sources of informal observational learning
    - If children watch two to four hours of television a day, they will have seen 8,000 murders and another 100,000 acts of violence by the time they have finished elementary school

CONSENSUS ON THE EFFECTS OF VIOLENCE IN THE MEDIA

- **Observational learning**: Children learn from observation. Television violence supplies models of aggressive "skills," which children may acquire. In fact, children are more likely to imitate what others do than to heed what they say.
- **Disinhibition**: Punishment inhibits or discourages behavior. Conversely, media violence may disinhibit aggressive behavior, especially when media characters "get away" with violence or are rewarded for it.
- **Increased emotional arousa**l: Media violence and aggressive video games increase viewers' level of emotional arousal. It "works them up." We are more likely to be aggressive—and less likely to consider right and wrong—when we are highly aroused.
- **Priming of aggressive thoughts and memories**: Media violence triggers aggressive ideas and memories.
- **Habituation**: We become used to repeated stimuli. Repeated exposure to media violence may decrease viewers' sensitivity to real violence. If children perceive violence as the norm, they may be more tolerant of it and place less value on restraining aggressive urges.
- **Provision of aggressive scripts**: Media violence provides aggressive scripts—or roles—as ways to manage various social situations.
- Violent video games are also connected with juvenile delinquency and dating violence).
    - Playing violent video games increases aggressive thoughts and behavior in the laboratory.
    - However, males are relatively more likely than females to act aggressively after playing violent video games and are more likely to see the world as a hostile place.
- Media violence and violent video games contribute to aggressive behavior, but aggressive youngsters are also more likely to seek it out.
- The family also affects the likelihood that children will imitate media violence.

TEACHING CHILDREN NOT TO IMITATE MEDIA VIOLENCE

- Children who watch violent shows may act less aggressively when they are informed that:
    - The violent behavior they observe in the media does not

represent the behavior of most people.

- The apparently aggressive behaviors they watch are not real. They reflect camera tricks, special effects, and stunts.
- Most people resolve conflicts by nonviolent means.
- The real-life consequences of violence are harmful to the victim and often to the aggressor.

INTRODUCTION TO PSYCHOLOGY

Year 1 | First Semester | Midterms

CHAPTER 7: REMEMBRANCE OF THINGS PAST—AND FUTURE

KINDS OF EXPERIENCE

- **EPISODIC MEMORIES** - memories of events that take place in one's life
  - Events that happen to a person or that take place in person's presence
- **PROSPECTIVE MEMORIES** - remembering to do something in the future
  - Ability to remember to perform intended actions in the future

EXPLICIT MEMORY

- Memory that clearly and distinctly expresses a specific information
- Also referred to as **declarative memory**
- Memory for specific information
- Things that are **explicit** are clear, or clearly stated or explained
- Use of the term *declarative* indicates that these memories state or reveal a specific information
- Information may be **autobiographical** (own life) or refer to general knowledge
- Its types are identified according to the type of information they hold

EPISODIC MEMORY (I REMEMBER…)

- Memories of the things that happen to us or take place in our presence
- Also referred to as **autobiographical memory**

SEMANTIC MEMORY (I KNOW…)

- General knowledge
- Semantic concern meanings
- Knowing something even though you have not experienced it or saw it yourself

IMPLICIT MEMORY

- Memory that is suggested (implied) but not plainly stated or verbally expressed, as illustrated in the things that people do but do not state clearly
- Also referred to as **nondeclarative memory**
- Memory of how to perform a task
- The act itself; doing something
- Involve procedures and skills, cognitive, and physical, and are also referred to as **procedural** or **skill memories**
- Can endure even when not used for years
- **PRIMING** - activation of specific associations in memory, often as result of repetition and without making a conscious effort to access the memory

RETROSPECTIVE MEMORY VS PROSPECTIVE MEMORY

RETROSPECTIVE MEMORY

- Recalling of information that has been previously learned
- Episodic, semantic, and implicit involve remembering things that were learned
- Memory for past events, activities, and learning experiences, as shown by explicit (episodic and semantic) and implicit memories

PROSPECTIVE MEMORY

- Remembering to do things in the future
- May fail when we are preoccupied, distracted, or "stressed out" about time
- Memory to perform an act in the future, as at a certain time or when a certain event occurs

Motivation also plays a role, since you are more likely to remember meeting someone who excites you.

- **Event-based tasks** are prospective memories triggered by events
- **Time-based tasks** are prospective memories that are to be performed at a certain time or after a certain amount of time has elapsed between occurrences
- **Age-related decline** takes place in retrospective and prospective memories
  - In the case of **prospective memory**, older adults appear about equally aware of specific cues or reminders as young adults; however, it takes them longer to respond to the cues or reminders
- **Moods** and **attitudes** affect prospective memory

PROCESSES OF MEMORY

- When we perceive information, we must change it into a form that can be remembered if we are to place it in our memory

ENCODING

- First stage of processing physical and chemical stimuli that reach our senses
- Modifies information so that it can be placed in memory
- Transform sensory input and our own ideas into psychological formats that can be represented mentally
- **Visual Code** - mentally representing an information as a picture or a mental image
- **Acoustic Code** - representing the stimuli as a sequence of sounds
- **Semantic Code** - represent stimuli in terms of their meaning

STORAGE

- Second stage of information processing
- Maintaining information over time
- **Maintenance rehearsal** - mental repetition of information to keep it in memory
- **Elaborative rehearsal** - the kind of coding in which new information is related to information that is already known

RETRIEVAL

- Third stage of information processing
- Locating information and returning it to consciousness
- With well-known information, retrieval is effortless
- Remembering large amounts of information, or information that is not perfectly understood, remembering can be difficult or can fail
- Requires knowledge of proper **retrieval cues**

MEMORY

- Process by which information is encoded, stored, and retrieved
- **Richard Atkinson** and **Richard Shiffrin** proposed three stages of memory that determine whether (and how long) information is retained

SENSORY MEMORY

- When we look at visual stimulus, our impressions may seem fluid enough, but, actually, they consist

of a series of eye fixations referred to as **SACCADIC EYE MOVEMENTS,** which jump from one point to another about 4 times each second
- Visual sensations seem continuous, or stream like, because of sensory memory

SENSORY MEMORY
- Type or stage of memory that is first encountered by a stimulus
- Although it holds impressions briefly, it is long enough so that a series of perceptions seem to be connected or are psychologically continuous

MEMORY TRACE
- Assumed change in the nervous system that reflects the impression made by a stimulus

WILLIAM MCDOUGALL (1904)
- Engaged in research in which he showed people 1 to 12 letters arranged in rows, just long enough to allow a single eye fixation
- Under these conditions, people could typically remember only four or five letters.
- Used a **whole-report procedure,** in which people were asked to report every letter they saw in the array

GEORGE SPERLING (1960)
- Modified McDougall's experimental method, and showed that there is a difference between what people can see and what they can report
- Used a modified **partial-report procedure,** in which people were asked to report the contents of one of three rows of letters
- He found that the amount of time that elapsed before he pointed to the row to be reported affected people's memory
- He concluded that the memory trace of visual stimuli decays within a second

ICONIC MEMORY

- Psychologists believe we possess a sensory register for each sense

ICONS
- Mental representation of a visual stimulus that is held briefly in sensory memory

ICONIC MEMORY
- Sensory register that briefly holds mental representations of visual stimuli
- Accurate, photographic memories, but brief

EIDETIC MEMORY
- What most people usually think as photographic memory—the ability to retain exact mental representations of visual stimuli over long periods
- Maintenance of detailed visual memories over several minutes

ICONIC MEMORY AND SACCADIC EYE MOVEMENT

- **SACCADIC EYE MOVEMENTS** occur about 4x a second, while **ICONIC MEMORY** holds icons for up to a second
  - This is the reason why the flow of visual information seems smooth and continuous

ECHOIC MEMORY

ECHOES
- Mental representations of sounds, or auditory stimuli, that is held briefly in sensory memory

ECHOIC MEMORY
- Sensory register that briefly holds mental representations of auditory stimuli
- Memory traces of auditory stimuli can last for several seconds, many times longer than the traces of visual stimuli (icons)

SHORT-TERM MEMORY

- Type or stage of memory that can hold information for up to a minute or so after the trace of the stimulus decays
- Also called **working memory**
- If you focus on a stimulus in the sensory register, you will tend to retain it in your own **short-term memory**
- According to **Goldman-Rakic,** "Working memory is the mental glue that links a thought through time from its beginning to its end" (1995)
- In short-term memory, the image tends to fade significantly after 10 to 12 seconds if it is not repeated or rehearsed

THE SERIAL POSITION EFFECT

- Tendency to recall the first and last items in a series
- May occur because we pay more attention to the first and last stimuli in a series
- They serve as the **visual** or **auditory boundaries** for the other stimuli

CHUNKING

CHUNK

- stimulus or group of stimuli that are perceived as a discrete piece of information

INTERFERENCE IN SHORT-TERM MEMORY

LLOYD AND MARGARET PETERSON (1959)

- Demonstrated how prevention of rehearsal can wreak havoc with short-term memory.
- **DISPLACE,** in memory theory, to cause information to be lost from short-term memory by adding new information

LONG-TERM MEMORY

- The third stage of information processing capable of relatively permanent storage
- **REPRESSION**, in Freud's psychodynamic theory, the ejection of anxiety-evoking ideas from conscious awareness

HOW ACCURATE ARE LONG-TERM MEMORIES?

- **ELIZABETH LOFTUS** noted that memories are distorted by our biases and needs, by the ways in which we conceptualize our worlds
- We represent much of our world in the form of **schemas**, which are ways of mentally representing the world, which can influence our perception of persons, objects, and situations.

HOW MUCH INFORMATION CAN BE STORED IN LONG-TERM MEMORY?

- human ability to store information is, for all practical purposes, unlimited
- Your long-term memory is a biochemical "hard drive" with no known limits on the amount of information it can store.
- New information may replace older information in short-term memory, but there is no evidence that long-term memories—those in "storage"—are lost by displacement

LEVELS OF PROCESSING INFORMATION

- People who use **elaborative rehearsal**, the kind of coding in which new information is related to information that is already known, to remember things are processing information at a deeper level than people who use **maintenance rehearsal**, mental repetition of information to keep it in memory)
- **FERGUS CRAIK** and **ROBERT LOCKHART** pioneered th**e levels-of-processing model of memory,** which holds that memories tend to endure when information is processed deeply—attended to, encoded carefully, pondered, and related to things we already know
    - Remembering relies on how carefully they attend to information and how deeply they process it, not on whether memories are transferred from one stage of memory to another

FLASHBULB MEMORIES

- We tend to remember events that are surprising, important, and emotionally stirring more clearly.
- One factor is the distinctness of the memory.
    - It is easier to discriminate stimuli that stand out. Such events are striking in themselves. The feelings caused by them are also special. It is thus relatively easy to pick them out from the storehouse of memories.

ORGANIZATION IN LONG-TERM MEMORY

- The storehouse of long-term memory is usually well organized
- We tend to categorize information based on their similarities
    - It allows us to make predictions about specific instances and to store information efficiently.
- We tend to organize information according to a **hierarchical structure,** which is an arrangement of items into groups or classes according to common or distinct features
- As we work our way up the hierarchy shown in the figure, we find more encompassing, or superordinate, classes to which the items below them belong

TIP-OF-THE-TONGUE PHENOMENON

- also known as the **feeling-of-knowing experience**
- Feeling that information is stored in memory although it cannot be readily retrieved

CONTEXT-DEPENDENT MEMORY

- information that is better retrieved in the context in which it was encoded and stored, or learned

STATE-DEPENDENT MEMORY

- information that is better retrieved in the physiological or emotional state in which it was encoded and stored, or learned

FORGETTING

NONSENSE SYLLABLES

- meaningless sets of two consonants, with a vowel sandwiched in between, that are used to study memory
- First used by **HERMANN EBINGHAUS (1850-1909)** to study memory and forgetting

MEMORY TASKS USED IN MEASURING FORGETTING

RECOGNITION

- One aspect of forgetting is failure to recognize something we have experienced.
- Recognition is the easiest type of memory task.
- We can recognize correct answers more easily than we can recall them unaided.

RECALL

- Specific memory task that involves retrieving information without direct cues
- Psychologists also often use lists of pairs of nonsense syllables, called **paired associates**, or nonsense syllables presented in pairs in experiments to measure recall.
- More difficult than recognition
- In a recall task, the person must retrieve a syllable, with another syllable serving as a cue.

RELEARNING

- Third method of measuring retention.
- Even when we cannot recall or recognize material that had once been learned, we can relearn it more rapidly the second time.
- To study the efficiency of relearning, **Ebbinghaus** devised the **method of savings**.
- **METHOD OF SAVINGS** - measure of retention in which the difference between the number of repetitions originally required to learn a list and the number of repetitions required to relearn the list after a certain amount of time has elapsed is calculated
    - **SAVINGS** - the difference between the number of repetitions originally required to learn a list and the number of repetitions required to relearn the list after a certain amount of time has elapsed
- We continue to forget material as time elapses but at a slower pace.

INTERFERENCE THEORY

- Material in short-term memory can be lost through decay. It can also be lost through displacement

INTERFERENCE THEORY

- the view that we may forget stored material because other learning interferes with it

RETROACTIVE INTERFERENCE

- the interference of new learning with the ability to retrieve material learned previously
- new learning interferes with the retrieval of old learning.

PROACTIVE INTERFERENCE

- the interference by old learning with the ability to retrieve material learned recently
- older learning interferes with the capacity to retrieve more recently learned material.

REPRESSION

- According to Freud, we are motivated to forget painful memories and unacceptable ideas because they produce anxiety, guilt, and shame.
- According to Freud, **REPRESSION** is the automatic ejection of painful memories and unacceptable urges from conscious awareness, motivated by the desire to avoid facing painful memories and emotions.
- **DISSOCIATIVE AMNESIA** - loss of memory of personal information that is thought to stem from psychological conflict or trauma

DO PEOPLE REALLY RECOVER REPRESSED MEMORIES OF CHILDHOOD?

- Despite shaky scientific support, there are cases of so-called recovered memories, especially memories of childhood sexual abuse by a relative, teacher, or friend.
- The question is whether these memories are induced by therapists who foster beliefs that become so deeply ingrained they seem like authentic memories.
- **Psychiatrist HAROLD LIEF** noted that "We don't know what percent of these recovered memories are real and what percent are pseudomemories," becoming one of the first to challenge such memories.
- **Psychologist Elizabeth Loftus** cites numerous studies that demonstrated how easy it was to implant a false memory.

INFANTILE AMNESIA

- Inability to recall events that occur prior to the age of three or so
- Also termed **childhood amnesia**
- Freud believed that young children have aggressive impulses and perverse lusts toward their parents.
  - He attributed infantile amnesia to repression of these impulses.
- There are also cognitive explanations for infantile amnesia:
  - Infants are not particularly interested in remembering the past.
  - Infants, in contrast to older children, tend not to weave episodes together into meaningful stories of their own lives.
  - Infants do not make reliable use of language to symbolize or classify events.

ANTEROGRADE AND RETROGADE AMNESIA

ANTEROGRADE AMNESIA

- Failure to remember events that occurred after physical trauma because of the effects of the trauma
- The ability to pay attention, the encoding of sensory input, and rehearsal are all impaired.

RETROGRADE AMNESIA

- failure to remember events that occurred prior to physical trauma because of the effects of the trauma
- In terms of stages of memory, it may be that our perceptions and ideas need to consolidate, or rest undisturbed for a while, if they are to be transferred to long-term memory

THE BIOLOGY OF MEMORY

- Early in the 20th century, many psychologists used the concept of the **engram** *(viewed as electrical circuits in the brain that corresponded to memory traces—neurological processes that paralleled experiences)* in their study of memory.

NEURAL ACTIVITY AND MEMORY

- Information received through other senses is just as likely to lead to corresponding changes in the cortical regions that represent them.
- Experiences perceived by several senses are apparently stored in numerous parts of the cortex.
- The recall of sensory experiences apparently involves neural activity in related regions of the brain.

- Research with sea snails, such as Aplysia and Hermissenda, offers more insight into the biology of memory.
- **LONG-TERM POTENTIATION** - enhanced efficiency in synaptic transmission that follows brief, rapid stimulation
- dendrites can also participate in LTP by sprouting new branches that attach to the transmitting axon.
- Serotonin and many other naturally occurring chemical substances, including **adrenaline, noradrenaline, acetylcholine, glutamate, antidiuretic hormone**, even the **sex hormones estrogen** and **testosterone** have also been shown to play roles in memory.

BRAIN STRUCTURES AND MEMORY

- Memory relies on complex neural networks that draw on various parts of the brain
- **The hippocampus** is vital in storing new information
  - involved in relaying sensory information to parts of the cortex.
  - Where and when of things
- The brain stores parts of memories in the appropriate areas of the **sensory cortex**.
  - Where memories are stored
- The **limbic system** is largely responsible for integrating these pieces of information
  - Essential for emotional aspects of memories
- **frontal lobes** apparently store information about where and when events occur
  - For organizing and contextualizing memories
- The **prefrontal cortex** is the executive center in memory
  - Ability to mentally travel through time
- The **thalamus** is involved in the formation of verbal memories.

CHAPTER 8: THINKING, LANGUAGE, AND INTELLIGENCE

THINKING

- Greek philosopher **Aristotle** argued that people differ from lower organisms in their capacity for **rational thinking**
- **THINKING** - paying attention to information, representing it mentally, reasoning about it, and making judgements and decisions about it
  - It refers to conscious, planned attempts to make sense of and change the world
  - Dreaming and daydreaming do not represent thinking because they are unplanned and proceed more or less their own

CONCEPTS

- **CONCEPTS** are mental categories used to group together objects, relations, events, abstractions, or qualities that have common properties.
- **COGNITIONS** is the mental activity involved in understanding, processing, and communicating information
- We tend to organize concepts in **hierarchies**
- **PROTOTYPES,** a concept of category of objects or events, is a good example of category.
  - Many simple prototypes, such as dog and red, are taught by means of examples or **exemplars** (specific examples)
- In language development, such over-inclusion of instances in category is labeled **over-extension**
- **Abstract concepts** such as bachelor or square root tend to be formed through explanation that involve more basic concepts

PROBLEM SOLVING

- **PROBLEM SOLVING** is the process of finding solutions to obstacles or challenges you encounter in life or work

METHODS OF PROBLEM SOLVING

UNDERSTANDING THE PROBLEM

- The parts of our mental representation of the problem relate to one another in a meaningful way.
- The elements of our mental representation of the problem correspond to the elements of the problem in the outer world.
- We have a storehouse of background knowledge that we can apply to the problem.

THE USE OF ALGORITHMS

- **ALGORITHM** is a specific procedure for solving a type of problem
  - It invariably leads to the solution if it is used properly
- **ANAGRAMS** are scrambled words, wherein the task is to try to reorganize jumbles or groups of letters into words
- **SYSTEMATIC RANDOM SEARCH** is an algorithm for solving problems in which each possible solution is tested according to a particular set of rules
- If you were to use **systematic random search** in solving anagram problems, you would list every possible letter combination. You could use a dictionary or a spell-checking program to see whether each result is a word. The method

might take a while, but it would work.

THE USE OF HEURISTIC DEVICES

- People sometimes use shortcuts to "jump to conclusions," and these are often correct conclusions.
- Shortcuts are called **HEURISTIC** or **HEURISTIC DEVICES,** which are rules of thumb that help us simplify and solve problems
  - Often based on strategies that worked in the past
  - Do not guarantee a correct solution, but when they work, they permit more rapid solutions
- **MEANS-END ANALYSIS** is one type of heuristic device, wherein we assess the difference between our current situation and our goals and do what we can to reduce this difference

THE USE OF ANALOGIES

- **ANALOGY** is a partial similarity among things that are different in other ways
- The **analogy heuristic** applies the solution of an earlier problem to the solution of a new one
  - We use analogy heuristic whenever we try to solve a new problem by referring to a previous problem
  - We tend to look for things that have helped us in similar situations

FACTORS THAT AFFECT PROBLEM SOLVING

- They way you approach a problem is central to how effective you are at solving it

EXPERTISE

- Experts solve problems more efficiently and rapidly than novices do
  - They know the particular area well.
  - They have a good memory for the elements in the problems.
  - They form mental images or representations that facilitate problem solving.
  - They relate the problem to similar problems.
  - They are more goal-directed and have efficient methods for problem solving.
- The experts apparently used more efficient methods than the novices.
- Experts seemed to use **parallel processing**, wherein they dealt simultaneously with two or more elements of the problems
- Novices were more likely to engage in **serial processing**—that is, to handle one element of the problem at a time.

MENTAL SETS

- Tendency to respond to a new problem with the same approach that helped solve similar problems
- Usually make our work easier, but they can mislead us when the similarity between problems is illusory.

ILLUSORY

- In Gestalt psychology, **Insight** is a sudden perception of relationships among elements of the mentally represented elements of a problem that permits its solution

INCUBATION

- In problem solving refers to standing back from the problem for a while as some process within may continue to work on it.
- Later, the answer may come to us in a flash of insight.
- Standing back from the problem may help by distancing us from unprofitable but persistent mental sets

FUNCTIONAL FIXEDNESS

- The tendency to view an object in terms of its name or familiar usage
- May hinder problem solving
- It can be similar to a mental set in that it makes it difficult to use familiar objects to solve problems in novel ways.

JUDGEMENT AND DECISION MAKING

- People make most of their decisions on the basis of limited information.
  - They take shortcuts. They use heuristic devices—rules of thumb—in judgments and decision making just as they do in problem solving

HEURISTIC IN DECISION MAKING

- According to **REPRESENTATIVE HEURISTIC,** people make judgments about events (samples) according to the populations of events that they appear to represent
- **AVAILABILITY HEURISTIC** is a decision-making heuristic in which our estimates of frequency or probability of events are based on how easy it is to find examples
- **ANCHORING AND ADJUSTMENT HEURISTIC** suggests that there can be a good deal of inertia in our judgments.
  - A decision-making heuristic in which a presumption or first estimate serves as a cognitive anchor; as we receive additional information, we make adjustments but tend to remain in the proximity of the anchor

THE FRAMING EFFECT

- The way in which wording, or the context in which information is presented, affects decision making

OVERCONFIDENCE

- Whether our decisions are correct or incorrect, most of us tend to be overconfident about them.
- There are several reasons for overconfidence:
  - We tend to be unaware of how flimsy our assumptions may be.
  - We tend to focus on examples that confirm our judgments and ignore those that do not.
  - Because our working memories have limited space, we tend to forget information that runs counter to our judgments.
  - We work to bring about the events we believe in, so they sometimes become self-fulfilling prophecies.

MOTIVATED REASONING

- Making decisions and judgments on the basis of emotion rather than careful evaluation of all the available evidence

LANGUAGE

COMMUNICATION BY NONHUMANS

- A language is a system of symbols along with rules that are used to manipulate the symbols.
- Symbols such as words stand for or represent objects, events, or ideas.
- Chimpanzees are our closest genetic relatives, sharing an estimated **98.42%** of their genetic code with humans.
- **Magnetic resonance imaging (MRI)** studies with chimpanzees and gorillas show that most of them, like humans, show enlargement in the left hemisphere of the cerebral cortex, in part of **Broca's area**.
  - The differences that remain between humans and chimps

are at least in part associated with capabilities such as fine control of the mouth and larynx that are not found in apes.
    - The genetic differences between chimps and humans probably explain why chimps cannot articulate speech but also apparently give chimps and other apes some meaningful ability to use language.

DO APES REALLY USE LANGUAGE?

- Although apes do not speak, they have been taught to use American Sign Language and other symbol systems.
- Critics of the view that apes can learn to produce language, such as Herbert Terrace and Steven Pinker, assert that:
    - Apes can string together signs in a given sequence to earn rewards, but animals lower on the evolutionary ladder, such as pigeons, can also peck buttons in a certain sequence to obtain a reward.
    - It takes apes longer to learn new signs than it takes children to learn new words.
    - Apes are unreliable in their sequencing of signs, suggesting that by and large they do not comprehend rules of grammar.
    - People observing apes signing may be subject to observer bias—that is, they may be seeing what they want to see.

WHAT IS LANGUAGE?

- **LANGUAGE** is the communication of thoughts and feelings by means of symbols that are arranged according to rules of grammar.
- True language is distinguished from the communication systems of lower animals by properties:
    - **SEMANTICITY:** the quality of language in which words are used as symbols for objects, events, or ideas
    - **INFINITE CREATIVITY:** The capacity to create rather than imitate sentences.
    - **DISPLACEMENT:** the quality of language that permits one to communicate information about objects and events in another time and place
- **Jean Piaget** believed that language reflects knowledge of the world but that much knowledge can be acquired without language.

LANGUAGE AND CULTURE

- Different languages have different words for the same concepts, and concepts do not necessarily overlap.

THE LINGUISTIC–RELATIVITY HYPOTHESIS

- Was proposed by **Benjamin Whorf**
- **Whorf** believed that language structures the way we perceive the world.
- The categories and relationships we use to understand the world are derived from our language.
- People who use only a few words to distinguish among colors seem to perceive the same color variations as people with more words.

LANGUAGE DEVELOPMENT: THE 2-YEAR EXPLOSION

PRELINGUISTIC VOCALIZATIONS

- newborn children have one inborn, highly effective form of verbal expression: crying—and more crying.

- But crying does not represent language; it is a **prelinguistic event**.
- During the second month, babies begin cooing, another form of prelinguistic expression that appears to be linked to feelings of pleasure.
- By the fifth or sixth month, children begin to babble. Children babble sounds that occur in many languages.
  - Babies' babbling frequently combines consonants and vowels, as in "ba," "ga," and, sometimes, the much-valued "dada".
  - "Dada" at first is purely coincidental (sorry, dads), despite the family's delight over its appearance.
- Babbling, like crying and cooing, is inborn and prelinguistic.
- Deaf children babble, and children from cultures whose languages sound very different all seem to babble the same sounds.
- But children single out the sounds used in the home within a few months.
- By the age of 9 or 10 months, they are repeating the sounds regularly, and foreign sounds are dropping out.
  - In fact, early experience in acquiring the phonemes (i.e., meaningful units of sound) native to one's own language can make it difficult to pronounce and even discriminate the phonemes used in other languages later in life.
- Children tend to utter their first word at 11 to 13 months, but a range of 8 to 18 months is normal.
- Parents often miss the first word because it is not pronounced clearly.
- The growth of vocabulary is slow at first. It may take children 3 to 4 months to achieve a 10-word vocabulary after they have spoken their first word.
- By about 18 months, children are producing a couple of dozen words.

DEVELOPMENT OF GRAMMAR

- The first linguistic utterances of children around the globe are single words that can express complex meanings, which are referred to as **HOLOPHRASES**
  - **HOLOPHRASES** are words used to express complex meanings
- Toward the end of the second year, children begin to speak two-word sentences, which was termed **TELEGRAPHIC SPEECH** because they resemble telegrams.
  - Telegrams cut out the "unnecessary" words.

OVERREGULARIZATION

- An important development for understanding the roles of nature and nurture in language development
- It is the application of regular grammatical rules for forming inflections to irregular verbs and nouns
- **Overregularization** reflects knowledge of grammar, not faulty language development.

OTHER DEVELOPMENTS

- By the age of six, children's vocabularies have expanded to 10,000 words, give or take a few thousand.
- By seven to nine, most children realize that words can have more than one meaning, and they are entertained by riddles and jokes that require some sophistication with language.

- Between the elementary school and high school years, language grows more complex, and children rapidly add to their vocabularies.
  - Vocabulary, in fact, can grow for a lifetime, especially in one's fields of specialization and interest.

NATURE AND NURTURE IN LANGUAGE DEVELOPMENT

LEARNING THEORY AND LANGUAGE DEVELOPMENT

- Learning theorists see language as developing according to laws of learning.
  - They usually refer to the concepts of imitation and reinforcement.
- From a social cognitive perspective, parents serve as models.
  - Children initially repeat the irregular verb forms they have heard accurately, apparently as a result of **modeling**.

THE NATIVIST APPROACH TO LANGUAGE DEVELOPMENT

- Holds that the innate factors—which make up children's nature—cause children to attend to and acquire language in certain ways.
- From this perspective, children bring neurological "prewiring" to language learning
- **PSYCHOLINGUISTIC THEORY** - the view that language learning involves an interaction between environmental factors and an inborn tendency to acquire language
- **Noam Chomsky** refers to the inborn tendency as a **language acquisition device (LAD)**.
- **LANGUAGE ACQUISITION DEVICE (LAD)** - in psycholinguistic theory, neural "prewiring" that facilitates the child's learning of grammar
- **"universal grammar"**—an underlying set of rules for turning ideas into sentences.

THEORIES OF INTELLIGENCE

- Thinking involves the understanding and manipulating of information, **intelligence** is the underlying ability to understand the world and cope with its challenges.
- **INTELLIGENCE** - a general mental capability that involves the ability to reason, plan, solve problems, think abstractly, comprehend complex ideas, learn quickly, and learn from experience

FACTOR THEORIES

- **Factor theories** argue that intelligence is made up of a number of mental abilities, ranging from one kind of ability to hundreds.
- In 1904, British psychologist **Charles Spearman** suggested that the behaviors we consider intelligent have a common underlying factor that he labeled ***g***, for *"general intelligence"* or broad reasoning and problem-solving abilities.
- ***g*** - Spearman's symbol for general intelligence, which he believed underlay more specific abilities
- Spearman supported his view by noting that people who excel in one area are also likely to excel in others.
  - But he also noted that even the most capable people are relatively superior in some areas—such as music or business or poetry.
  - For this reason, he suggested that specific, or ***s***, factors account for *specific abilities*.

- ***s*** - Spearman's symbol for specific factors, or s factors, which he believed accounted for individual abilities

- American psychologist **Louis Thurstone** (1938) analyzed tests of specific abilities and concluded that Spearman had oversimplified intelligence.
  - Thurstone's data suggested the presence of eight specific factors, which he labeled **primary mental abilities**

TABLE 7.1 PRIMARY MENTAL ABILITIES, ACCORDING TO THURSTONE

Ability	Definition
Visual and spatial abilities	Visualizing forms and spatial relationships
Perceptual speed	Grasping perceptual details rapidly, perceiving similarities and differences between stimuli
Numerical ability	Computing numbers
Verbal meaning	Knowing the meanings of words
Memory	Recalling information (e.g., words and sentences)
Word fluency	Thinking of words quickly (e.g., rhyming and doing crossword puzzles)
Deductive reasoning	Deriving examples from general rules
Inductive reasoning	Inferring general rules from examples

  - **PRIMARY MENTAL ABILITIES** - according to Thurstone, the basic abilities that make up intelligence examples include word fluency and numerical ability

THE THEORY OF MULTIPLE INTELLIGENCES

- **Howard Gardner's** theory of multiple intelligences proposes, instead, that there are a number of intelligences, not just one.
- Two of these "intelligences" are familiar ones: **language ability** and **logical–mathematical ability**.
- Gardner also refers, however, to **bodily–kinesthetic talents** (of the sort shown by dancers and athletes), **musical talent, spatial–relations skills, and two kinds of personal intelligence**: awareness of one's own inner feelings and sensitivity to other people's feelings.
- Gardner more recently added **"naturalist intelligence"** and **"existential intelligence."**
- **Naturalist intelligence** refers to the ability to look at natural events, such as kinds of animals and plants, or the stars above, and to develop insights into their nature and the laws that govern their behavior.
- **Existential intelligence** means dealing with the larger philosophical issues of life.

THE TRIARCHIC THEORY OF INTELLIGENCE

- Psychologist **Robert Sternberg** has constructed a three-pronged or triarchic theory of intelligence that includes analytical, creative, and practical intelligence
- **Analytical intelligence** can be defined as academic ability.
  - It enables us to solve problems and acquire new knowledge.
  - It is the type of intelligence measured by standard intelligence tests.
- **Creative intelligence** is defined by the ability to cope with novel situations and generate many possible solutions to problems.
- **Practical intelligence** ("street smarts") enables people to deal with other people, including difficult people, and to meet the demands of their environment.

EMOTIONAL INTELLIGENCE AND SOCIAL INTELLIGENCE

- **John Mayer** and his colleagues developed the theory of **emotional intelligence**, which holds that

social and emotional skills are a form of intelligence, just as academic skills are.

- **Emotional intelligence** resembles two of Gardner's "intelligences"—intrapersonal skills and interpersonal skills (including insight into the feelings of other people).
- Failure to develop emotional intelligence is connected with poor ability to cope with stress, depression, and aggressive behavior.
- **Linda Gottfredson**'s definition: Intelligence is "a very general mental capability that, among other things, involves the ability to reason, plan, solve problems, think abstractly, comprehend complex ideas, learn quickly and learn from experience.

CREATIVITY AND INTELLIGENCE

- **CREATIVITY** - the ability to generate novel and useful solutions to problems

- Creative people share a number of qualities:
  - They take chances.
  - They refuse to accept limitations.
  - They appreciate art and music.
  - They use common materials to make unique things.
  - They challenge social norms and take unpopular stands.
  - They challenge ideas that other people accept at face value.
- Creative problem solving demands divergent rather than **convergent thinking**.
- **CONVERGENT THINKING -** a thought process that narrows in on the single best solution to a problem
- **DIVERGENT THINKING** - a thought process that attempts to generate multiple solutions to problems
- Intelligence test questions usually require analytical, convergent thinking to focus in on the one right answer.
- Tests of creativity determine how flexible a person's thinking is.

THE MEASUREMENT OF INTELLIGENCE

THE STANFORD-BINET INTELLIGENCE SCALE

- The commonsense notion that academic achievement depends on children's intelligence led Alfred Binet and Theodore Simon to invent measures of intelligence.
- The current version is the Stanford–Binet Intelligence Scale (SBIS).
- Binet assumed that intelligence increases with age, so older children should get more items right than younger children.
- Binet therefore included a series of age-graded questions, arranged in order of difficulty.
- The **Binet–Simon scale** yielded a score called a **mental age (MA)**.
- The MA shows the intellectual level at which a child is functioning.
- The SBIS also yielded an **intelligence quotient (IQ)** rather than an MA.
- The IQ reflects the relationship between a child's mental age and his or her actual or **chronological age (CA)**.
- Use of this ratio reflects the fact that the same MA score has different implications for children of different ages.
- In 1912, German psychologist Wilhelm Stern suggested the IQ as a way to deal with this problem.

Stern computed IQ using this formula:

$$IQ = \frac{\text{Mental age (MA)}}{\text{Chronological age (CA)}} \times 100.$$

THE WECHSLER SCALES

- **David Wechsler** developed a series of scales for use with children and adults.
- The Wechsler scales group test questions into a number of separate subtests
- Each subtest measures a different intellectual task.
  - In this way, the Wechsler scales highlight people's relative strengths and weaknesses, as well as measure overall intellectual functioning.
- Wechsler's scales permit the computation of verbal and performance IQs.
- Wechsler also introduced the concept of the deviation IQ.

GROUP TESTS

- Group tests for children were first developed during World War I.

THE RELIABILITY AND VALIDITY OF INTELLIGENCE TESTS

- Over the years, the SBIS and the Wechsler scales have been shown to be reliable and valid.
- In terms of reliability, the scores are rather consistent from testing to testing.
  - This kind of reliability is called test–retest reliability.
- The tests also show validity in that the scores correlate moderately to highly with the variables they are supposed to predict, such as school performance, even though motivation and adjustment to the school setting are also involved

DIFFERENCES IN INTELLECTUAL FUNCTIONING

SOCIOECONOMIC AND ETHNIC DIFFERENCES

- There is a body of research suggestive of differences in intelligence—or, more precisely, intelligence test scores—between socioeconomic and ethnic groups.
- Many studies of IQ confuse the factors of social class and ethnicity because disproportionate numbers of African Americans, Latin Americans, and Native Americans are found among the lower socioeconomic classes.
- When we limit our observations to particular ethnic groups, we still find an effect for social class.
- Most psychologists believe that such ethnic differences reflect cultural attitudes toward education rather than inborn racial differences
- **Cattell's Culture-Fair Intelligence Test** evaluates reasoning through the child's ability to understand and use the rules that govern a progression of geometric designs
- **Goodenough's Draw-A-Person test** is based on the premise that children from all cultural backgrounds have had the opportunity to observe people and note the relationships between the parts and the whole.

GENDER DIFFERENCES

- Reviews of the research suggest that girls are somewhat superior to boys in verbal abilities, such as vocabulary, ability to generate sentences and words that are similar in meaning to other words,

spelling, knowledge of foreign languages, and pronunciation
- Males seem to do somewhat better at manipulating visual images in working memory.
- Males as a group excel in visual–spatial abilities of the sort used in math, science, and map-reading

NATURE AND NURTURE IN INTELLIGENCE

GENETIC INFLUENCES IN INTELLIGENCE

- Research on genetic influences has employed kinship studies, twin studies, and adoptee studies.
- **HERITABILITY** - the degree to which the variations in a trait from one person to another can be attributed to, or explained by, genetic factors

ENVIRONMENTAL INFLUENCES ON INTELLIGENCE

THE HOME ENVIRONMENT

- Children of parents who are emotionally and verbally responsive, furnish appropriate play materials, are involved with their children, encourage independence, and provide varied daily experiences obtain higher IQ scores later on

EDUCATION

- The findings on intelligence, the home environment, and educational experiences show that much indeed can be done to enhance intellectual functioning in children.

THE FLYNN EFFECT

- Philosopher and researcher J**ohn Flynn** (2007) found that IQ scores in the Western world increased substantially between 1947 and 2002, some 18 points in the United States.
- Psychologist **Richard Nisbett** (2009) argues that our genetic codes could not possibly have changed enough in half a century to account for this enormous difference and concludes that social and cultural factors such as the effects of improved educational systems and the penetration of the mass media must be among the reasons for the change.

INTRODUCTION TO PSYCHOLOGY

Year 1 | First Semester | Midterms

MOTIVATION AND EMOTION

THE PSYCHOLOGY OF MOTIVATION

- Concerned with the **why** of behavior.
- **MOTIVATION** is the state in which an organism experiences an inducement or incentive to do something
- To answer these whys, psychologists use concepts such as motives, needs, drives, and incentives

MOTIVES

- Hypothetical states that activate behavior, propelling us toward goals
- Inferred from behavior
- May take the form of needs, drives, and incentives, which are also inferred from behavior

NEEDS

- A state of deprivation
- Physiological and psychological
- We must meet physiological needs to survive
- Psychological needs include needs for achievement, power, self-esteem, social approval, and belonging
  - Not necessarily based on states of deprivation
  - Shaped by personal or social factors
- Needs give rise to drives

DRIVES

- A condition of arousal in an organism that is associated with a need
- Internal pushes
- Created by needs, which push us to satisfy the need
- Longer deprivation strengthens these drives
- **Physiological drives** are the counterparts of physiological needs.
- Psychological needs for approval, achievement, and belonging also give rise to drives
- We can also be driven to obtain incentives

INCENTIVE

- An object, person, or situation perceived as capable of satisfying a need or as desirable for its own sake
- External pulls
- External rewards/goals we desire which can satisfy a need or be appealing

THEORIES OF MOTIVATION

THE EVOLUTIONARY PERSPECTIVE

- Notes that many animals are neurally "**prewired**" (born with preprogrammed tendencies) to respond to certain situations in certain ways

INSTINCT

- Involving an inherited disposition to activate specific behavior patterns that ebay an organism to reach specific goals
- They are genetically transmitted from generation to generation
- **William James** and **William McDougall** asserted that humans have instincts that foster survival and social behavior
  - James numbered love, sympathy, and modesty as social instincts
  - McDougall compiled 12 "basic" instincts. Including hunger, sex, and self-assertion

CHAPTER 9: MOTIVATION AND EMOTION

DRIVE-REDUCTIONISM AND HOMEOSTASIS

- Sigmund Freud believed that tension motivates us to behave in ways that restore us to a resting state
  - This was similar to those of the **drive-reduction theory** of learning, as set forth by psychologists **Clark Hull** in the 1930s
- **DRIVE-REDUCTION THEORY** - the view that organisms learn to engage in behaviors that have the effect of reducing drives
- According to Hull, **primary drives** trigger arousal (tension) and activate behavior
- We learn to engage in behaviors that reduce the tension
- We also acquire drives called **acquired drives** through experience
- In all cases, reduction of tension is the goal
- Some people appear to acquire what could be considered excessive drives for money or affiliation
- Primary drives like hunger are triggered when we are in a state of deprivation.
  - Sensations of hunger motivate us to act in ways that will restore the bodily balance
  - Tendency to maintain a stray state is called **Homeostasis**

HOMEOSTASIS

- tendency of the body to maintain a steady state
- work like a thermostat
- when the temperature in a room drops below the set point, the heating system turns on. The heat stays on until the set point is reached

THE SEARCH FOR STIMULATION

- Physical needs give rise to drives like hunger and thirst
- In such cases, we are motivated to reduce the tension of stimulation that impinges on us
- However, in the cause of stimulus motives, organisms seek to increase stimulation
- Humans and other animals appear motivated to seek novel stimulation

HUMANISTIC THEORY

- Humanistic psychologists like Abraham Maslow suggest that human behavior is not just mechanical and aimed toward survival and the reduction of tension
- Maslow believed that people are also motivated by a conscious desire for personal growth
- Humanists note that people tolerate pain, hunger, and many other kinds of tension to obtain personal fulfillment
- Maslow believed that we are separated from other animals by our capacity for **Self-actualization,** or self-initiated striving to become what we believe we are capable of being

SELF-ACTUALIZATION

- According to Maslow and other humanistic psychologists, self-initiated striving to become what one is capable of being
- Maslow considered self-actualization to be as important a need in humans as hunger
- The need for self-actualization pushes people to strive to become concert pianists, chief executive officers, or best-selling authors—

wven when they have plenty of money to live on
- Maslow organized human needs into a hierarchy

MASLOW'S HIERARCHY OF NEEDS

- Maslow's ordering of needs from the most basic (physiological needs) to most elaborate and sophisticated
- Ranges from physiological needs such as hunger and thirst, through self-actualization
- He believed that we naturally strive to climb this hierarchy
- Critics of Maslow's theory argue that there is too much individual variation for the hierarchy of motives to apply to everyone
- Some people whose physiological, safety, and love needs are met show little interest in achievement and recognition

COGNITIVE PERSPECTIVES ON MOTIVATION

- Cognitive theorists note that people represent their worlds mentally.
- As in Piaget's cognitive developmental theory, they see people as natural scientists who strive to understand the world so that they can predict and control events.
    - Therefore, people try to eliminate inconsistencies and discrepancies in information so that their ability to make sense of the world remains whole.
- According to **cognitive-dissonance theory**, people are generally motivated to hold consistent beliefs and to justify their behavior. That is why we are generally more likely to appreciate what we must work to obtain.
- **Drive-reduction theory** may explain why we drink when thirsty, but stimulus motives might explain why we go clubbing and drink alcohol.

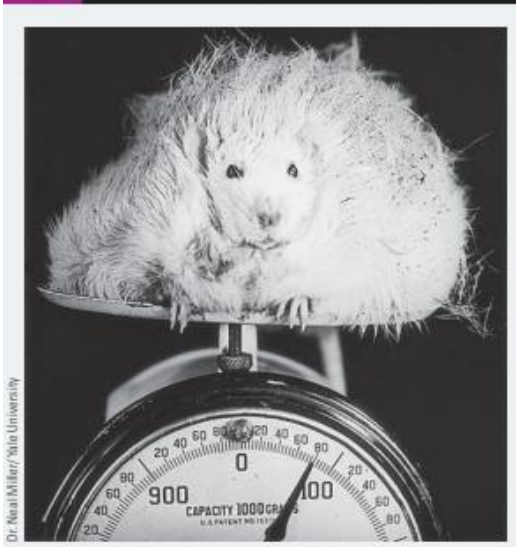
HUNGER

- We need food to survive, but for many of us, food means more than survival.
- We associate food with the nurturance of the parent–child relationship and with visits home on holidays.

BIOLOGICAL INFLUENCES ON HUNGER

- **SATIETY** - the state of being satisfied; fullness
- Mouth is where we get signals of **satiety** that regulate our eating
- We also get signals of satiety from the digestive tract, although it takes longer for these signals to reach the brain
- An empty stomach leads to stomach contractions, which we call **hunger pangs**.
- **HYPOTHALAMUS** - pea-sized structure in the brain that's involved in many aspects of motivation, including sex, aggression, and hunger.
    - Two parts of it that involve the hunger drive are of particular interest.
    - **VENTROMEDIAL NUCLEUS (VMN)** - a central area on the underside of the hypothalamus that appears to function as a stop-eating center
    - When the VMN is destroyed, the rat becomes **hyperphagic** (excessive eating)

FIG.8.2 A HYPERPHAGIC RAT



- **LATERAL HYPOTHALAMUS** - an area at the side of the hypothalamus that appears to function as a start-eating center
  - If you destroy the lateral hypothalamus, the rat may stop eating altogether—that is, become **aphagic** (undereating).

PSYCHOLOGICAL INFLUENCES ON HUNGER

- Research has confirmed that sitting around watching television increases the amount of food we eat.
- The more time we spend sitting rather than moving, the more weight we tend to gain

BEING OVERWEIGHT: A SERIOUS AND PERVASIVE PROBLEM

- Studies using data obtained by the federal government find that extremely obese people, whose BMIs are greater than or equal to 40, live shorter lives than people who are normal in weight

FACTORS IN BECOMING OVERWEIGHT

- Many biological and psychological factors are involved in obesity.
- Today, it is widely accepted that heredity plays a role in obesity in humans and other animals, including monkeys and rats
- The efforts of overweight people to maintain a slender profile may be sabotaged by an adaptive mechanism that would help preserve life in times of famine—**adaptive thermogenesis**.
  - This mechanism causes the body to produce less energy (burn fewer calories) when someone goes on a diet.
- Fatty tissue in the body also metabolizes (burns) food more slowly than muscle does.
  - For this reason, a person with a high fat-to-muscle ratio metabolizes food more slowly than a person of the same weight with more muscle.
- Meanwhile, the normal distribution of fat cells is different for men and women.
- We also live in an "**obesogenic environment**".
  - Foods high in sugar and fat are everywhere.
- **Situations** contribute as well.
  - Family celebrations, watching television, arguments, and tension at work can all lead to overeating or going off a diet.
- Efforts to diet may also be impeded by negative emotions such as depression and anxiety, which can lead to binge eating

EATING DISORDERS

- **EATING DISORDERS** - characterized by persistent, gross disturbances in eating patterns.

ANOREXIA NERVOSA

- life-threatening eating disorder characterized by extreme fear of being too heavy, dramatic weight loss, a distorted body image, and resistance to eating enough to reach or maintain a healthful weight.
- Mostly afflicts women during adolescence and young adulthood
- Affluent females may be at greater risk because they have greater access to fitness centers and health clubs.
  - They are also more likely to subscribe to magazines that idealize slender bodies and shop in boutiques that cater to svelte women.
- **FEMALE ATHLETE TRIAD** - describes women who have low availability of energy (as from eating poorly), menstrual problems, and lessened bone density, a cluster of health problems that usually afflicts people in late adulthood
- Distortion of the body image—seeing oneself as overweight—is a feature of anorexia

BULIMIA NERVOSA

- Entails repeated cycles of binge eating and purging.
- Like anorexia, it tends to afflict women during adolescence and young adulthood
- Binge eating often follows on the heels of food restriction—as in dieting
- Individuals with eating disorders tend to be perfectionists about their bodies.
  - They will not settle for less than their idealized body shape and weight

SEXUAL MOTIVATION AND SEXUAL ORIENTATION

- It is now widely understood that the adolescent pre-occupation with sex is strongly related to the surge in sex hormones that occurs at puberty.

HORMONES AND SEXUAL MOTIVATION

- Sex hormones can be said to fuel the sex drive.
- The most common sexual problem among women is lack of sexual desire or interest, and the sex drive in women is also connected to testosterone levels
- Sex hormones promote the development of male and female sex organs and regulate the menstrual cycle.
  - They affect the sex drive and promote sexual response; these are **activating effects**.
  - **ACTIVATING EFFECTS** - the arousal-producing effects of sex hormones that increase the likelihood of sexual behavior
- **ESTRUS** - the periodic sexual excitement of many female mammals as governed by levels of sex hormones

SEXUAL RESPONSE AND SEXUAL BEHAVIOR

- Men show more interest in sex than women do
- The sexual response cycle is characterized by vasocongestion and myotonia.
  - **Vasocongestion** is the swelling of the genital tissues with blood, causing erection of

the penis and swelling of the area surrounding the vaginal opening. The testes and the nipples swell as blood vessels dilate in these areas.
    - **MYOTONIA** - muscle tension, which causes grimaces, spasms in the hands and feet, and the spasms of orgasm.

EXCITEMENT PHASE

- The first phase of the sexual response cycle, which is characterized by muscle tension, increases in the heart rate, and erection in the male and vaginal lubrication in the female
- Vasocongestion during the excitement phase causes erection in men.
    - The scrotal skin thickens, becoming less baggy.
    - The testes increase in size and become elevated.
- In the female, excitement is characterized by vaginal lubrication.
    - Vasocongestion swells the clitoris, flattens and spreads the vaginal lips, and expands the inner part of the vagina.
- The nipples may erect in both men and women. Heart rate and blood pressure increase.

PLATEAU PHASE

- The second phase of the sexual response cycle, which is characterized by increases in vasocongestion, muscle tension, heart rate, and blood pressure in preparation for orgasm
- The level of sexual arousal remains somewhat stable during the plateau phase of the cycle.
- **EJACULATION** - the process of propelling seminal fluid (semen) from the penis
- In women, vasocongestion swells the outer part of the vagina and the inner vagina expands further.
- Heart rate may increase to 100 to 160 beats per minute.
- Blood pressure continues to rise.

ORGASMIC PHASE

- **ORGASM** - the height or climax of sexual excitement, involving involuntary muscle contractions, release of sexual tensions, and usually, subjective feelings of pleasure

RESOLUTION PHASE

- The fourth phase of the sexual response cycle, during which the body gradually returns to its pre-aroused state
- After orgasm, the body returns to its unaroused state.
- Men enter a **refractory period** during which they cannot experience another orgasm or ejaculate.
- Women do not undergo a refractory period and therefore can become quickly re-aroused if they so desire.

SEXUAL ORIENTATION AND GENDER IDENTITY

- **LGBT ACTIVISTS** - activist a person who fights for the rights of lesbians, gay males, bisexuals, and transgendered people
- **SEXUAL ORIENTATION** - concerns the direction of one's romantic interests and erotic attractions—toward members of one's own sex, the other sex, or both
- **HOMOSEXUAL** - referring to people who are sexually aroused by, and interested in forming romantic relationships with, people of the same gender

- **GAY MALE** - a male homosexual
- **LESBIAN** - a female homosexual

- **HETEROSEXUAL** - referring to people who are sexually aroused by, and interested in forming romantic relationships with, people of the other gender
  - erotically attracted to people of the other sex.
- Homosexuals and heterosexuals also prefer to develop romantic relationships with, respectively, people of their own sex or people of the other sex.
- **GENDER IDENTITY** - refers to one's sense of whether a person sees herself or himself as being female, male, or somewhere in between.
  - One's psychological sense of being female, male, or somewhere in between

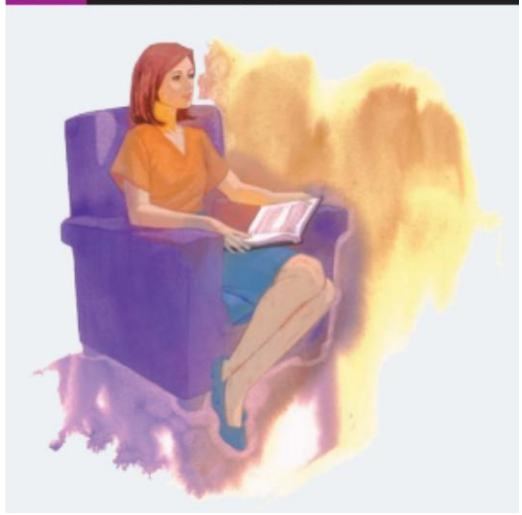
GENDER NONCONFORMITY

- Not behaving in a way that is consistent with the gender-role stereotype associated with one's sex in a given culture.

ACHIEVEMENT MOTIVATION

- Psychologist **David McClelland** (1958) helped pioneer the assessment of achievement motivation through evaluation of fantasies.
- One method involves the **Thematic Apperception Test (TAT)**, developed by **Henry Murray.**
  - The TAT contains cards with pictures and drawings that are subject to various interpretations.
  - Individuals are shown one or more TAT cards and asked to construct stories about the pictured theme: to indicate what led up to it, what the characters are thinking and feeling, and what is likely to happen.

FIG.8.5 WHAT IS GOING ON HERE?



- Classic studies find that people with high achievement motivation earn higher grades than people with comparable learning ability but lower achievement motivation.
  - They are more likely to earn high salaries and be promoted than less motivated people with similar opportunities
- People with high achievement motivation seem to prefer challenges and are willing to take moderate risks to achieve their goals.

EXTRINSIC VS. INTRINSIC MOTIVES

- **Carol Dweck** (2017) finds that achievement motivation can be driven by performance or learning goals, or both.
- Performance goals are usually met through extrinsic rewards such as praise and income.
  - Parents of children who develop performance goals are likely to respond to good

grades with rewards such as toys or money and to respond to poor grades with anger and removal of privileges.

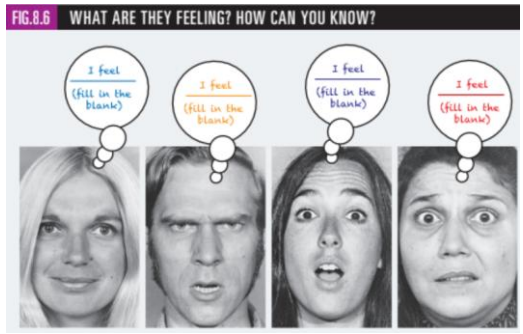
- Learning goals usually lead to intrinsic rewards, such as self-satisfaction.
  - Students who develop learning goals often have parents with strong achievement motivation who encourage their children to think and act independently.
  - Parents and teachers help children develop learning goals by showing warmth and praising them for their efforts to learn, exposing them to novel and stimulating experiences, and encouraging persistence

EMOTION

- Defined as feeling states with physiological, cognitive, and behavioral components.
- Emotions are associated with arousal of the **autonomic nervous system (ANS)**
- **PARASYMPATHETIC NERVOUS SYSTEM** - the branch of the autonomic nervous system that is most active during processes that restore reserves of energy to the body, such as relaxing and eating
- **Depression** usually involves cognitions of helplessness and hopelessness, parasympathetic arousal, and tendencies toward inactivity—or sometimes self-destruction.
- Happiness, grief, jealousy, disgust, embarrassment, liking—all have cognitive, physiological, and behavioral components.
- **HAPPY PEOPLE** - believe in their ability to effect change; are open to new experience; are willing to risk developing new relationships

THE EXPRESSION OF EMOTIONS

- Smiling is apparently a universal sign of friendliness and approval.
- Baring the teeth, as noted by Charles Darwin (1872) in the 19th century, may be a universal sign of anger.
- As the originator of the theory of evolution, Darwin believed that the universal recognition of facial expressions would have survival value.
- Most investigators agree that certain facial expressions suggest the same emotions in people who are reared in diverse cultures.
- Moreover, people in diverse cultures recognize the emotions indicated by certain facial expressions.
- **Paul Ekman** (1980) took photographs of people exhibiting anger, disgust, fear, happiness, sadness, and surprise



- Facial expressions sometimes occur in the absence of the emotion they are thought to accompany.
- The voice, posture, and gestures also provide clues to what people are feeling and are about to do.

POSITIVE PSYCHOLOGY

- The field of psychology that is about personal well-being and satisfaction; joy, sensual pleasure, and happiness; and optimism and hope for the future
- Genetic factors apparently play a powerful role in happiness.
- Similarly, negative life events, such as the loss of a loved one, financial reverses, or injuries can sadden us—and understandably so. Yet we may tend to bounce back to a more or less characteristic level of happiness
- Happy people are also more open to new experiences and new relationships.
- Research has also suggested that religious people are happier than those who are not
- There are the attitudinal aspects of happiness.
  - People at any income level can make themselves miserable when they compare themselves to people with more
- Happiness also tends to be accompanied by optimism—a cognitive bias toward assuming that things will work out
- But the "bias" is not groundless, because happy people often believe in their ability to effect change and then try harder.

THE FACIAL-FEEDBACK HYPOTHESIS

- Facial expressions reflect emotional states, and our ability to "read" these expressions enables us to interact appropriately with other people.
- **FACIAL-FEEDBACK HYPOTHESIS** - the view that stereotypical facial expressions can contribute to stereotypical emotions
- Psychological research has yielded some interesting findings concerning the facial-feedback hypothesis.
  - Inducing people to smile, for example, leads them to report more positive feelings and to rate cartoons as more humorous

THEORIES OF EMOTION

- The "commonsense theory" of emotions is that something happens (a situation) that is cognitively appraised (interpreted) by the person, and the feeling state (a combination of arousal and thoughts) follows.

THE JAMES-LANGE THEORY

- **William James** suggested that our emotions follow, rather than cause, our behavioral responses to events.
- At about the same time this view was also proposed by the Danish physiologist **Karl G. Lange**.
  - It is therefore termed the James–Lange theory of emotion.
- According to **James and Lange** (1922), certain external stimuli instinctively trigger specific patterns of arousal and action, such as fighting or fleeing
- Emotions are simply the cognitive representations (or by-products) of automatic physiological and behavioral responses.
- James–Lange theory is consistent with the facial-feedback hypothesis.
  - That is, smiling apparently can induce pleasant feelings, even

if the effect may not be strong enough to overcome deep feelings of sadness.

- The theory also suggests that we may be able to change our feelings by changing our behavior.
- Changing one's behavior to change one's feelings is one aspect of behavior therapy.

THE CANNON-BARD THEORY

- Walter Cannon (1927) and Philip Bard (1934) suggested that an event might simultaneously trigger bodily responses (arousal and action) and the experience of an emotion.
- When an event is perceived, the brain stimulates autonomic and muscular activity (arousal and action) and cognitive activity (experience of the emotion).
- According to the Cannon–Bard theory of emotion, emotions accompany bodily responses.
  - They are not produced by bodily changes, as in the James–Lange theory.

THE THEORY OF COGNITIVE APPRAISAL

- Cognitive appraisal is based on many factors, including our perception of events and the ways other people respond to those events
- When other people are present, we engage in social comparison to arrive at a response.

EVALUATION

- Research suggests that the patterns of arousal connected with various emotions are more specific than suggested by Schachter and Singer—although less so than suggested by James and Lange
- Research with brain imaging suggests that different emotions, such as happiness and sadness, involve different structures in the brain
- The emotion of disgust apparently has the most specific location, the **primary taste cortex**.
  - It makes sense that an emotion related to "distasteful" experiences would be centered here.
- Regarding the Schachter and Singer study, Zimbardo and his colleagues (1993) note that a lack of control over our feelings and a lack of understanding of what is going on are disturbing experiences.
  - Thus, our cognitive appraisal of situations does affect our emotional responses, but perhaps not as Schachter and Singer believed.

10: THE VOYAGE THROUGH THE LIFE SPAN

PRENATAL DEVELOPMENT

- The most dramatic gains in height and weight occur during prenatal development.
- Within nine months, the newly conceived organism develops from a nearly microscopic cell to a neonate (newborn) about 20 inches long.
- During the months following conception, the single cell formed by the union of sperm and egg—the **zygote**—multiplies, becoming two cells, then four, then eight, and so on.
- **ZYGOTE** - A fertilized ovum (egg cell)
- Following **conception** (which takes place in the fallopian tube), the zygote divides repeatedly as it proceeds on its three- to four-day voyage to the uterus.
    - The ball-like mass of multiplying cells wanders about the uterus for another three to four days before beginning to implant in the uterine wall.
- Implantation takes another week or so.
- The period from conception to implantation is called the **germinal stage**.
- The **embryonic stage** lasts from implantation until about the eighth week of development.
    - During this stage, the major body organ systems take form.
    - The growth of the head precedes that of other parts of the body.
    - The growth of the organs—heart, lungs, and so on—also precedes the growth of the extremities.
    - The relatively early maturation of the brain and the organ systems allows them to participate in the nourishment and further development of the embryo.



By the end of the first trimester, formation of all the major organ systems is complete. Fingers and toes are fully formed, and the sex of the fetus can be determined visually.



At this midway point between conception and birth, the fetus is covered with fine, downy hair, called *lanugo*.

    - The heart will continue to beat without rest every minute of every day for most of a century, perhaps longer.
- During the second month, the nervous system begins to transmit messages.
- By five to six weeks, the embryo is only a quarter-inch to a half-inch

long, yet nondescript sex organs have formed.
- By about the seventh week, the genetic code (XY or XX) begins to assert itself, causing the sex organs to differentiate.
  - If a Y sex chromosome is present, testes form and begin to produce androgens (male sex hormones), which further masculinize the sex organs.
  - In the absence of these hormones, the embryo develops sex organs typical of the female, regardless of its genetic code.
- By the end of the second month, the head has become rounded and the facial features distinct—all in an embryo that is about one-inch long and weighs one-thirtieth of an ounce.
  - As it develops, the embryo is suspended within a protective amniotic sac in the mother's uterus.
    - **AMNIOTIC SAC** - a sac within the uterus that contains the embryo or fetus
  - The sac is surrounded by a clear membrane and contains amniotic fluid.
    - The fluid serves as a natural air bag, allowing the child to move around without injury.
    - It also helps maintain an even temperature.
- From now until birth, the embryo exchanges nutrients and wastes with the mother through the placenta.
  - **PLACENTA** - a membrane that permits the exchange of nutrients and waste products between the mother and her developing child but does not allow the maternal and fetal bloodstreams to mix
- The embryo is connected to the placenta by the umbilical cord.
  - **UMBILICAL CORD** - a tube between the mother and her developing child through which nutrients and waste products are conducted
- The placenta is connected to the mother by blood vessels in the uterine wall.
- The **fetal stage** lasts from the beginning of the third month until birth.
- By the end of the third month, the major organ systems and the fingers and toes have formed.
- In the middle of the fourth month, the mother usually detects the first fetal movements.
- By the end of the sixth month, the fetus moves its limbs so vigorously that mothers often feel that they are being kicked.
- The fetus opens and shuts its eyes, sucks its thumb, and alternates between periods of wakefulness and sleep.
- During the three months prior to birth, the organ systems of the fetus continue to mature.
  - The heart and lungs become increasingly capable of sustaining independent life
  - The fetus gains about 51/2 pounds and doubles in length.
- Newborn boys average about 71/2 pounds and newborn girls about 7 pounds.

CHILDHOOD: PHYSICAL DEVELOPMENT

- During infancy (first 2 years of childhood), dramatic gains in height and weight continue

- Babies usually double their birth weight in about five months and triple it by their first birthday
  - Their height increases by about 10 inches in the first year.
- Children grow another four to six inches during the second year and gain some four to seven pounds.
- After that, they gain about two to three inches a year until they reach the adolescent growth spurt.
- Weight gains also remain fairly even at about four to six pounds per year until the spurt.

REFLEXES

REFLEXES

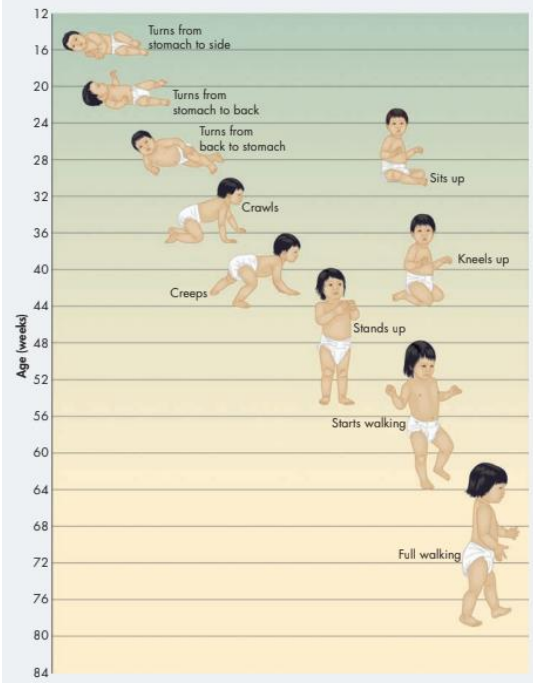
- Simple, inborn responses elicited by specific stimuli.
- Reflexes are essential to survival and occur automatically
- Newborn children do not know that it is necessary to eat to survive. Fortunately, rooting and sucking reflexes cause them to eat.
  - They turn their head toward stimuli that prod or stroke the cheek, chin, or corner of the mouth. This is termed **rooting**.
  - They also automatically suck objects that touch their lips.
- **ROOTING** - the turning of an infant's head toward a touch, such as by the mother's nipple
- Newborns use the **withdrawal reflex** to avoid painful stimuli.
- In the **startle**, or **Moro**, **reflex**, they draw up their legs and arch their backs in response to sudden noises, bumps, or loss of support while being held
- They grasp objects that press against the palms of their hands (the grasp, or **palmar reflex**)
- They fan their toes when the soles of their feet are stimulated (the **Babinski reflex**)
- Pediatricians test these reflexes to assess babies' neural functioning
- Babies also breathe, sneeze, cough, yawn, blink, defecate, and urinate reflexively

MOTOR DEVELOPMENT

- Refers to the progression from simple acts like lifting the head to running around
- **Maturation** and **experience** both play key roles in motor development.
- Maturation of the brain is a key to motor development.
- Motor development provides some of the most fascinating changes in infants, in part because so much seems to happen so quickly—and so much of it during the first year.
- Children go through a sequence that includes rolling over, sitting up, crawling, creeping, walking, and running.
- The ages at which infants first engage in these activities vary, but the sequence generally remains the same
- Invariant sequences suggest an unfolding of the genetic code (**maturation**)
- Role of maturation in areas such as physical development (height and weight gain), language development, and motor development is clear
- Environmental factors are also involved
  - Children may have certain genetic potentials for body size and growth rates, but they do not reach them unless environmental factors, such as

nutrition, and relatively clean air are available

FIG.9.2 MOTOR DEVELOPMENT: FROM LYING AROUND TO RUNNING AROUND—ALL IN LESS THAN TWO YEARS



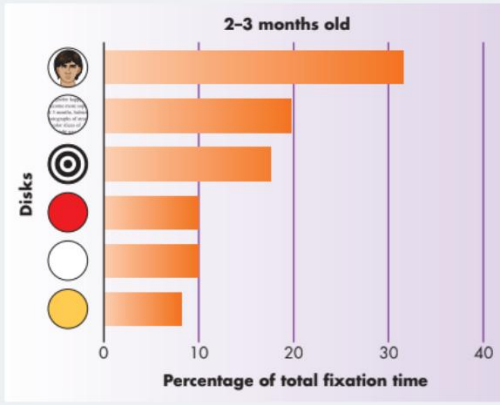
- Children do not understand or produce language until their genetic codes spark the development of certain structures and processes in the brain.
  - But the environment is also involved.
  - Children learn the languages used in their homes and communities. They do not speak foreign tongues without being exposed to them

PERCEPTUAL DEVELOPMENT

- Newborn children spend about 16 hours a day sleeping and do not have much opportunity to learn about the world. Yet they perceive the world reasonably well soon after birth.
- Within a couple of days, infants can follow, or "track," a moving light with their eyes.
- By three months, they can discriminate most colors.
- Newborns are nearsighted, meaning they focus better on nearby than distant objects but by about four months, infants focus on distant objects about as well as adults do
- The visual preferences of infants are measured by the amount of time, termed **fixation time**, they spend looking at one stimulus instead of another.

In classic research by **Robert Fantz (1961)**, two-month-old infants preferred visual stimuli that resembled the human face to newsprint, a bull's-eye, and featureless red, white, and yellow disks

FIG.9.3 TWO-MONTH-OLDS' PREFERENCES FOR VISUAL STIMULI: WHAT DO TWO-MONTH-OLDS LIKE TO LOOK AT?



- Classic research has shown that infants tend to respond to cues for depth by the time they are able to crawl (at about six to eight months).
- Most also have the good sense to avoid crawling off ledges and tabletops into open space
- Normal newborns hear well.
  - Most newborns reflexively turn their heads toward unusual sounds.
  - This finding, along with findings about visual tracking,

suggests that infants are preprogrammed to survey their environments.
    - Speaking or singing softly in a low-pitched tone soothes infants. This is why lullabies help infants fall asleep.
    - Research indicates that neonates prefer their mother's voice to those of other women, but they do not show a preference for their father's voice
    - Babies, of course, have had months of "experience" in the uterus.
    - For at least two or three months before birth, they have been able to hear. Because they are predominantly exposed to sounds produced by their mother, learning may contribute to newborn preferences.
- Neonates can also discriminate distinct odors, and their preferences are quite similar to those of older children and adults
- Maternal–infant recognition of each other's odor apparently contributes to mother–infant bonding

CHILDHOOD: COGNITIVE DEVELOPMENT

- Cognitive functioning develops over many years, young children have ideas about the world that differ considerably from those of adults.
- Many of these ideas are charming but illogical—at least to adults.

JEAN PIAGET'S COGNITIVE-DEVELOPMENT THEORY

- **Jean Piaget** (1896-1980) earned his Ph.D. in biology.
- His first task was to adapt English verbal reasoning items for use with French children. To do so, he had to try out the items on children in various age groups and see whether they could arrive at correct answers. The task was boring until Piaget became intrigued by the children's wrong answers.
- Another investigator might have shrugged them off, but Piaget perceived patterns in the children's "mistakes."
    - The wrong answers reflected consistent, if illogical, cognitive processes. Piaget's observations led to his theory of cognitive development.

ASSIMILATION

- Piaget described human thought, or intelligence, in terms of the concepts of assimilation and accommodation.
- **Assimilation** means responding to a new stimulus through existing cognitive structures.
    - According to Piaget, the inclusion of a new event into an existing schema
- Infants, for example, usually try to place new objects in their mouth to suck, feel, or explore.
- Piaget would say that the child is assimilating a new toy to the sucking **schema**.
- A **schema** is a pattern of action or a "mental structure" involved in acquiring or organizing knowledge.
    - According to Piaget, a hypothetical mental structure that permits the classification and organization of new information

ACCOMODATION

- It is the creation of new ways of responding to objects or looking at the world.
- According to Piaget, the modification of schemas so that information inconsistent with existing schemas can be integrated or understood
- With accommodation, children transform existing schemas to incorporate new events.
- For example, children who study biology learn that whales cannot be assimilated into the "fish" schema.
- They accommodate by constructing new schemas, such as "mammals without legs that live in the sea."
- The ability to accommodate to novel stimuli advances as a result of maturation and experience.

PIAGET'S STAGES OF COGNITIVE DEVELOPMENT

- Piaget hypothesized that children's cognitive processes develop in an orderly sequence.
- Some children may be more advanced than others, but the sequence remains the same
- Piaget (1963) identified four major stages of cognitive development:

THE SENSORIMOTOR STAGE

- Newborn infant is capable of assimilating novel stimuli only to existing reflexes (or ready-made schemas), such as the rooting and sucking reflexes. But by the time an infant reaches the age of one month, he or she already shows purposeful behavior by repeating behavior patterns that are pleasurable, such as sucking his or her hand.
- During the first month or so, an infant apparently does not connect stimuli perceived through different senses.
- Reflexive turning toward sources of auditory and olfactory stimulation cannot be considered purposeful searching.
- But within the first few months, the infant begins to coordinate vision with grasping to look at the object being held or touched.
- A three- or four-month-old infant may be fascinated by her own hands and legs.
    - The infant may become absorbed in watching herself open and close her fists.
    - The infant becomes increasingly interested in acting on the environment to make interesting results (such as the sound of a rattle) last longer or occur again.
    - Behavior becomes increasingly intentional and purposeful.
- Between four and eight months of age, the infant explores cause-and-effect relationships such as the thump made by tossing an object or the swinging that results from kicking a hanging toy.
- For most infants younger than six months, objects are not yet represented mentally.



A child makes no effort to search for an object that has been removed or placed behind a screen. By the age of 8 to 12 months, however, infants realize that objects removed from sight still exist and attempt to find them. In this way, they show what is known as object permanence, thereby making it possible to play peekaboo.

- **OBJECT PERFORMANCE** - recognition that objects removed from sight still exist, as demonstrated in young children by continued pursuit



- Between one and two years of age, children begin to show interest in how things are constructed.
  - It may be for this reason that they persistently touch and finger their parents' faces and their own.
- Toward the end of the second year, children begin to engage in mental trial and error before they try out overt behaviors
- Because the first stage of development is dominated by learning to coordinate perception of the self and of the environment with motor activity, Piaget termed it the **sensorimotor stage**.
  - This stage comes to a close with the acquisition of the basics of language at about age two.

THE PREOPERATIONAL STAGE

- Characterized by the use of words and symbols to represent objects and relationships among them.
- The second of Piaget's stages, characterized by illogical use of words and symbols, spotty logic, and egocentrism
- But be warned—any resemblance between the logic of children between the ages of two and seven and your own logic may be coincidental.
  - Children may use the same words as adults, but this does not mean their views of the world are the same.
- Preoperational children tend to think one-dimensionally—to focus on one aspect of a problem or situation at a time.
- One consequence of one-dimensional thinking is **egocentrism**.
  - **EGOCENTRISM,** according to Piaget, the assumption that others view the world as one does oneself
  - Preoperational children cannot understand that other people do not see things the same way they do
  - To egocentric preoperational children, all the world's a stage that has been erected to meet their needs and amuse them.
  - When asked, "What are television sets for?", they may answer, "To watch my favorite shows and cartoons."
- Preoperational children also show **animism**.
  - They attribute life and consciousness to physical objects like the sun and the moon.

- In addition, they also show **artificialism**.
  - They believe that environmental events like rain and thunder are human inventions.
  - Asked what causes thunder, a four-year-old may reply "A man grumbling."

TABLE 9.1 EXAMPLES OF PREOPERATIONAL THOUGHT

Type of Thought	Sample Questions	Typical Answers
Egocentrism	Why does it get dark out?	So I can go to sleep.
	Why does the sun shine?	To keep me warm.
	Why is there snow?	For me to play in.
Animism	Why do trees have leaves?	To keep them warm.
	Why do stars twinkle?	Because they're happy and cheerful.
Artificialism	What makes it rain?	Someone emptying a watering can.
	Why is the sky blue?	Somebody painted it.
	What is the wind?	A man blowing.

- **LAW OF CONCENTRATION** - according to Piaget, recognition that basic properties of substances such as weight and mass remain the same when superficial features change
  - Holds that basic properties of substances such as mass, weight, and volume remain the same—that is, are conserved—when one changes superficial properties such as their shape or arrangement.
  - Conservation requires the ability to think about, or center on, two aspects of a situation at once, such as height and width.
  - Conserving the mass, weight, or volume of a substance requires the recognition that a change in one dimension can compensate for a change in another.



- The preoperational child focuses on the most apparent dimension of the situation—in this case, the greater height of the thinner glass
  - He does not realize that the increased width of the squat glass compensates for the decreased height.
- Piaget (1948) found that the moral judgment of preoperational children is also one-dimensional.
  - Five-year-olds tend to be slaves to rules and authority.
  - When you ask them why something should be done in a certain way, they may insist, "Because that's the way to do it!" or "Because Mommy says so!" Right is right and wrong is wrong. Why? "Because!"—that's why.
- According to most older children and adults, an act is a crime only when there is criminal intent.
- But in the court of the one-dimensional, pre-operational child, there is **objective responsibility**.
  - **OBJECTIVE RESPONSIBILITY** - according to Piaget, the assignment of blame according to the amount of damage done rather than the motives of the actor

- People are sentenced (and harshly!) on the basis of the amount of damage they have done, not their motives or intentions.
- To demonstrate objective responsibility, Piaget would tell children stories and ask them which character was naughtier and why.

John, for example, accidentally breaks 15 cups when he opens a door. Henry breaks one cup when he sneaks into a kitchen cabinet to find forbidden jam.

- The preoperational child usually judges John to be naughtier. Why? Because he broke more cups.

THE CONCRETE OPERATIONAL STAGE

- By about **age seven**, the typical child is entering the **concrete operational stage**.
- Piaget's third stage, characterized by logical thought concerning tangible objects, conservation, and subjective morality
- In this stage, which **lasts until about age 12**, children show the beginnings of the capacity for adult logic.
- However, their logical thoughts, or operations, generally involve tangible objects rather than abstract ideas.
- Concrete operational children are capable of **decentration**; they can center on two dimensions of a problem at once.
  - This attainment has implications for moral judgments, conservation, and other intellectual undertakings.
- **DECENTRATION** - simultaneous focusing on more than one dimension of a problem, so that flexible, reversible thought becomes possible
- Children now show **subjective moral judgments**.
  - **SUBJECTIVE MORAL JUDGEMENT** - according to Piaget, moral judgment that is based on the motives of the perpetrator
  - When assigning guilt, they center on the motives of wrongdoers as well as on the amount of damage done.
  - Concrete operational children judge Henry more harshly than John because John's misdeed was an accident.
- Concrete operational children understand the laws of conservation.
- **REVERSIBILITY** - recognition that many processes can be reversed or undone so that things are restored to their previous condition.
  - Centering simultaneously on the height and the width of the glasses, the boy recognizes that the loss in height compensates for the gain in width.
- Children in this stage are less egocentric.
  - They are able to take on the roles of others and to view the world, and themselves, from other people's perspectives.
  - They recognize that people see things in different ways because of different situations and sets of values.
- During the concrete operational stage, children's own sets of values begin to emerge and acquire stability.
  - Children come to understand that feelings of love between them and their parents can

endure even when someone is temporarily angry or disappointed.

EVALUATION OF PIAGET'S THEORY

- **Was Piaget's timing accurate?** Some critics argue that Piaget's methods led him to underestimate children's abilities
  - Other researchers using different methods have found, for example, that preschoolers are less egocentric and that children are capable of conservation at earlier ages than Piaget thought.
- **Does cognitive development occur in stages?** Cognitive events such as egocentrism and conservation appear to develop more continuously than Piaget thought—that is, they may not occur in stages
  - Although cognitive developments appear to build on previous cognitive developments, the process may be more gradual than stage like.
- **Are developmental sequences always the same?** Here, Piaget's views have fared better.
  - It seems there is no variation in the sequence in which cognitive developments occur.
- In sum, Piaget's theoretical edifice has been rocked, but it has not been reduced to rubble.
- **Lev Vygotsky (1896–1934)**, a Russian psychologist whose work was banned in communist Russia.
  - Unlike Piaget, Vygotsky was not a stage theorist.
  - Instead, he saw the transmission of knowledge from generation to generation as cumulative, and focused on the ways in which children's interactions with their elders enhance their cognitive development.

LEV VYGOTSKY'S SOCIOCULTURAL THEORY

- **Vygotsky's sociocultural theory** focuses instead on the ways in which children's cognitive development is influenced by the cultures in which they are reared and the people who teach them.
- Vygotsky's theory (1978) focuses on the transmission of information and cognitive skills from generation to generation.
  - The transmission of skills involves teaching and learning, but Vygotsky did not view learning as a mechanical process that can be described in terms of conditioning.
  - Rather, he focused more generally on how the child's social interaction with adults, largely in the home, organized a child's learning experiences in such a way that the child can obtain cognitive skills—such as computation or reading skills—and use them to acquire information.
- Like Piaget, Vygotsky saw the child's functioning as adaptive and the child adapts to his or her social and cultural interactions.
- Key concepts in Vygotsky's theory include the **zone of proximal development** and **scaffolding**.
- **ZONE OF PROXIMAL DEVELOPMENT (ZPD)** - Vygotsky's term for the situation in which a child carries out tasks with the help of someone who is more skilled, frequently an adult

who represents the culture in which the child develops
    - The word proximal means "nearby" or "close," as in the words approximate and proximity.
    - It refers to a range of tasks that a child can carry out with the help of someone who is more skilled.
    - The "zone" refers to the relationship between the child's abilities and what she or he can do with help from others.
    - Adults or older children best guide the child through this zone by gearing their assistance to the child's capabilities
    - Within the zone, the child works with and learns from others
    - When learning with others, the child tends to internalize—or bring inward—the conversations and explanations that help him or her gain skills
    - Children not only learn the meanings of words from teachers but also learn ways of talking to themselves about solving problems within a cultural context.
    - Outer speech becomes inner speech. What was the teacher's becomes the child's.
    - What was a social and cultural context becomes embedded within the child; thus we have the term, **sociocultural theory**.
- **SCAFFOLDING** - Vygotsky's term for temporary cognitive structures or methods of solving problems that help the child as he or she learns to function independently
    - A scaffold is a temporary skeletal structure that enables workers to fabricate a building, bridge, or other, more permanent, structure.
    - Cognitive scaffolding refers to the temporary support provided by a parent or teacher to a child who is learning to perform a task.
    - The amount of guidance decreases as the child becomes more skilled and self-sufficient.
    - In Vygotsky's theory, teachers and parents provide children with problem-solving methods that serve as cognitive scaffolding while the child gains the ability to function independently.
- Vygotsky focused on the processes in the teacher–learner relationship.

LAWRENCE KOHLBER'S THEORY OF MORAL DEVELOPMENT

- Cognitive-developmental theorist Lawrence Kohlberg (1981) used the following tale in his research into children's moral reasoning:

  In Europe a woman was near death from a special kind of cancer. There was one drug that the doctors thought might save her. It was a form of radium that a druggist in the same town had recently discovered. The drug was expensive to make, but the druggist was charging 10 times what the drug cost him to make. He paid $200 for the radium and charged $2,000 for a small dose of the drug. The sick woman's husband, Heinz, went to everyone he knew to borrow the money, but

he could only get together about $1,000, which was half of what it cost. He told the druggist that his wife was dying and asked him to sell it cheaper or let him pay later. But the druggist said: "No, I discovered the drug, and I'm going to make money from it." So Heinz got desperate and broke into the man's store to steal the drug for his wife (Kohlberg, 1969).

- As a stage theorist, Kohlberg argues that the stages of moral reasoning follow a specific sequence.
- Children progress at different rates, and not all children (or adults) reach the highest stage. But the sequence is always the same:
  - Children must go through stage 1 before they enter stage 2, and so on.
- According to Kohlberg, there are three levels of moral development and two stages within each level.
- When it comes to the dilemma of Heinz, Kohlberg believed that people could justify Heinz's theft of the drug or his decision not to steal it by the reasoning of any level or stage of moral development.
  - In other words, Kohlberg was not as interested in the eventual "yes" or "no" as he was in how a person reasoned to arrive at yes or no.

THE PRECONVENTIONAL LEVEL

- According to Kohlberg, evaluating importance of individual needs and personal conscience in regard to maintenance of the social order
- Applies to most children through about the age of nine.
- Children at this level base their moral judgments on the consequences of behavior.
- **STAGE 1:** Obedience and Punishment orientation
  - Right and wrong are judged by consequences
  - Motivation is to avoid punishment
- **STAGE 2:** Individualism and Exchange (Instrumental orientation)
  - Morality is based on self-interest and reciprocity

THE CONVENTIONAL LEVEL

- According to Kohlberg, a period during which moral judgments largely reflect social conventions; a "law and order" approach to morality
- Right and wrong are judged by conformity to conventional standards of right and wrong.
- Moral behavior is what is "normal"—what the majority does.
- Many people do not mature beyond the conventional level.
- **STAGE 3:** Interpersonal relationships
  - Decisions are guided by social approval and maintaining relationships
- **STAGE 4:** Maintaining social order
  - Morality is tied to obeying laws, rules, and authority to maintain order

THE POSTCONVENTIONAL LEVEL

- Postconventional moral reasoning is more complex and focuses on dilemmas in which individual needs are pitted against the need to maintain the social order and on personal conscience.
- **STAGE 5:** Social contract and individual rights
  - Laws are seen as social contracts that should promote the greatest good, but they can be changed if unjust
- **STAGE 6:** Universal ethical principles

- Morality is based on abstract principles like justice, equality, and human dignity, even if they conflict with laws

EVALUATION OF KOHLBERG'S THEORY

- As Kohlberg's theory predicts, evidence supports the view that the moral judgments of children develop in an upward sequence
- Postconventional thought, if found at all, first occurs in adolescence.
- Formal operational thinking is apparently a prerequisite, and education is likely to play a role
- Postconventional reasoning involves understanding of abstract moral principles and empathy with the views and feelings of others.
- Kohlberg believed that the stages of moral development were universal, but he may have underestimated the influence of social, cultural, and educational institutions
- Parents are also important.
  - Using reason in discipline and discussing the feelings of others advance moral reasoning.

CHILDHOOD: SOCIAL AND EMOTIONAL DEVELOPMENT

- Social relationships are crucial to us as children.
- When we are infants, our very survival depends on them.
- Later in life, they contribute to our feelings of happiness and satisfaction.

ERIK ERIKSON'S STAGES OF PSYCHOSOCIAL DEVELOPMENT

- According to Erik Erikson, we undergo several stages of psychosocial development.
- During Erikson's first stage, trust versus mistrust, we depend on our primary caregivers (usually our parents) and come to expect that our environments will—or will not—meet our needs.
- Toddlers through about the age of three are said to be in the stage of autonomy versus shame and doubt.
- During this period, their relationships with parents and friends can encourage the development of self-direction and initiative, or feelings of shame and guilt.
- Children in this stage need to develop feelings of self-control over physical functions—such as toileting—and a sense of independence.
- One of the ways that many children demonstrate their growing autonomy, much to the dismay of their parents, is by refusing to comply with parental requests or commands.
- Erikson believed that children are in the stage of **initiative versus guilt** through about the age of five, in which they begin to assert control over the environment and strive to master adult skills.
- Erikson labeled the years of about 6 to 12 the stage of **industry versus inferiority**, during which children meet academic and social challenges in school.
  - A positive outcome contributes to a sense of industry, whereas setbacks can lead to feelings of inferiority.

ATTACHMENT

- Psychologist **Mary D. Salter Ainsworth (1913–1999)** defined attachment as an emotional tie that is formed between one animal or person and another specific individual.
- It is the enduring affectional tie that binds one person to another
- Attachment keeps organisms together—it is vital to the survival of the infant—and it tends to endure.
- The behaviors that define attachment include:
  (1) attempts to maintain contact or nearness, and
  (2) shows of anxiety when separated.
- Babies and children try to maintain contact with caregivers to whom they are attached.
  - They engage in eye contact, pull and tug at them, ask to be picked up, and may even jump in front of them in such a way that they will be "run over" if they are not picked up.

THE STRANGE SITUATION AND PATTERNS OF ATTACHMENT

- Ways in which infants behave in strange situations are connected with their bonds of attachment with their caregivers.
- Given this fact, Ainsworth and her colleagues (1978) innovated the strange situation method to learn how infants respond to separations and reunions with a caregiver (usually the mother) and a stranger.
- Using this method, Ainsworth and her colleagues identified three major types of attachment, including secure attachment and two types of insecure attachment:

1. **SECURE ATTACHMENT.** Securely attached infants mildly protest their mother's departure, seek interaction upon reunion, and are readily comforted by her.
2. **AVOIDANT ATTACHMENT**. Infants who show avoidant attachment are least distressed by their mother's departure. They play by themselves without fuss and ignore their mothers when they return.
3. **AMBIVALENT/RESISTANT ATTACHMENT**. Infants with ambivalent/resistant attachment are the most emotional. They show severe signs of distress when their mother leaves and show ambivalence upon reunion by alternately clinging to and pushing their mother away when she returns.

- Attachment is connected with the quality of care that infants receive.
  - The parents of securely attached children are more likely to be affectionate and reliable caregivers
- Insecure attachment in infancy predicts psychological disorders during adolescence

PHASES OF ATTACHMENT

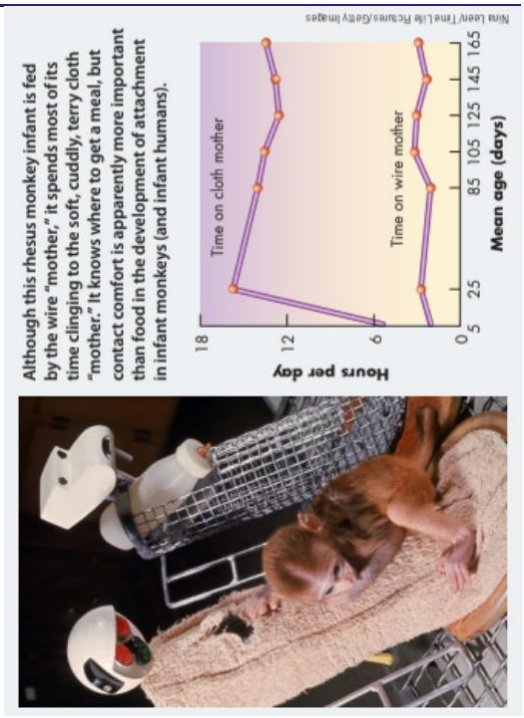
1. **THE INITIAL PRE-ATTACHMENT PHASE**, which lasts from birth to about three months and is characterized by indiscriminate attachment. That is, babies prefer being held or being with someone to being alone, but they are generally willing to be held by unfamiliar people.
2. **THE ATTACHMENT - IN - THE - MAKING PHASE**, which occurs at about three or four months and is characterized by preference for familiar figures.
3. **THE CLEAR-CUT-ATTACHMENT PHASE**, which occurs at about six or seven months and is characterized by intensified

dependence on the primary caregiver.

- **John Bowlby** (1988), a colleague of Ainsworth, believed that attachment is also characterized by fear of strangers ("stranger anxiety").
  - That is, at about 8 to 10 months of age, children may cry and cling to their parents when strangers try to befriend them.
  - But not all children develop fear of strangers.
  - It therefore does not seem necessary to include fear of strangers as an essential part of the process of attachment.

THEORETICAL VIEWS OF ATTACHMENT

- Early in the 20th century, behaviorists argued that attachment behaviors are learned through experience.
- Caregivers feed their infants and tend to their other physiological needs.
  - Thus, infants associate their caregivers with gratification of needs and learn to approach them to meet their needs
  - The feelings of gratification associated with the meeting of basic needs generalize into feelings of security when the caregiver is present.
- Classic research by psychologist **Harry F. Harlow** suggests that skin contact may be more important than learning experiences.
- Harlow noted that infant rhesus monkeys reared without mothers or companions became attached to pieces of cloth in their cages.
  - They maintained contact with them and showed distress when separated from them.



- In one study, Harlow placed infant rhesus monkeys in cages with two surrogate mothers.
  - One "mother" was made of wire mesh from which a baby bottle was extended.
  - The other surrogate mother was made of soft, cuddly terry cloth.
  - The infant monkeys spent most of their time clinging to the cloth mother, even though "she" did not gratify their need for food. Harlow concluded that monkeys—and perhaps humans—have an inborn need for **contact comfort** that is as basic as the need for food.
  - **CONTACT COMFORT** - a hypothesized primary drive to seek physical comfort through contact with another
- Gratification of the need for contact comfort, rather than food, might be why infant monkeys (and humans) cling to their mothers.

- Skin-to-skin contact in humans also increases the secretion of the hormone **oxytocin**, and oxytocin is connected with cuddling, caring, and development of the parent–infant relationship
- Other researchers, such as **ethologist Konrad Lorenz**, argue that for many animals, attachment is an instinct—inborn.
  - Ethologists study the behavioral characteristics of various species of animals.
- Attachment, like other instincts, is theorized to occur in the presence of a specific stimulus and during a **critical period** of life—that is, a period during which the animal is sensitive to the stimulus.
  - **CRITICAL PERIOD** - a period of time when an instinctive response can be elicited by a particular stimulus
- Some animals become attached to the first moving object they encounter.
  - The formation of an attachment in this manner is therefore called **imprinting**.
  - **IMPRINTING** - a process occurring during a critical period in the development of an organism, in which that organism responds to a stimulus in a manner that will afterward be difficult to modify
- **Lorenz** became well known when pictures of his "family" of goslings were made public.
  - He was present when the goslings hatched and during their **critical period**, and he allowed them to follow him.
  - The critical period for geese and some other animals is bounded, at the younger end, by the age at which they first walk and, at the older end, by the age at which they develop fear of strangers.
  - The goslings followed Lorenz persistently, ran to him when frightened, honked with distress at his departure, and tried to overcome barriers between them.
  - If you substitute crying for honking, it all sounds rather human.
- **Ainsworth and Bowlby** (1991) consider attachment to be instinctive in humans.
  - However, among humans attachment is less related to issues such as locomotion and fear of strangers (which is not universal).
  - Moreover, the critical period with humans is quite extended.

PARENTING STYLES

- **Diana Baumrind** has been particularly interested in the connections between parental behavior and the development of children's instrumental competence—the ability to manipulate the environment to achieve one's goals.
- Baumrind has focused largely on four aspects of parental behavior: **(1)** restrictiveness; **(2)** demands for the child to achieve intellectual, emotional, and social maturity; **(3)** communication ability; and **(4)** warmth and involvement.
- She labeled the three parenting styles the authoritative, authoritarian, and permissive styles.
- Other researchers also speak of the uninvolved style. These four styles are defined in the following ways:

1. **AUTHORITATIVE PARENTS**. The parents of the most competent children rate high in all four areas of behavior.
- Parents who are strict and warm; authoritative parents demand mature behavior but use reason rather than force in discipline
- Such authoritative parents are strict (restrictive) and demand mature behavior.
- However, they temper their strictness with a desire to reason with their children and with love and support.
- They expect much, but they explain why and offer help.

2. **AUTHORITARIAN PARENTS**. Have strict guidelines about what is right and wrong, and they demand that their children follow those guidelines.
- Parents who are rigid in their rules and who demand obedience for the sake of obedience
- Both authoritative and authoritarian parents adhere to strict standards of conduct, but authoritative parents explain their demands and are supportive, whereas authoritarian parents rely on force and communicate poorly with their children
- Authoritarian parents do not respect their children's points of view, and they may be cold and rejecting.

3. **PERMISSIVE PARENTS**. are generally easygoing with their children.
- Parents who impose few, if any, rules and who do not supervise their children closely
- As a result, the children do pretty much what the children want
- Permissive parents are warm and supportive, but poor at communicating.

4. **UNINVOLVED PARENTS**. Tend to leave their children on their own.
- Parents who generally leave their children to themselves
- They make few demands and show little warmth or encouragement.

- Research shows that the children of warm parents are more likely to be socially and emotionally well adjusted
  - They are also more likely to internalize moral standards—that is, to develop a conscience.
- Strictness seems to pay off, provided it is tempered with reason and warmth.
- Children of authoritative parents have greater self-reliance, self-esteem, social competence, and achievement motivation than other children
- Children of authoritarian parents are often withdrawn or aggressive and usually do not do as well in school as children of authoritative parents.
- Children of permissive parents seem to be less mature.
  - They are often impulsive, moody, and aggressive.
- In adolescence, lack of parental monitoring is often linked to delinquency and poor academic performance.
- Children of uninvolved parents tend to obtain poorer grades than children whose parents make demands on them.
  - The children of uninvolved parents also tend to be more likely to hang out with crowds who "party" a good deal and use drugs.
- Children profit when parents make reasonable demands, show

warmth and encouragement, and spend time with them.

ADOLESCENCE

- The period of life bounded by puberty and the assumption of adult responsibilities
- Except for infancy, more changes occur during adolescence than during any other time.
- Like childhood, adolescence entails physical, cognitive, social, and emotional changes.

PHYSICAL DEVELOPMENT

PUBERTY

- The period of physical development during which sexual reproduction first becomes possible
- Period during which the body becomes sexually mature.
- It heralds the onset of adolescence.
- Puberty begins with the appearance of **secondary sex characteristics** such as body hair, deepening of the voice in males, and rounding of the breasts and hips in females.
    - **SECONDARY SEX CHARACTERISTICS** - characteristics that distinguish females from males, such as distribution of body hair and depth of voice, but that are not directly involved in reproduction
- In boys, **pituitary hormones** stimulate the testes to increase the output of **testosterone**, which in turn causes enlargement of the penis and testes and the appearance of body hair.
    - By the early teens, erections become common, and boys may ejaculate.
    - **Ejaculatory** ability usually precedes the presence of mature sperm by at least a year.
    - Ejaculation thus is not evidence of reproductive capacity.
- In girls, increases in body weight are believed to trigger a cascade of hormonal secretions in the brain that cause the ovaries to secrete higher levels of the female sex hormone, **estrogen**.
    - Estrogen stimulates the growth of breast tissue and tissue in the hips and buttocks.
    - The pelvis widens, rounding the hips.
    - Small amounts of **androgens** produced by the adrenal glands, along with estrogen, spur the growth of pubic and underarm hair.
    - Estrogen and androgens also stoke the development of female sex organs.
    - Estrogen production becomes cyclical during puberty and regulates the menstrual cycle.
    - The beginning of menstruation, or **menarche**, usually occurs between 11 and 14.
    - Girls cannot become pregnant until they ovulate, however, and ovulation may begin a couple of years after menarche.

BRAND DEVELOPMENT

- *What happens to the brains of adolescents who spend hours a day practicing the piano or the violin?* - Their learning translates physically into increases in the

thickness of the parts of the cerebral cortex that are being used
- The gains in thickness of the cerebral cortex represent increases in gray matter, which
- consists of associative neurons that transmit messages back and forth in the brain when we are engaged in thought and sensorimotor activities.
- Many adolescents show poor judgment, at least from time to time, and take risks that most adults would avoid
- Brain immaturity may play a role.

Deborah Yurgelun-Todd and her colleagues showed pictures of people with fearful expressions to adolescents ranging in age from 11 to 17 while the adolescents' brains were scanned by functional magnetic resonance imaging (fMRI).

- Compared to adults, the adolescents' **frontal lobes** (the seat of executive functioning) were less active, and their **amygdalas** (a part of the limbic system involved in discriminating emotions, including fear) were more active.
- The adolescents often misread the facial expressions, with those younger than 14 more often inferring sadness, anger, or confusion rather than fear.
- The researchers suggest that one reason many adolescents fail to show the judgment, insight, and reasoning ability of adults is immaturity of the frontal lobes

COGNITIVE DEVELOPMENT

- The adolescent thinker approaches problems differently from the elementary school child.
  - A younger child sticks to the facts, to concrete reality. Speculating about abstract possibilities and what might be is very difficult.
  - The adolescent, on the other hand, is able to deal with the abstract and the hypothetical.

FORMAL OPERATIONAL STAGE

- Piaget's fourth stage, characterized by abstract logical thought and deduction from principles
- According to Piaget, children undergo three stages of cognitive development prior to adolescence: sensorimotor, pre-operational, and concrete operational.
- The formal operational stage is the final stage in Piaget's theory, and it represents cognitive maturity.
- For many children in Western societies, formal operational thought begins at about the beginning of adolescence—the age of 11 or 12.
- Some people enter this stage later, however, and some never do.
- The major achievements of the stage of formal operations involve classification, logical thought, and the ability to hypothesize.
- Central features are the ability to think about ideas as well as objects and to group and classify ideas—symbols, statements, and entire theories.
- Adolescents can generally follow arguments from premises to conclusions and back again.
  - They can generally appreciate both the outer environment and the world of the imagination: they engage in hypothetical thinking and deductive reasoning.
- Formal operational adolescents (and adults) think abstractly.

- They solve geometric problems about circles and squares without reference to what the circles and squares may represent in the real world.

- Adolescents in this stage derive rules for behavior from general principles and can focus, or center, on multiple aspects of a situation at once to solve problems.
- During the stage of formal operations, adolescents become capable of dealing with hypothetical situations
  - They realize that situations can have different outcomes, and they think ahead, imagining those outcomes.
- Adolescents also conduct social "experiments" to test their hypotheses.
  - They may try out various tones of voice and ways of treating others to see what works best for them.

ADOLESCENT EGOCENTRISM

- Adolescents in the formal operational stage can reason deductively.
  - They classify objects or people and then draw conclusions about them.
- Adolescents can be proud of their new logical abilities, leading to a new sort of egocentrism: they demand acceptance of their logic without recognizing the exceptions or practical problems that may be considered by adults.
- Adolescent thought is marked by an egocentrism in which they can understand the thoughts of others, but still have trouble separating things that are of concern to others and those that are of concern only to themselves

- **IMAGINARY AUDIENCE** - belief that other people are as concerned with our thoughts and behavior as we are.
  - An aspect of adolescent egocentrism; the belief that other people are as concerned with our thoughts and behaviors as we are
  - May drive the intense adolescent desire for privacy.
  - It helps explain why adolescents are so self-conscious
  - Self-consciousness seems to peak at about 13 and then decline. Girls tend to be more self-conscious than boys
- **PERSONAL FABLE** - belief that our feelings and ideas are special, even unique, and that we are invulnerable.
  - Another aspect of adolescent egocentrism; the belief that our feelings and ideas are special and unique and that we are invulnerable
  - The personal fable seems to underlie adolescent showing off and risk taking

THE POSTCONVENTIONAL LEVEL OF MORAL REASONING

- Kohlberg's theory of moral reasoning involves three levels: preconventional, conventional, and postconventional.
- None of Kohlberg's levels is tied to a person's age.
  - Although postconventional reasoning is the highest level, most adolescents and adults never reach it.
  - But if postconventional reasoning emerges, it usually does so in adolescence.
- Kohlberg's (1969) research found postconventional moral judgments

were absent among 7- to 10-year-olds.
    - But by age 16, stage 5 reasoning is shown by about 20% of adolescents, and stage 6 reasoning by about 5%.
- At the **postconventional level**, moral reasoning is based on the person's own moral standards.
- **POSTCONVENTIONAL LEVEL** - according to Kohlberg, a period during which moral judgments are derived from moral principles and people look to themselves to set moral standards
- Moral judgments are derived from personal values, not from conventional standards or authority figures.
- People at the postconventional level see their conscience as the highest moral authority.
- This point has created confusion.
    - To some it suggests that it is right to break the law when it is convenient, but this interpretation is wrong.
    - Kohlberg means that people at this level feel they must do what they think is right even if they break the law or must sacrifice themselves.

SOCIAL AND EMOTIONAL DEVELOPMENT

- In the 19th century, psychologist **G. Stanley Hall** described adolescence as a time of storm and stress (*Sturm und Drang*)
- Current views challenge the assumption that "storm and stress" is the norm
- Many adolescents experience a rather calm and joyous period of development.
- We need to consider individual differences and cultural variations

RELATIONSHIPS WITH PARENTS AND PEERS

- Adolescents spend much less time with their parents than they did as children
- Adolescents interact more with their mothers than their fathers and thus have more conflict with their mothers.
    - But they also view their mothers as more supportive and more likely to accept their opinions
- The decrease in time spent with family may reflect a striving for independence, yet most adolescents maintain love, loyalty, and respect for their parents
    - Adolescents who feel close to their parents are more likely to show greater independence, higher self-esteem, better school performance, and fewer psychological problems
- Parent–adolescent conflict is greatest during puberty and declines in later adolescence
- Although relationships with parents generally remain positive, the role of peers as a source of activities, influence, and support increases during the teen years
- Adolescents stress the importance of acceptance, self-disclosure, mutual understanding, and loyalty in their friendships
- Adolescents and their friends are typically the same age and the same race
- Friends are also often alike in their school attitudes, educational aspirations, school achievement, and attitudes about drinking, drug use, and sexual activity

Research shows that most adolescents' friends are not bad influences. Many peers reinforce positive behaviors, such as academic achievement.

EGO IDENTITY VERSUS ROLE DIFFUSION

- According to **Erik Erikson (1963)**, we undergo eight stages of psychosocial development, each of which is characterized by a certain "crisis."
  - Four of them, beginning with trust versus mistrust, occur in childhood.
  - The fifth, that of ego identity versus role diffusion, occurs in adolescence.
- **Ego identity** is a firm sense of who one is and what one stands for.
  - It is Erikson's term for a firm sense of who one is and what one stands for
  - It can carry one through difficult times and lend meaning to achievements.
  - Adolescents who do not develop ego identity may experience **role diffusion**.
  - **ROLE DIFFUSION** - Erikson's term for lack of clarity in one's life roles (due to failure to develop ego identity)

EMERGING ADULTHOOD

- Period of development, roughly spanning the ages of 18 to 25, during which time many young people in affluent nations attend college and graduate school, sort out identity issues, and create life plans
- **Jeffrey Arnett** (2014) hypothesizes that five features distinguish the stage of emerging adulthood that is sandwiched between the stages of adolescence, which precedes it, and young adulthood, which follows it:
  - **The age of identity exploration**. Many people of the ages from 18 or 20 to about 25 or 30 are on the path to making vital choices in terms of their love lives and their career lives.
    - They are experimenting with romantic partners and career possibilities.
  - **The age of instability**. Americans today average about seven different jobs between 20 and 29.
    - They frequently change romantic partners—sometimes by choice, sometimes because the partner decides to move on.
    - They may move from place to place with little if any furniture.
    - And they frequently change educational directions, finding what they like, finding what they can do, and finding what is available.
  - **The age of self-focus.** People are exceptionally self-focused during emerging adulthood.
    - They are freer to make decisions than they are as children or adolescents.
    - They are more mature, more independent from parental influences, and usually have more resources.
  - **The age of feeling in-between**. Emerging adults are similar to adolescents in

one way: Whereas adolescents may feel that they exist somewhere between childhood and adulthood, emerging adults are likely to think that they are swimming between adolescence and "real" adulthood.

- They are likely to be out of school—that is, high school or undergraduate college—but obtaining further training or education.
- They are beyond the sometimes silly flirtations of adolescence but not yet in long-term, or at least permanent, relationships.
- They may not be completely dependent on caregivers, but they are just as unlikely to be self-supporting.
- They may be between roommates or apartments.
- When they are asked whether they think they have become adults, Arnett (2000) found the most common answer of 18- to 25-year-olds to be something like "In some respects yes and in other respects no."
- Many think that they have developed beyond the conflicts and exploratory voyages of adolescence, but they may not yet have the ability—or desire—to assume the financial and interpersonal responsibilities they associate with adulthood.
  - **The age of possibilities.** Emerging adults typically feel that the world lies open before them.
  - Like the majority of adults, they have an optimistic bias.
  - They believe that things will work out.
  - Yet some of them have a "revolving door" existence: they leave home and then return, according to the ebb and flow of financial and emotional resources.

ADULTHOOD

PHYSICAL DEVELOPMENT

- The most obvious aspects of development during adulthood are physical.

EARLY ADULTHOOD

- Most people are at their height of sensory sharpness, strength, reaction time, and cardiovascular fitness.
- Young adults are at their tallest, and height remains stable through middle adulthood, declining somewhat in late adulthood.
- Physical strength in both men and women peaks in the 20s and early 30s and then slowly declines.
- Most athletes experience a decline in their 30s, and most athletes retire by age 40.
- Sexually speaking, most people in early adulthood become readily aroused.
  - They tend to attain and maintain erections as desired and to lubricate readily.

MIDDLE ADULTHOOD

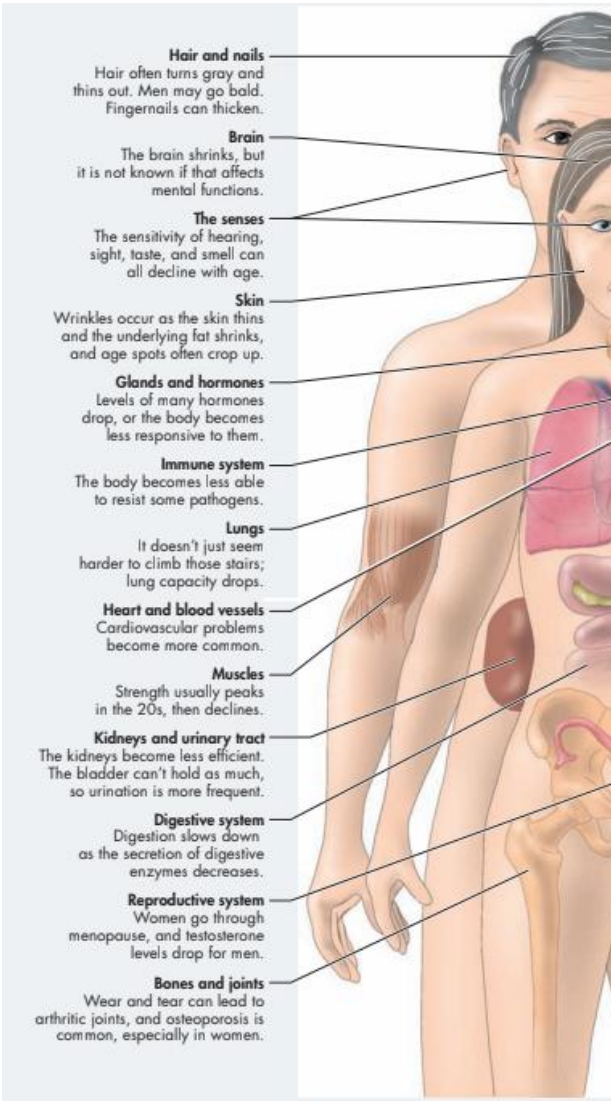
- In our middle years, we are unlikely to possess the strength,

coordination, and stamina that we had during our 20s and 30s.

- Still, the years between 40 and 60 are reasonably stable.
- There is gradual physical decline, but it is minor and likely to be of concern only if a person competes with young adults— or with idealized memories of one's self.
- Because the physical decline in middle adulthood is gradual, people who begin to eat more nutritious diets and to exercise may find themselves looking and feeling better than they did in early adulthood.
- For women, **menopause** is usually considered to be the single most important change of life that occurs during middle adulthood.
- **MENOPAUSE** - the cessation of menstruation
  - Menopause usually occurs during the late 40s or early 50s.
  - Menopause is the final phase of the climacteric, which is caused by a decline in secretion of female sex hormones.
  - Ovulation comes to an end, and there is some loss of breast tissue and skin elasticity.
- Loss of bone density can lead to osteoporosis (brittle bones).
- During the climacteric, many women experience hot flashes, loss of sleep, and some anxiety and depression.
- Women's experiences during and following the climacteric reflect the intensity of their physical symptoms—which vary considerably—and the extent to which their self-concept was wrapped up with their reproductive capacity

LATE ADULTHOOD

- Various changes—some of them troublesome—do occur during the later years.
- Changes in calcium metabolism increase the brittleness of the bones and heighten the risk of breaks due to falls.
- The skin becomes less elastic and subject to wrinkles and folds.
- Older people see and hear less acutely.
  -



  - Because of a decline in the sense of smell, they may use more spice to flavor their food.
- Older people need more time to respond to stimuli.
- Older drivers, for example, need more time to respond to changing road conditions.
- As we grow older, our immune system functions less effectively, leaving us more vulnerable to disease.
- Age-related changes impact sexual functioning, yet most people can enjoy sex for a lifetime if they remain generally healthy and adjust their expectations.

COGNITIVE DEVELOPMENT

- Cognitive development in adulthood has many aspects—creativity, memory functioning, and intelligence.
- Memory functioning does decline with age. But declines in memory are not usually as large as people assume and are often reversible.
  - Memory tests usually measure ability to recall meaningless information.
  - Older people show better memory functioning in areas in which they can apply their experience, especially their specialties, to new challenges.
- **CRYSTALLIZED INTELLIGENCE** - defined as a cluster of knowledge and skills that depend on accumulated information and experience, awareness of social conventions, and the capacity to make good decisions.
- one's lifetime of intellectual achievement, as shown largely through vocabulary and knowledge of world affairs
- **FLUID INTELLIGENCE** - mental flexibility, demonstrated by the ability to process information rapidly, as in learning and solving problems in new areas.
- It is the sort of intellectual functioning that is typcally measured on intelligence tests, especially with problems that have time limits.
- In the absence of senile dementias, crystallized intelligence commonly increases throughout the life span, along with scores on verbal subtests of standardized intelligence tests.
- The same studies that indicate that crystallized intelligence tends to increase throughout adulthood tend to show a decline for fluid intelligence
- **ALZHEIMER'S DISEASE** - progressive and irreversible brain disease affecting 4 to 5 million Americans.
- a progressive form of mental deterioration characterized by loss of memory, language, problem solving, and other cognitive functions

SOCIAL AND EMOTIONAL DEVELOPMENT

- Changes in social and emotional development during adulthood are probably the most "elastic" or fluid.
- These changes are affected by cultural expectations and individual behavior patterns.

EARLY ADULTHOOD

- Many theorists suggest that early adulthood is the period of life during which people tend to establish themselves as independent members of society.

- Many young adults adopt what theorist **Daniel Levinson** and his colleagues (1978) call the Dream—the drive to "become" someone, to leave their mark on history—which serves as a tentative blueprint for life.
- Erik Erikson (1963) characterized young adulthood as the stage of **intimacy versus isolation**.
  - Erikson saw the establishment of intimate relationships as central to young adulthood.
  - Young adults who have evolved a firm sense of identity during adolescence are ready to "fuse" their identities with those of other people through marriage and abiding friendships.
- People who do not reach out to develop intimate relationships risk retreating into isolation and loneliness.
- The later 30s are often characterized by settling down—planting roots.
  - They become focused on career advancement, children, and long-term mortgages.

MIDDLE ADULTHOOD

- Erikson (1963) labeled the life crisis of the middle years **generativity versus stagnation**.
  - **Generativity** involves doing things that we believe are worthwhile, such as rearing children or producing on the job.
  - He theorized that generativity enhances and maintains self-esteem.
  - Generativity also involves making the world a better place through joining church or civic groups.
  - **Stagnation** means treading water, as in keeping the same job at the same pay for 30 years or neglecting to rear one's children.
  - Stagnation damages self-esteem.
- According to **Levinson** and colleagues (1978), whose research involved case studies of 40 men, there is a midlife transition at about 40 to 45 characterized by a shift in psychological perspective from how many years one has lived to how many years one has left.
- Research suggests that women may undergo a midlife transition earlier than men do
  - Why? Much of it is related to the winding down of the "biological clock"—their ability to bear children.
- In both women and men, according to Levinson, the midlife transition may trigger a midlife crisis.
  - The middle-level, middle-aged businessperson looking ahead to another 10 to 20 years of grinding out accounts in a cubbyhole may encounter depression.

EMPTY-NEST SYNDROME

- Sense of depression and loss of purpose felt by some parents when the youngest child leaves home
- It was assumed that women experience a sense of loss when their youngest child goes off to college, gets married, or moves out of the home
  - However, research does not find that parents suffer from an empty-nest syndrome when the youngest child leaves home.

MIDDLE-AGED CHILDREN AND AGING PARENTS: THE SANDWICH GENERATION

- The relationships between middle-aged and older parents can grow quite close, especially as tensions and expectations from earlier years tend to slip into history.
- If the aging parents require assistance, the task usually falls to a middle-aged daughter, who then becomes what has been dubbed part of the sandwich generation.
- She is sandwiched between caring for or contributing to the support of her children or grand-children at the same time she is caring for her parents
- **SANDWICH GENERATION** - people in middle adulthood who are responsible for meeting the needs of their children yet also responsible for aging parents

LATE ADULTHOOD

- Research suggests that many individuals in late adulthood continue to be creative and also to maintain a firm sense of who they are and what they stand for.
- The Greek philosopher Plato was so optimistic about late adulthood that he argued that one could achieve great pleasure in one's later years, engage in meaningful public service, and also achieve wisdom.
- According to psychologist Erik Erikson, late adulthood is the stage of ego integrity versus despair.
- The basic challenge is to maintain the belief that life is meaningful and worthwhile as one ages and faces the inevitability of death.
- Erikson, like Plato, spoke of the importance of wisdom.
    - He believed that ego integrity derives from wisdom, which can be defined as expert knowledge about the meaning of life, balancing one's own needs and those of others, and pushing toward excellence in one's behavior and achievements
- Erikson also believed that wisdom enables people to accept their lifespan as occurring at a certain point in the sweep of history and as being limited.
- We spend most of our lives accumulating objects and relationships, and Erikson suggests that adjustment in the later years requires the ability to let go of them.

SUCCESSFUL AGING

- Erikson focused on letting go and accepting one's place in the sweep of history, successful aging focuses on maximizing one's life experiences at any age.
- Researchers who study successful aging focus on factors such as overall good physical health, including the absence or control of chronic diseases such as arthritis and diabetes, engaging in physical activity, social networking, and the absence of serious cognitive impairment and depression.
- Research shows that sustained physical exercise, such as jogging or bicycling, helps the development of new cells in the hippocampus, which improves overall brain health and possibly memory.
- Sexual activity remains desirable, even if some aspects of performance are not what they were.
- According to researchers **Margaret Baltes and Laura Carstensen**, successful agers tend to seek emotional fulfillment

by **selective optimization with compensation**: they reshape their lives to concentrate on what they find to be important and meaningful.
    - **SELECTIVE OPTIMIZATION WITH COMPENSATION** - reshaping of one's life to concentrate on what one finds to be important and meaningful in the face of physical decline and possible cognitive impairment.
- Successful agers form emotional goals that bring them satisfaction.
- They may no longer compete in certain athletic or business activities.
    - Instead, they focus on matters that allow them to maintain a sense of control over their own lives.
- Retaining social contacts and building new ones also contributes to a positive outlook, as does continuing with one's athletic activities, when possible, and one's artistic and cultural activities.

"LYING DOWN TO PLEASANT DREAMS"

- American poet **William Cullen Bryant** is best known for his poem "**Thanatopsis,**" which he composed at the age of 18. "**Thanatopsis**" expresses Erik Erikson's goal of ego integrity—optimism that we can maintain a sense of trust through life.
    - By meeting squarely the challenges of our adult lives, perhaps we can take our leave with dignity. And when our time comes to "join the innumerable caravan"—the billions who have died before us—perhaps we can depart life with integrity as well.

11: PERSONALITY—THEORY AND MEASUREMENT

PERSONALITY

- Reasonably stable patterns of **emotions, motives,** and **behavior** that distinguish one person from another

THE PSYCHODYNAMIC PERSPECTIVE

- Sigmund Freud presented many psychodynamic theories of personality

PSYCHODYNAMIC THEORY

- Sigmund Freud's perspective, which emphasizes the importance of unconscious motives and conflicts as forces that determine behavior
- Personality is characterized as conflict
  - Conflicts start off as **external** (drives like sex, aggression, need for superiority vs. laws and morals)
  - At some point, laws and social norms are **internalized** (conflicts between inner forces)
  - Behavior, thoughts, and emotions reflect these internal struggles

SIGMUND FREUD'S THEORY OF PSYCHOSEXUAL DEVELOPMENT

SIGMUND FREUD

- Believed that much of the mind is unconscious
- Was a mass of contradictions
- One of the greatest thinkers of 20th century, but also criticized as overrated
- Preached liberal views on sexuality, but was himself a model of sexual restraint
- Created a form of psychotherapy, but was also faced psychological problems
- Realized that although some things are hidden underneath awareness, it could still have an effect on our behavior

CONSCIOUSNESS

- **Conscious** part of the mind region that is part of awareness
  - Thoughts that we are aware of (Ex. I am hungry right now)
- **Preconscious** and **Unconscious** parts of the mind that are out of awareness
  - **Preconscious mind** contains ideas that are out of awareness but can be made conscious by focusing on them (ex. What I ate for dinner three days ago)
  - **Unconscious mind** - hidden instincts like sex and aggression
    - Contains repressed urges kept below awareness (Ex. something you do not know that is happening)

REPRESSION

- Automatic ejection of anxiety-evoking ideas from awareness
- Other investigators allow that forgetting and distortion of memory occurs, but view the **Freudian concept** of repression as little more than a myth
- In the unconscious mind, primitive drives seek expression, while learned values try to keep them in check

THE PSYCHOANALYTIC THEORY

- When people are made comfortable to talk about anything

that pops in their mind when they are comfortable, as a way to explore our unconscious

- **PSYCHOANALYSIS** - method to explore the unconscious by encouraging people to talk freely about thoughts and feelings

THE STRUCTURE OF PERSONALITY

- We have mental or psychic structures that describe the clashing forces of our personality
  - Presence is suggested in our behavior, expressed thoughts, and emotions
- **ID** - present at birth
  - Represents biological drives
  - Fully unconscious
  - Follows **PLEASURE PRINCIPLE**—demands instant gratification without consideration for law, social custom, or other people
  - Ex. You're very hungry and pass by a convenience store. The id impulsive urges you to grab a candy bar and eat it immediately without paying
- **EGO** - begins to develop during first year of life, largely because a child's demands for gratification cannot all be met immediately
  - Represents reason and self-awareness
  - Stands for reason and good sense, for rational ways of coping with frustration
  - Guided by the **REALITY PRINCIPLE** — curbs the appetites of the ID and seeks ways to find gratification yet avoid social disapproval
  - Uses **defense mechanisms**, such as repression to control impulses
  - Ex. You want the candy bar, but ego steps in and recognizes that stealing is illegal. It decides to walk into the store, locate the candy bar, and take it to the register to pay for it.
- **SUPEREGO** - Develops through **identification** with parents and moral figures
  - Functions according to **MORAL PRINCIPLE** — judges right and wrong and causes guilt when standards are not met

Freud believed that a healthy personality balances the id's demands with the superego's morals through the ego's rational control.

STAGES OF PSYCHOSEXUAL DEVELOPMENT

DEFINITION OF TERMS	
TERM	**DEFINITION**
EROS	Major instinct that aims to preserve and perpetuate life — life instinct (drive to live, sex drive, basic instinctual impulses)
LIBIDO	Person's sex drive
LIBIDINAL ENERGY	involves sexual impulses, which Freud considered to be psychosexual
EROGENOUS ZONE	Part of the body sensitive to touch; where the libidinal energy travels as a child develops
GRATIFI-CATION	Satisfaction of drive or libido
FIXATION	Excessive or insufficient gratification in a stage

- **PSYCHOSEXUAL DEVELOPMENT** - in psychodynamic theory, the

process by which libidinal energy is expressed through different erogenous zones during different stages of development
    - Freud believed sexual impulses drive personality development through 5 stages
- **Oral stage (0-1 year)** - first stage
    - Gratification is attained primarily through oral activities like sucking and biting
    - Conflict centers on the nature and extent of oral gratification
    - Early weaning (cessation of breastfeeding) can lead to frustration
    - Excessive gratification can lead infant to expect that it will routinely obtain anything it wants
    - Oral traits include dependency, gullibility, and excessive optimism or pessimism (depending on the child's experiences with gratification)
    - Fixation may lead to dependency or oral habits (Smoking, overeating)
- **ANAL STAGE (2 years)** - the second stage
    - gratification is attained through control of elimination
    - Conflict over self-control
    - **Anal fixations** - may stem from conflict between parent and child and lead to either two sets of traits in adulthood
        - **Anal retentive traits** involve excessive use of self-control: perfectionism, strong need for order, and exaggerated neatness and cleanliness
        - **Anal-expulsive traits** include carelessness, messiness, and sadism
- **PHALLIC STAGE (3-6 years)**- focus on genitals.
    - Major erogenous zone is the phallic region; Penis in boys and clitoris in girls
    - Parent-child conflict over masturbation, to which parents may respond with threats or punishment
    - Children may develop strong sexual attachments to the parent of the other gender and begin to view the parent of the same gender as a rival for the other parent's affections
    - **Oedipus complex** (for boys) - desire for mother, rivalry with father
    - **Electra complex** (for girls) - desire for father, resentment toward mother
    - Resolution occurs through **identification** with same-sex parents
- **LATENCY STAGE (5 or 6 years)** - repression of sexual urges
    - Sexual feelings remain unconscious
    - Prefer playmates of their own gender
    - Focus on schoolwork
    - Libido is successfully sublimated (channeled) into non-sexual, socially acceptable activities
    - No major conflict
- **GENITAL STAGE (puberty onwards)** - Sexual impulses reawaken, but is displaced onto other adults or adolescents of the other gender
    - Goal is to redirect sexual urges away from parent or towards socially acceptable partners

- In Freud's view, oral or anal stimulation, masturbation, and sexual activity with people of the same gender all represent pregenital fixations and immature forms of sexual conduct.

NEO-FREUDIANS

- Freud's intellectual heirs who accepted his focus on inner conflict but emphasized other factors

CARL JUNG (1875-1961)

- Developed **Analytical Psychology**
- Downplayed the importance of sex, which he saw as just one of several important instincts
- Added **Collective unconscious** containing primitive images, or **archetypes**, that reflect history of our species
- Archetypes—shared store of ancestral memories remain unconscious, but Jung believed they affect our thoughts and feelings, and cause us to respond to cultural themes in the media

ALFRED ADLER (1870-1937)

- Believed that people are motivated by an **inferiority complex**, which give rise to a **drive for superiority**
- Believed that self-awareness plays a major role in formation of personality
- Introduced **CREATIVE SELF** - self-aware aspect of personality that strives to overcome obstacles and develop the person's potential
- Developed **INDIVIDUAL PSYCHOLOGY**

KAREN HORNEY (1885-1952)

- Argued that little girls do not feel inferior to boys (penis envy)
- Believed that genuine and consistent love can alleviate the effects of a traumatic childhood
- Emphasized social relationships over sexual and aggressive drives

ERIK ERIKSON (1902-1994)

- Believed that social relationships are more important determinants of personality than sexual urges
- Proposed stages of **psychosocial developments**
- First stage is labeled that stage of **trust vs. mistrust**
- Goal of adolescence is the attainment of **ego identity**, not genital sexuality
  - **EGO IDENTITY** - firm sense of who one is and what one stands for

EVALUATION OF THE PSYCHODYNAMIC PERSPECTIVE

STRENGTHS

- Highlighted importance of early childhood experiences
- Introduced idea of the unconscious mind influencing behavior
- Pioneered psychotherapy and talk therapy

WEAKNESSES

- Concepts (id, ego, superego) are vague and hard to test scientifically
- Overemphasis on sex and aggression
- Based mostly on case studies, not empirical research

THE TRAIT PERSPECTIVE

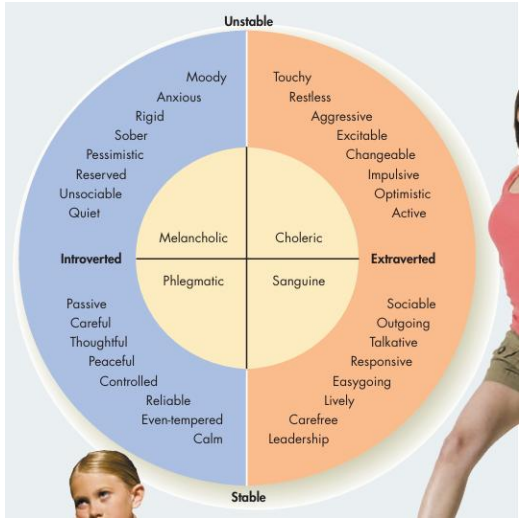
- **TRAITS** - Reasonably stable elements of personality that are inferred from behavior
- Freud linked development of certain traits to children's experiences in each stage of psychosexual development

FROM HIPPOCRATES TO THE PRESENT

- Hippocrates believed that traits were embedded in bodily fluids
- Viewed that personality depends on balance of four basic fluids, or "humors"
  - **Yellow Bile** - associated with choleric (quick-tempered)
  - **Blood** - sanguine (warm, cheerful)
  - **Phlegm** - phlegmatic (sluggish, calm, cool)
  - **Black bile** - melancholic (gloomy, pensive)
- Diseases was believed to reflect an imbalance among the humors
  - Ex. Depression represented excess of black bile
- Bloodletting and vomiting were recommended to restore balance
- Although speculative, these terms were still used in descriptions of personality
- Modern theories assume that traits are heritable and are embedded in the nervous system
  - Use **factor analysis** by Charles Spearman to intelligence, to determine which traits are basic to others
- **Gordon Allport** and colleagues catalogued some 18,000 human traits—physical, behavioral, and moral—from a search through word lists like dictionaries, forming the foundation for modern personality studies

HANS EYSENCK'S TRAIT THEORY

- British psychologist **Hans J. Eysenck (1916-1997)** focused of his research on relationships between two personality traits: **introversion-extraversion** and **emotional stability-instability**
  - **Introversion** - intense imagination and the tendency to inhibit impulses
  - **Extraversion** - tendency to be socially outgoing and to express feelings
- **Emotional instability** is also called neuroticism
- According to Eysenck:
  - **Choleric** - extraverted + unstable
  - **Sanguine** - extraverted + stable
  - **Phlegmatic** - introverted + stable
  - **Melancholic** - introverted + unstable



THE "BIG FIVE": THE FIVE-FACTOR MODEL

- Developed by many theorists, especially **Robert McCrae** and **Paul T. Costa Jr.**
- Five core factors and traits include:
  - Extraversion
  - Neuroticism
  - Conscientiousness
  - Agreeableness
  - Openness to experience

TABLE 10.1 THE "BIG FIVE": THE FIVE-FACTOR MODEL

Factor Name	Traits
I. Extraversion	Contrasts talkativeness, assertiveness, and activity with silence, passivity, and reserve
II. Agreeableness	Contrasts kindness, trust, and warmth with hostility, selfishness, and distrust
III. Conscientiousness	Contrasts organization, thoroughness, and reliability with carelessness, negligence, and unreliability
IV. Neuroticism	Contrasts nervousness, moodiness, and sensitivity to negative stimuli with coping ability
V. Openness to experience	Contrasts imagination, curiosity, and creativity with shallowness and lack of perceptiveness

- These factors appear across many cultures, including Philippines, and are largely hereditary through shaped by environment
  - Agreeable people tend to have fewer accidents
  - Liberals score higher on openness; conservatives on conscientiousness
  - Happier individuals score high on stability, agreeableness, conscientiousness, and extraversion

BIOLOGY AND TRAITS

- Researchers estimate that the heritability of the extraverted personality is 40% to 60%
- Research suggests that brain levels of the neurotransmitter dopamine tend to be higher in extraverts than in introverts
- Genetic factors are part of a child's basic temperament and are involved in shyness and behavioral inhibition
- **Antisocial personality** - frequent conflict with society and lack of feelings of guilt or anxiety
  - Tied to low emotional response and reduced activity in the **ventral prefrontal cortex** and **amygdala**

EVALUATION OF THE TRAIT PERSPECTIVE

- Trait theory helped develop personality tests and link traits to job suitability
- However, it is mainly descriptive, not explanatory—defines traits but does not explain their origins or how to change them

LEARNING THEORY PERSPECTIVES

- Learning theorists tend not to theorize in terms of traits
- Focus instead on behaviors and presume that those behaviors are largely learned
- What is learned can also be unlearned
- Both behaviorists and social cognitive theorists have contributed to the discussion of personality

BEHAVIORISM

- **John B. Watson (1924)**: Give me a dozen healthy infants, well-formed, and my own specified world to bring them up in, and I'll guarantee to take any one at random and train him to become any type of specialist I might suggest—doctor, lawyer, merchant-chief and, yes, even beggar-man and thief, regardless of his talents, penchants, tendencies, abilities, vocations, and the race of his ancestors
- Personality is **plastic** - situational or environmental influences shape personality, not innate traits
  - Watson argued that unseen, undetectable mental structures must be rejected in favor of that which can be seen and measured

- B.F. Skinner emphasized reinforcements as shapers of behavior. He agreed that psychologists should avoid trying to see into the black box of the organism and instead emphasize the effects of reinforcements.
- Ignored personal freedom and internal choice; behavior shaped by **social approval and customs**

- Criticisms
  - Plays down consciousness, choice, and self-direction
  - Humans may choose pain, hardship, or death over pleasure—challenging behaviorist ideas

SOCIAL COGNITIVE THEORY

- Developed by **Albert Bandura**
- Focuses on learning by observation and on the cognitive processes that underlie personal differences
- See people as influencing their environment just as their environment affects them
- Includes both **situational variables** (rewards and punishments) and **person variables** (variables within people, ex. Knowledge and skills)
- Behavior depends on both situation, **expectancies** (beliefs about outcomes), and the **subjective values** (perceived values of outcomes)
- **SELF-EFFICACY EXPECTATIONS** - beliefs that we can accomplish certain things
  - Positive self-efficacy → high self-esteem and achievement motivation
  - Psychotherapy often motivates people to try new things by changing their self-efficacy expectations

OBSERVATIONAL LEARNING

- Also termed modeling or cognitive learning
- Acquiring knowledge by observing others when there is no direct reinforcement

EVALUATION OF THE LEARNING PERSPECTIVE

CONTRIBUTIONS

- Emphasized scientific study of observable behavior
- Highlighted environmental environmental and situational influences on learning and personality
- Explained learning through reinforcement and observation

LIMITATIONS

- Behaviorism overlooks internal factors like self-awareness and genetics
- Social cognitive theory underplays biological influences on behavior

HUMANISTIC-EXISTENTIAL PERSPECTIVE

- Puts self-awareness at the cen ter of consideration and argues that people are capable of free choice, self-fulfillment, and ethical behavior
- **HUMANISM** - view that people are capable of free choice, self fulfillment, and ethical behavior
  - Emphasizes positive human potential and the drive for self-actualization
- **EXISTENTIALISM** - view that people are completely free and responsible for their own behavior
  - Focuses on giving personal meaning to life despite pain, death, or suffering

- Jean-Paul Sartre and Martin Heidegger saw human life as trivial in the grand scheme of things
- Viktor Frankl, Ludwig Binswanger, and Medard Boss argued that seeing human existence as meaningless could give rise to withdrawal and apathy—even suicide
- **Psychological salvation** requires giving personal meaning to things and making personal choices

- **GENDER-TYPING -** the process by which males and females come to display behavior patterns consistent with stereotypical masculine and feminine gender roles

ABRAHAM MASLOS AND THE CHALLENGE OF SELF-ACTUALIZATION

- **SELF-ACTULIZATION** - in humanistic theory, the innate tendency to strive to realize one's potential
- **GENDER-SCHEMA THEORY** - a cognitive view of gender-typing that proposes that once girls and boys become aware of their anatomic sex, they begin to blend their self-expectations and self-esteem with the ways in which they fit the gender roles prescribed in a given culture

CARL ROGER'S SELF-THEORY

- People shape themselves through free choice and action
- **SELF** - center of experience
  - Ongoing sense of who and what you are

THE SELF-CONCEPT AND FRAMES OF REFERENCE

- **SELF-CONCEPTS** - consists of our impressions of ourselves and our evaluations of our adequacy
- **FRAMES OF REFERENCE** Unique ways we look at ourselves

SELF-ESTEEM AND POSITIVE REGARD

- At first, self-esteem reflects the esteem in which the others hold us
- **SELF-IDEAL** - image of what we want to become
- **CONGRUENCE** - the closer our self-concept is to our self-ideal, the healthier we are
- **UNCONDITIONAL POSITIVE REGARD** - a persistent expression of esteem for the value of a person, but not necessarily an unqualified acceptance of all of the person's behaviors
  - Accepting someone for who they are, not just what they do
- **CONDITIONAL POSITIVE REGARD** - judgment of another person's value on the basis of the acceptability of that person's behaviors
  - Accepting others only when they behave "properly"
- **CONDITIONS OF WORTH** - standards by which the value of a person is judged
- **ANXIETY** - often stems from recognition that people have feelings and desires that are inconsistent with their distorted self-concept

EVALUATION OF THE HUMANISTIC-EXISTENTIAL PERSPECTIVE

STRENGTHS

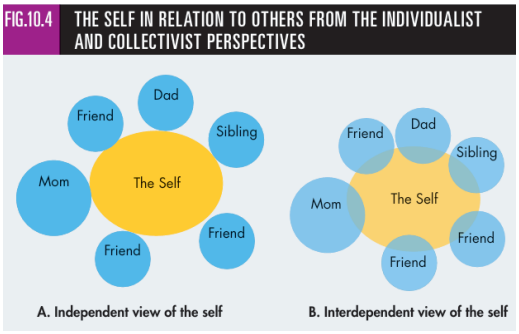
- Emphasizes conscious experience, free will, and personal meaning
- Positive, growth-oriented view of human nature
- Inspires personal responsibility and authenticity

WEAKNESSES

- Too subjective — based on private experiences, hard to measure scientifically
- Lacks explanation for how traits and personalities develop

THE SOCIOCULTURAL PERSPECTIVE

- The view that focuses on the roles of ethnicity, gender, culture, and socioeconomic status in personality formation, behavior, and mental processes



INDIVIDUALISM VERSUS COLLECTIVISM: WHO AM I (IN THIS CULTURAL SETTING)?

- **INDIVIDUALISTS** - define themselves in terms of their personal identities and to give priority to their personal goals
- **COLLECTIVISTS** - define themselves in terms of the groups to which they belong and to give priority to the group's goals
- **CULTURAL INFLUENCE** - culture shapes whether one values independence (individualism) or belonging (collectivism)
  - Modernization can make some Asians more individualistic

ACCULTURATION, ADJUSTMENT, & SELF-ESTEEM

- Self-esteem is connected with patterns of acculturation
- **ACCULTURATION** - the process of adaptation in which immigrants and native groups identify with a new, dominant culture by learning about that culture and making behavioral and attitudinal changes
- **Patterns of Acculturation:**
  - **ASSIMILATED** - fully adopt new culture; lose original customs
  - **SEPARATED** - keep old culture; avoid new one
  - **BICULTURAL** - blend both cultures; switch between them as needed
- Biocultural individuals often have higher self-esteem
- English proficiency reduces anxiety and depression among immigrants

EVALUATION OF THE SOCIOCULTURAL PERSPECTIVE

STRENGTHS

- Highlights how ethnicity, gender, culture, and social class shape personality and behavior
- Promotes cultural awareness and appreciation of human diversity

WEAKNESS

- May overemphasize culture, overlooking biological or personal factors

MEASUREMENT OF PERSONALITY

- **VALIDITY** - extent to which it measures what it is supposed to measure
  - Assessed by comparing test results to external criteria or standards
- **RELIABILITY** - stability of one's test results from one testing to another

  - Tested by comparing testing results on different occasions or at different ages
- **STANDARDIZATION** - process that checks out the scores, validity, and reliability of a test with people of various ages and from various groups
  - Comparing scores to people of the same age/group
- **BEHAVIOR-RATING SCALES** - assess behavior in settings such as classrooms or mental hospitals

OBJECTIVE TESTS

- **OBJECTIVE TESTS** - tests whose items must be answered in a specified, limited manner
- Tests whose items have concrete answers that are considered correct
- Administered to people with previously identified symptoms, such as depressive or schizophrenic symptoms

PROJECTIVE TESTS

- Have no clear specified answers
- Use ambiguous stimuli
- People project their personality through responses

RORSCHACH INKBLOT TEST

- Test by Hermann Rorschach
- Says what inkblot looks like → reveals thought patterns, perceptions

THEMATIC APPERCEPTION TEST

- By Henry Murray and Christiana Morgan
- Make up stories about drawings → shows needs, emotions, and relationships

12: STRESS, HEALTH, AND COPING

STRESS: WHAT IT IS, WHERE IT COMES FROM

- In physics, **stress** is defined as a pressure or force exerted on a body
- Psychologists define **stress** as the demand made on an organism to adapt, cope, or adjust
  - Some is healthful and necessary to keep us alert and occupied
- According to **Hans Selye** (1907-1982), healthful stress is referred to as **EUSTRESS**
- **EUSTRESS** - positive form of stress that can be beneficial and motivating
  - Occurs when stressors are perceived as challenges that one can manage/overcome, leading to feelings of excitement or fulfillment
- Intense or prolonged stress, can overtax our ability to adjust, affect our moods, impair our ability to experience pleasure, and harm the body
- **DISTRESS** - negative form of stress that occurs when a situation is perceived as threatening, harmful, or overwhelming
  - Often leads to feelings of anxiety, worry, and can have negative effect on both mental and physical health

HEALTH PSYCHOLOGY

- Field of psychology that studies relationships between psychological factors and prevention and treatment of physical health problems
- Health psychologists investigate how:
  - Psychological factors can lead to or aggravate illness
  - People can cope with stress
  - Stress and pathogens (microscopic organisms that can cause disease) interact to influence the immune system
  - People decide whether or not to seek health care
  - Psychological interventions such as health education and behavior modification can contribute to physical health

DAILY HASSLES AND LIFE CHANGES

- **Daily hassles** are regularly occurring experiences that can harm our well-being, including our sex lives
  - Daily hassles are notable conditions and experiences that are threatening or harmful to a person's well-being
  - Source of stress that include irritating, frustrating, and distressing demands that people face on a day-to-day basis
    - **Household hassles**: preparing meals, shopping, and home maintenance
    - **Health hassles**: illness, obtaining medical treatment, and side effects of medication
    - **Time-pressure hassles**: having too many things to do and not enough time
    - **Inner concern hassles**: being socially isolated, lonely
    - **Environmental hassles**: crime, weather, noise, and pollution
    - **Financial responsibility hassles**: not affording things, and owing money

- **Work hassles**: job dissatisfaction and problems with coworkers
- **Security hassles**: job security, terrorism, investments, and retirement

- Life changes differ from daily hassles in two ways:
  - Many life changes are positive, whereas all hassles are negative
  - Hassles occur regularly, whereas life changes occur at irregular intervals

SCALE OF "LIFE CHANGING UNITS"

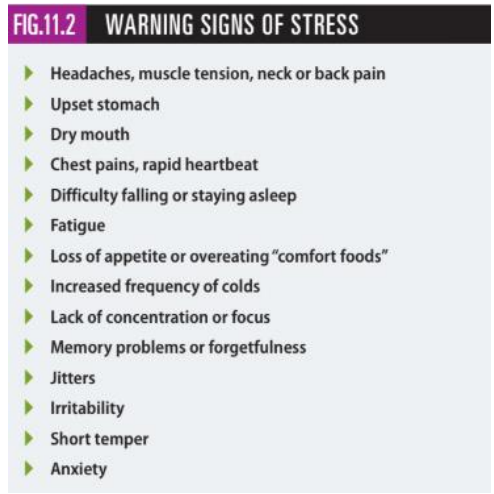
TABLE 11.1 LIFE-CHANGE "UNITS" CONNECTED WITH VARIOUS EVENTS

Life Change	Units
Death of a spouse	94
Beginning college	84
Unwanted pregnancy	80
Academic failure	77
Graduating college	68
Prepping for a big test	65
Major financial problems	65
Academic success	54
Change of address	43
Going on vacation	30

STRESS IN AMERICA

- Perennial stressors for two Americans out of three include money (62%) and work (61%)
- Health (46%) is an ongoing area of concern as are intimate relationships (58%)
- In the wake of the 2016 election, nearly ⅔ of Americans say that future (63%) of nation is a significant source of stress, citing social divisiveness (59%), health care (32%), trust in government, and hate crimes (31%)
- Stress causes us to worry and affects our moods
  - It also predicts physical and psychological health problems
- Respondents reported many physical and psychological symptoms of stress
  - Nearly ½ reported that stress made them irritable (45%) or angry
  - Stress made more than 2 in 5 tired (41%), and more than ⅓ reported having headaches (36%)
  - More than 1 in 4 had indigestion (26%), and almost 1 in 4 felt tense (23%)
  - About ⅓ felt depressed (34%) or as they they could cry

FIG.11.2 WARNING SIGNS OF STRESS



- Respondents list listening to music (47%) and exercise (46%) or walking among most common stress coping mechanisms
  - Other methods include watching TV or movies (40%), surfing the internet, praying (29%), napping (27%), eating (23%), drinking alcohol, and smoking
- Stress of an event also reflects meaning of the event to an individual
  - We appraise events, and our responses depend on their perceived danger, our values and goals, our beliefs in our coping ability, and our social situations

CONFLICT

- Feeling of being pulled in 2 or more directions by opposing motives
- Being torn in different directions by opposing motives

APPROACH-APPROACH CONFLICT

- Least stressful type
- Choice between two desirable options
- Each of the two goals is desirable, and both are within reach
- Usually resolved by making a decision
- Those who experience this may vacillate until they make a decision

AVOIDANCE-AVOIDANCE CONFLICT

- Choice between two undesirable outcomes
- More stressful because you are motivated to avoid each of two negative goals
- Avoiding one requires approaching the other
- Each potential outcome is undesirable
- When conflict is highly stressful and no resolution is in sight, some people withdraw from the conflict by focusing on other matters or doing nothing

APPROACH-AVOIDANCE CONFLICT

- Single goal has both positive and negative aspects
- When same goal produces both approach and avoidance motives
- Goals that produce mixed motives may seem more attractive from a distance but undesirable from up close

MULTIPLE APPROACH-AVOIDANCE CONFLICT

- Most complex form of conflict
- Each of several alternative courses of action has pluses and minuses
- Each alternative has both positive and negative aspects

IRRATIONAL BELIEFS

- Psychologists **Albert Ellis** noted that our beliefs about events, not just the events themselves, can be stressors
- A -> B -> C approach: Activating events -> Beliefs -> Consequences
- Anxieties about the future and depression over a loss are normal and to be expected
- **CATASTROPHIZE** - to interpret negative events as being disastrous; to "blow out of proportion"
- Ellis proposed that many of us carry with us the irrational belief, which are our personal doorways to distress
  - They can give rise to problems in themselves

COMMON IRRATIONAL BELIEFS

1. Need for approval
2. Punishing mistakes
3. Catastrophizing
4. External Control
5. Obsessing over danger
6. Avoidance over responsibility
7. Dependence on others
8. Demanding perfection
9. Permancent impact of the past
10. Absolute control
11. Happiness through inaction
12. Helplessness over emotion

THE TYPE A BEHAVIOR PATTERN

- Behavior characterized by a sense of time urgency, competitiveness, impatience, and hostility
- Referred as the **coronary-prone behavior pattern**
- Key traits:
  - Highly drive and time-conscious

    - Difficulty relaxing or delegating tasks
    - Prone to hostility and aggression
- Type A people are highly driven, competitive, impatient, hostile, and aggressive
- Difficulty relaxing or delegating tasks
- Prone to hostility and aggression

HEALTH RISKS

- Increased stress levels
- High risk of hypertension and coronary diseases

BEHAVIORAL PATTERNS

- Restlessness and impatience
- Striving for continuous self-improvement

TYPE B PEOPLE

- Relax more readily and focus more on the quality of life
- They show lower blood pressure than Type A people in response to stressors
    - Reason that type A behavior is also referred to as **CORONARY-PRONE BEHAVIOR PATTERN**
- Less ambitious and less impatient, and pace themselves

PSYCHOLOGICAL MODERATORS OF STRESS

- Psychological factors can influence, or moderate the effects that stressors have on us

SELF-EFFICACY EXPECTATIONS: "THE LITTLE ENGINE THAT COULD"

- **SELF-EFFICACY** is the ability to make things happen
- **SELF-EFFICACY EXPECTATIONS**, beliefs that we can bring about desired changes through our own efforts, affect our ability to withstand stress
- Adrenaline and noradrenaline are secreted when we are under stress
    - Excessive arousal can distract us from coping with the tasks at hand.
- People with higher self-efficacy expectations have biological as well as psychological reasons for remaining calmer
    - People who have higher self-efficacy expectations are less prone to be disturbed by adverse events

PSYCHOLOGICAL HARDINESS

- Also helps people resist stress
- Cluster of traits that buffer stress and are characterized by commitment, challenge, and control
- Derived from work of **Suzanne Kobasa**
- Psychologically hardy executives had 3 characteristics:
    - **COMMITMENT** - Psychologically hardy executives tended to involve themselves in, rather than feel alienated from, what they were doing or encountering.
    - **CHALLENGE** - Psychologically hardy executives believed that change, rather than stability is normal in life. They appraised change as an interesting incentive to personal growth, not a threat to security.
    - **CONTROL -** Psychologically hardy executives were high in perceived control over their lives. Psychologically hardy people tend to have what psychologist Julian B. Rotter (1990) termed an internal **locus of control**.
        - **LOCUS OF CONTROL** - place to which an

individual attributes control over the receiving of rienforcers—either inside or outside the self
- Psychologically hardy people may be more resistant to stress because they choose to face it, and they interpret stress as making life more interesting

SENSE OF HUMOR

- Can moderate effects of stress
- Laughter stimulates output of **endorphins**, which might enhance the functioning of the immune system
- Benefits of humor may be explained in terms of the positive cognitive shifts they entail and the positive emotions that accompany them

PREDICTABILITY AND CONTROL

- Ability to predict a stressor apparently moderates its impact
- **Predictability** allows us to brace ourselves for the inevitable and plan ways of coping with it
- **Control**—even the illusion of being in control—allows us to feel that we are not at the mercy of fates
- **Predictability** is of greater benefit to **INTERNALS** than to **EXTERNALS**
  - **INTERNALS** - people who perceive the ability to attain reinforcements as being largely within themselves
    - People who wish to exercise control over their situations
  - **EXTERNALS** - people who perceive the ability to attain reinforcements as being largely outside themselves

EMOTIONAL SUPPORT

- People are social beings, and emotional support also seems to act as a buffer against the effects of stress
- **Emotional concern:** listening to people's problems and expressing feelings of sympathy, caring, understanding, and reassurance
- **Instrumental aid**: the material supports and services that facilitate adaptive behavior
- **Information**: guidance and advice
- **Appraisal**: feedback from others about how one is doing
- **Socializing**: conversation, recreation, even going shopping with someone

STRESS AND THE BODY

- Stress is more than just as feeling; it has very definite effects on the body, which can lead to psychological and physical health problems
- **Dr. Hans Selye** - mid-20th century biochemist and endocrinologist
  - Stress researcher

THE GENERAL ADAPTATION SYNDROME

- Body's response to different stressors shows certain similarities whether the stressor is a bacterial invasion, perceived danger, or a major life change
- Selye labeled this response as **general adaptation syndrome (GAS)**
- **SELYE'S POSTULATES:**
  - Under stress, body is like a clock with an alarm; does not

shut off until energy is depleted

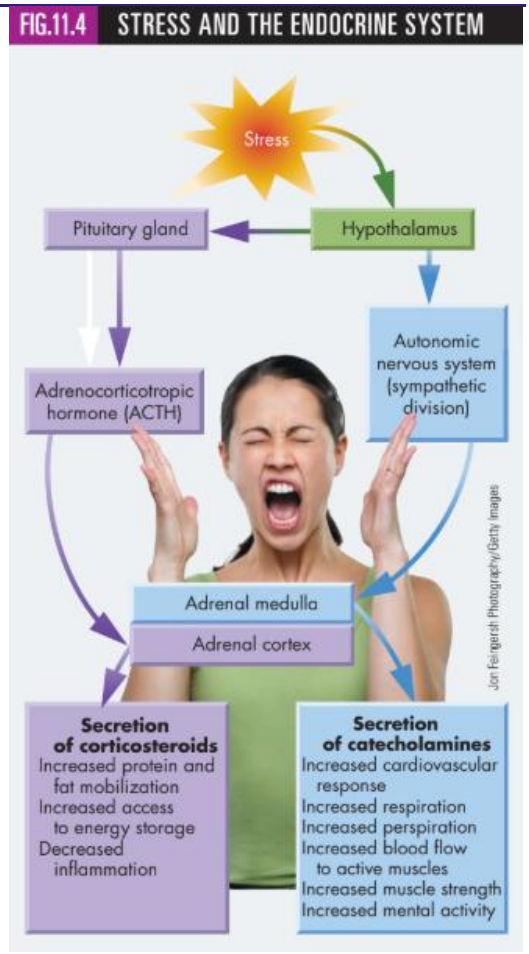
- **GAS** - Selye's hypothetical three-stage response to stress
- Group of bodily changes that occur in 3 stages:

THE ALARM REACTION

- First stage of GAS, triggered by perception of a stressor and characterized by sympathetic activity
- Cope with emergency short-term changes
- Characterized by sympathetic activity of Autonomic nervous system
- Mobilizes or arouses the body
- Basis for **fight-or-flight reaction**
  - **Fight-or-flight reaction** - an innate adaptive response to the perception of danger
    - Stressors were life-threatening
    - Removal of threat = lower state of arousal
- Endocrine, Cardiovascular, Musculoskeletal system: **SURVIVAL**

HORMONES:

- Hypothalamus secretes **corticotrophin-releasing** hormone which causes the pituitary gland to secrete **adrenocorticotrophic hormone**
  - **Adrenocorticotropic hormone** causes adrenal cortex to secrete **cortisol** and other corticosteroids
  - Corticosteroids help protect body by combating allergic reactions and reducing inflammation
  - EFFECTS OF CORTISOL:



- **RUNDOWN:**
  - Corticosteroids, adrenaline, and noradrenaline are secreted
  - Respiration rate increases
  - Heart rate increases
  - Blood pressure increases
  - Muscle tenses
  - Blood shifts from internal organs to skeletal musculature
  - Digestion is inhibited
  - Sugar is released from the liver
  - Coagulability of blood increases

THE RESISTANCE STAGE

- 2nd stage of GAS, characterized by prolonged sympathetic activity in an effort to restore lost energy

and repair damage. Also called the adaptation stage
- Continued release of stress hormones
- If alarm reaction mobilizes body and stressor is not removed, we enter adaptation or resistance stage
- Levels of endocrine and sympathetic activity are lower than in the alarm reaction, but still higher than normal
- Person feels tense, and body remains under a heavy burden
  - Elevated heart rate, blood pressure, muscle tension
  - Irregular blood sugar levels
  - Immune system is suppressed
- Body can cope with these elevated metabolic functions for some time

THE EXHAUSTION STAGE

- The third stage of the GAS, characterized by weakened resistance and possible deterioration
- If stressor is still not dealt with adequately
- Muscles become fatigued; body is depleted of the resources required for combating stress
- With exhaustion, parasympathetic division of ANS may predominate
  - Heart rate and respiration slows down
- Continued stress in this stage may lead to "**diseases of adaptation**"
  - Connected with constriction of blood vessels and alteration of the heart rhythm and can range from allergies to hives and heart disease, ultimately death

EFFECTS OF STRESS ON THE IMMUNE SYSTEM

- Psychologists, biologists, and medical researchers have combined their efforts in a field of study called **psychoneuroimmunology**
  - Addresses the relationships among psychological factors, the nervous system, the endocrine
  - Study effects of stress in the immune system
- Research show that stress suppresses immune system as measured by presence of substances in the blood that makes up part of the immune system

THE IMMUNE SYSTEM

- The system of the body that recognizes and destroys foreign agents (antigens) that invade the body
- Combats disease in several ways
  - Production of white blood cells, which engulf and kill pathogens
    - **LEUKOCYTES** engage in search-and-destroy missions which they recognize and eradicate foreign agents and unhealthy cells
    - white blood cells (derived from the Greek words *leukos*, meaning "white," and *kytos*, literally meaning "a hollow" but used to refer to cells)
  - Body reacts to **ANTIGENS** by generating specialized proteins or **ANTIBODIES**
    - **ANTIGENS** - substances that stimulate the body to mount an immune system response to it (short for antibody generator)
    - **ANTIBODIES** - substances formed by white blood cells that

recognize and destroy antigens
    - **INFLAMMATION** - increased blood flow to an injured area of the body, resulting in redness, warmth, and an increased supply of white blood cells
        - white blood cells combat invading microscopic life-forms such as bacteria, which otherwise might use the local damage as a port of entry into the body
        - Blood vessels in injury contract, then the blood vessles will dilate, increasing the flow of blood
        - First blood vessels in injury contract - stem bleeding
        - Dilate - increase flow of blood, cells, and natural chemicals

STRESS AND THE IMMUNE SYSTEM

- Stress exhaust because it produce steroids
- Steroids suppress the functioning of the immune system
- Suppression has negligible effects when steroids are secreted occasionally.
    - But persistent secretion of steroids decreases inflammation, interferes with the formation of antibodies, and leads to vulnerability
- **Walter Cannon:** humans are prewired for flight or fight
- **Shelly E. Taylor**: Women under stress care more for their children and seek social contact
- **Taylor**: Pituitary hormone will release oxytocin, promoting nurturing effects

Women tend to outlive men.

PSYCHOLOGY AND CHRONIC HEALTH PROBLEMS

- **BIOPSYCHOSOCIAL** - approach to health recognizes that there is no single, simple answer to these questions.
    - Having to do with the interactions of biological, psychological, and sociocultural factors
- The likelihood of contracting an illness—be it a case of the flu or cancer—can reflect the interaction of many factors, including biological, psychological, and sociocultural factors.

BIOLOGICAL FACTORS

- Pathogens, injuries, age, gender, and a family history of disease
- Some create predispositions toward the health problem
- Genetic predispositions interact with the environment

PSYCHOLOGICAL FACTORS

- Attitudes, emotions, and behaviors
- Psychological states affect immune system, making individuals susceptible to disorders like cancer

SOCIOCULTURAL FACTORS

- Socioeconomic status, lifestyle



CORONARY HEART DISEASE

- Heart's blood supply is blocked or interrupted by a build-up of fatty substances in the coronary arteries
- Leading cause of death in US
- **RISK FACTORS:**
  - Family history
  - Physiological conditions: obesity, high serum cholesterol levels, and **hypertension**
    - **Essential hypertensionism** - hypertension without identifiable cause
      - Has a genetic component
  - Patterns of consumption
  - Type A behavior
  - Job strain
  - Chronic fatigue, stress, anxiety, depression, and emotional strain
  - A physically inactive lifestyle
- **BLOOD PRESSURE** - connected with emotions like depression and anxiety
  - Rises when we inhibit expression of strong feelings or are angry or on guard against threats
  - Most people cannot sense hypertension
- **REDUCING RISK OF CHD:**
  - Stopping smoking
  - Eating healthier
  - Controlling blood pressure and LDL cholesterol
  - Management of anger and stress
  - Exercising regularly

CANCER

- development of abnormal, or mutant, cells that may take root anywhere in the body and can metastasize (spread to other parts): in the blood, bones, digestive tract, lungs, and sex organs.
- Second-leading cause of death in US
- If their spread is not controlled early, the cancerous cells may metastasize—that is, establish colonies elsewhere in the body.
- **RISK FACTORS:**
  - Smoking
  - Consuming more than a drink or two
  - High intake of animal fats
  - Genetic predisposition
  - Sunbathing
- **CULTURAL FACTORS**
  - Higher cancer death rate in nations with high daily fat intake
- **PSYCHOLOGICAL FACTORS** - Prolonged psychological conditions suppress immune functioning
- **ETHNIC DISPARITIES IN CANCER** - African Americans are more likely than European Americans to develop and die from cancer
- Disparities linked to lower socioeconomic status and lack of health care access
- **CARCINOGENIC GENES** - may remove the brakes from cell division, allowing cells to multiply wildly.
  - Or they may allow mutations to accumulate unchecked. Many behavior patterns however, markedly heighten the risk

REDUCING RISK OF CANCER

- Get regular medical checkups for early detection
- Stop smoking

- Healthy diet; eat fewer saturated fats; consume more fruits, vegetable, and whole grains
- Exercise regularly

13: PSYCHOLOGICAL DISORDERS

WHAT ARE PSYCHOLOGICAL DISORDERS

- Patterns of behavior or mental processes that are connected with emotional distress or significant impairment in functioning
- Are not predictable responses to events
- Behaviors of mental processes are suggestive of psychological disorders when they meet some combinations of the ff criteria:
  1. **They are unusual** – Posses behaviors that are of bizarre quality; includes unpredictable panic attacks
  2. **They suggest faulty perception or interpretation of reality** – Includes **hallucinations** and **ideas of persecution**
     - **HALLUCINATIONS** - perception in the absence of sensory stimulation
     - **IDEAS OF PERSECUTION** - erroneous beliefs that one is being victimized or persecuted
  3. **They suggest severe personal distress** – includes anxiety, exaggerated fears, and other psychological states - if severe can be considered abnormal
  4. **They are self-defeating** – Shows behaviors or mental processes that are hazardous to the self or others
  5. **The individual's behavior is socially unacceptable** – we must consider cultural context of a behavior pattern in judging whether it is normal

PERSPECTIVES ON PSYCHOLOGICAL DISORDERS

THE DEMONOLOGICAL MODEL

- People assumed behaviors associated with psychological disorders were caused by possession by the devil.
- Led to brutal treatments
- Ancient Greeks believed that Gods punished humans by causing madness, except **HIPPOCRATES**, who suggested that psychological disorders are caused by brain abnormalities

THE MEDICAL MODEL

- Age of reason, where many health professionals, including **Philippe Pinel**, began to view psychological disorders as diseases of the mind, and they encouraged humane treatment
- Assumed that illnesses have physical or biological causes that can be identified and that people afflicted by them are to be cured through treatment or therapy

CONTEMPORARY PSYCHOLOGICAL MODELS

- **DIATHESIS-STRESS MODEL** - assumes that there may be biological differences between individuals (**diathesis**) that explain why some people develop certain psychological disorders under stress, whereas others do not
- **BIOPSYCHOSOCIAL MODEL** - explains psychological disorders in the terms of a combination of:
  a. Biological vulnerabilities
  b. Psychological factors
  c. Sociocultural factors

CLASSIFYING PSYCHOLOGICAL DISORDERS

- Classification is important to scientific progress
- Enables clear communication among psychological investigators
- Most widely used classification system for psychological disorders is the **Diagnostic and Statistical Manual of Mental Disorders** (DSM-5), compiled by the American Psychiatric Association in 2013

PREVALENCE OF PSYCHOLOGICAL DISORDERS

- About **50%** of people will meet the criteria for a DSM disorder at some point in life, often starting in childhood or adolescence
- Slightly more than **25%** of people experience a psychological disorder in any given year
- Women are more likely to suffer from anxiety and mood disorders
- Men are more prone to substance use disorders
- When considering the ripple effects, psychological disorders affect everyone in some way.

ANXIETY DISORDERS

ANXIETY DISORDERS

- Disorders characterized by excessive worrying, fear of losing control, nervousness, and inability to relax
- Have psychological and physical symptoms
- Psychological symptoms: worrying, fear of the worst happening, fear of losing control, nervousness, inability to relax
- Physical symptoms reflect arousal of sympathetic branch of autonomic nervous system: trembling, sweating, fast heart rate, elevated blood pressure, faintness
- Anxiety is an appropriate response to a real threat, but can be abnormal when it is excessive and out of nowhere, and when it prevents us from doing important things

SPECIFIC PHOBIA DISORDERS

SPECIFIC PHOBIAS

- Persistent fear of a specific object or situation

CLAUSTROPHOBIA

- Fear of tights or enclosed spaces

ACROPHOBIA

- Fear of heights

AGORAPHOBIA

- Fear of being in open, busy areas

ARACHNOPHOBIA

- Fear of spiders

SOCIAL ANXIETY DISORDER

- Irrational, excessive fear of public scrutiny
- Extreme fear of social situations where the individual may be judged or humiliated
- Also known as **social phobia**

PANIC DISORDER

- Recurrent experiencing of attacks of extreme anxiety in the absence of external stimuli that usually elicit anxiety
- **PANIC ATTACK**: abrupt anxiety attack unrelated to specific objects or situations
  - May last minute or hours, which leaves the person drained afterwards

- People with panic disorder may experience: strong cardiac related sensations
  - Shortness of breath, heavy sweating, tremors, and pounding of heart
- Levels of cortisol (stress hormone) in the saliva are elevated
- People with the disorder may also experience:
  - Choking sensations
  - Nausea
  - Numbness or tingling
  - Flushes or chills
  - Fear of going crazy or losing control
- Diagnosis is reserved for those who undergo multiple attacks or live in fear of attacks

GENERALIZED ANXIETY DISORDER

- Feelings of dread and foreboding and sympathetic arousal of at least 6 months' duration
- Central symptom is persistent anxiety that cannot be attributed to a phobic object, situation, or activity
- Pervasive worrying about numerous stressors
- Symptoms include:
  - Motor tension (shakiness, inability to relax, furrowed brow, and fidgeting)
  - Autonomic over arousal (sweating, racing heart, frequent urination, and diarrhea)
  - Excessive vigilance
  - Tendency to be distracted

ORIGINS OF ANXIETY DISORDER

PSYCHOLOGICAL VIEWS

- **BEHAVIORISM:** Some learning theorists consider phobias to be conditioned fears that were acquired in early childhood
  - Children may develop fears through observational learning by imitation parents who react negatively to certain stimuli
  - Avoidance of feared stimuli is reinforced by reduction of anxiety
- **COGNITIVISM**: Cognitive theorists note that people's appraisals of the magnitude of threats help determine whether they are traumatic and can lead to PTSD
  - Interpretation of threats influences anxiety
  - Early anxieties can shape self-perception
  - Entertaining negative thoughts worsen anxiety by intensifying symptoms and disrupting focus
- Belief that we will not be able to handle a threat heightens anxiety
- Belief that we are in control reduces anxiety

BIOLOGICAL VIEWS

- **GENETICS:** Anxiety disorders tend to run in families
  - Identical twins are much more likely to both have the disorder than fraternal twins
  - Biological parent places the child at greater risk for anxiety and related traits than the adoptive parent
- **Evolutionary psychologists:** suggest that anxiety may reflect natural selection
  - Humans are genetically predisposed to fear stimuli that may have posed a threat to their ancestors
  - Inherited a reactive autonomic nervous system and high cortisol levels

- Autonomic nervous system may stimulate production of excessive quantities of adrenaline (epinephrine) and noradrenaline (norepinephrine)
  - These chemicals normally pump up the body in preparation for the fight-or-flight response

BIOPSYCHOSOCIAL VIEWS

- Anxiety disorders often result from a mix of biological, psychological, and social factors
- In panic disorder, biological imbalances may trigger initial attacks
- Fear of future attacks and sensitivity to bodily cues can worsen anxiety and create a sense of helplessness
- Cycle of fear and helplessness intensifies discomfort and panic
- Psychological treatments can help reduce physical symptoms
- These methods also empower individuals by showing they can manage and cope with panic attacks

COPING METHODS

- Breathing exercises
- Coping strategies

OBSESSIVE-COMPULSIVE AND RELATED DISORDERS

- Share a pattern of repetitive behavior associated with personal distress or impaired function in daily life

OBSESSIVE COMPULSIVE DISORDER

- **OBSESSION:** recurrent anxiety-provoking thoughts or images that seem irrational and beyond control
  - Compelling and recurrent that they disrupt daily life
  - Common types:
    - Contamination fears
    - Harm-related obsessions
    - Perfectionism
    - Religious or moral concerns
    - Sexual thoughts
    - Health concerns
- **COMPULSION**: irresistible urges to engage in thoughts or behaviors that tend to reduce the anxiety, often repeatedly
  - Behavior that tends to reduce anxiety
  - Typically time-consuming and only provides temporary relief, reinforcing cycle of OCD
  - EXAMPLES:
    - cleaning/washing
    - Checking
    - counting/repeating
    - Arranging
    - Reassurance-seeking
    - Mental rituals
- **HOW DOES IT HAPPEN?**
  - Repeating cycle of obsessions and compulsions
  - Involves:
    - Obsession: trigger
    - Anxiety: emotional response
    - Compulsion: attempt to fix
    - Relief: temporary reprieve

HOARDING DISORDER

- People with **hoarding disorder** feel an overpowering need to accumulate certain kinds of possessions and have difficulty discarding them
- Other related disorders include:
  - **body dysmorphic disorder**, in which individual is preoccupied with exaggerated or imagined physical defects
  - continuous hair pulling
  - compulsive skin-picking

TRAUMA-AND STRESSOR-RELATED DISORDERS

- Individual shows difficulty coming to terms with traumatic events

POSTTRAUMATIC STRESS DISORDER

- A disorder that follows a distressing event outside the range of normal human experience
- Characterized by features such as intense fear, avoidance
- Characterized by rapid heart rate, and feelings of anxiety and helplessness that are caused by a traumatic experience
- May occur months or years after the event
- Traumatic event is revisited in the form of intrusive memories, recurrent dreams, and flashbacks—feeling that event is recurring
- They often have sleep problems, irritable outbursts, difficulty concentrating, extreme vigilance, and an intensified "startle" response

ACUTE STRESS DISORDER

- A disorder , like PTSD, that is characterized by feelings of anxiety and helplessness and caused by a traumatic event
- Occurs within a month of the event and lasts from two days to four weeks

If ASD symptoms last beyond 4 weeks and meet criteria for PTSD, diagnosis may change to PTSD.

KEY FACTORS:
1.Severity of trauma
2. Personal coping mechanisms
3. Support systems
4. Co-occurring mental health conditions, like depression or anxiety

DISSOCIATIVE DISORDERS

- Disorders in which there are sudden, temporary changes in consciousness or self-identity
- Mental processes—processes that make a person feel whole—may seem to be split off from one another

DISSOCIATIVE IDENTITY DISORDER

- A.k.a multiple personality disorder, William's disorder
- Disorder in which a person appears to have two or more distinct identities or personalities that may alternately emerge
- Two or more identities or personalities, each with distinct traits and memories "occupy" the same person
- Each identity may or may not be aware of the other identities

DISSOCIATIVE AMNESIA

- Loss of memory or self-identity; skills and general knowledge are usually retained
- Person is suddenly unable to recall important personal information
- Not attributed to organic problems
- In generalized dissociative amnesia, people forget their entire lives
- Amnesia may last for hours or years

DEPERSONALIZATION - DEREALIZATION DISORDER

- Disorder in which one feels detached from oneself or that one's surroundings are not real

ORIGINS OF DISSOCIATIVE DISORDERS

LEARNING THEORISTS & COGNITIVE PSYCHOLOGISTS

- People with dissociative disorders have learned not to think about bad memories or disturbing impulses to avoid feelings of anxiety, guilt, and shame

PSYCHODYNAMIC AND LEARNING THEORIES

- Dissociative disorders help people keep disturbing memories or ideas out of mind

OTHER FACTS

- US and Canada are the places where DID has been publicized and symptoms were learned
- DID is a global phenomenon that is udnertreated

SOMATIC SYMPTOM AND RELATED DISORDERS

- People with **somatoform disorders** complain of physical problems or a persistent belief that they have a serious disease, yet no evidence of physical abnormality can be found

ILLNESS ANXIETY DISORDER

- Earlier called **hypochondria**
- Insist that they are suffering from a serious physical illness, even though no medical evidence of illness can be found
- Fear of illness may disrupt their work or home life

CONVERSION DISORDER

- Disorder in which anxiety or unconscious conflicts are "converted" into physical symptoms that often have the effect of helping the person cope with anxiety or conflict
- Characterized by a major change in/ loss of physical functioning, although there are no medical findings to explain the loss of functioning
- It appears to convert source of stress into a physical ability
- Some people with conversion disorder show indifference to their symptoms
  - **LA BELLE INDIFFERENCE** - lack of concern for their (imagine) medical problem
  - Lack of concern suggests awareness that physical problems have their benefits

ORIGINS OF SOMATIC SYMPTOM AND RELATED DISORDERS

BIOPSYCHOSOCIAL PERSPECTIVE

- Somatoform disorders illustrate how biological, psychological, and social factors interact

PSYCHOLOGICAL PERSPECTIVE

- These disorders related to what one focuses on
- **Illness Anxiety Disorder**: Sensitive to bodily sensations and tend to ruminate about them
- **Conversion Disorder:** Individuals may be highly susceptible to hypnosis
- Some researchers view it as a form of self-hypnosis, where intense focus on imagined symptoms blocks out contradictory information

BIOLOGICAL FACTORS

- Traits like perfectionism and rumination may be partially inherited

EVOLUTIONARY VIEW

- A moderate concern for bodily harm can be adaptive for survival
- However, excessive concern may lead to dysfunctional preoccupations and distress

MOOD DISORDERS

- Moods are feeling states that color our psychological lives
- People with mood disorders experience disturbances in mood that are unusually severe or prolonged and impair their ability to function

BIPOLAR DISORDER

- Former name: manic-depressive disorder
- Have mood swings from ecstatic elation to deep depression
- Disorder in which the mood alternates between two extreme poles (elation and depression)
- In **elated** or **manic phase**, person may show excessive excitement or silliness
  - Show **rapid flight of ideas** - rapid speech and topic changes, characteristic of manic behavior
  - Argumentative, show poor judgement, destroy property, make huge contributions to charity, give away expensive things
  - Oversexed and too restless
- In **bipolar depression** or **depressive phase,** persons often sleep more than usual and are lethargic
  - More likely to have insomnia and agitation
  - Exhibit social withdrawal and irritability
  - Attempt suicide when shift from elated to depressive state
- Imbalances in the neurotransmitter **serotonin,** and genetics play a role in bipolar disorder

DEPRESSIVE DISORDERS

- May complain of lack of energy, loss of self-esteem, difficulty in concentrating, loss of interest in activities and other people, pessimism, crying, and thoughts of suicide

MAJOR DEPRESSIVE DISORDER

- Affects 5% to 7% of us within any given year and one person in 6 or 7 over the course of our lives
- Unable to make decisions
- Faulty perception of reality
- Delusions, hallucinations, unworthiness, guilt for imagined wrongdoings, and feelings of rotting from disease
- Symptoms include:
  - Poor appetite
  - Weight loss
  - Agitation
  - **PSYCHOMOTOR RETARDATION** - slowness in motor activity and in thought
- Minority may display faulty perception of reality—**psychotic behaviors**

ORIGINS OF MAJOR DEPRESSIVE DISORDER

- **PSYCHOLOGICAL VIEWS** - Depressed people behave as though they cannot obtain reinforcement

- Many people with depressive disorders feel helpless and hopeless
  - **LEARNED HELPLESSNESS** - a possible explanation for some depressive behavior, based on findings that organisms in aversive situations learn to show inactivity when their behavior is not reinforced
  - **COGNITIVE FACTORS**, such as making irrational standards or demands for self, also contribute to depression
  - **ATTRIBUTIONAL STYLES** Tendency to attribute one's behavior to internal or external factors, stable or unstable factors, and global or specific factors
    - **INTERNAL** - involves self-blame
    - **EXTERNAL** - places blame elsewhere
    - **STABLE** - suggests a problem that cannot be changed
    - **UNSTABLE** - suggests a temporary condition
    - **GLOBAL** - suggests that problem is quite large
    - **SPECIFIC** - chops the problem down to a manageable size

- **BIOLOGICAL PERSPECTIVES**: Genetic factors are strongly implicated in depressive disorders
  - Depression caused by under-utilization of **serotonin** in the brain
  - People with severe depression often respond to drugs (SSRIs) that heighten the action of serotonin

- **BIOPSYCHOSOCIAL PERSPECTIVE:** Depression may be a reaction to losses and stress
  - Depression may be a reaction to losses and stresses
  - Stressors contribute to feelings of depression
  - We tend to be more depressed by things we bring on ourselves
  - People who remain depressed have lower self-esteem, are less likely to be able to solve social problems, and have less social support

SUICIDE

- Suicide is the second leading cause of death among 15 to 24 year olds
- Linked to feelings of depression and hopelessness
- **Suicidal adults** tend to experience confusions about who they are and where they are going, impulsiveness, emotional instability, and social problems
- **Suicidal teenagers** highly achieving, rigid perfectionists
- Contributors to suicidal behavior
  - Concerns over sexuality, sexual abuse, grades, problems at home, and substance abuse
  - Anxiety or fear of being found out about something negative
  - Tendency to run in families

SOCIOCULTURAL FACTORS IN SUICIDE

- Suicide is connected not only with feelings of depression and stressful events, but also with age, educational status, ethnicity, and gender
- Rates of suicide amd suicide attempts also vary among

different ethnic groups and according to gender

MYTHS ABOUT SUICIDE

- Individuals who threaten suicide are only seeking attention
  - Most people who commit suicide give warnings about their intentions or have made prior attempts
- Discussing suicide with a person who is depressed does not prompt the person to attempt suicide
- Suicidal thinking is not necessarily a sign of psychosis, neurosis, or personality disorder
- People may consider suicide when they think they ran out of options

SCHIZOPHRENIA AND OTHER PSYCHOTIC DISORDERS

- People experience a loss of contact with reality, see or hear things that are not actually there (hallucinations), have ideas of people planting "radios" in their walls that have no basis in fact (delusions)

DELUSIONAL DISORDER

- Psychotic disorder characterized by persistent false beliefs, often involving paranoid themes
- Delusional beliefs typically concern events that could possibly occur
- Behavior is generally not bizarre

SCHIZOPHRENIA

- Severe psychological disorder affecting thought, language, perception, motor activity, and social interaction
- Often considered the most debilitating psychological disorder
- Affects nearly 1% of the global population
- Typically begins early in life, with lasting negative effects

POSITIVE SYMPTOMS

- Those symptoms that indicate the presence of inappropriate behavior, such as hallucinations, delusions, agitation, and inappropriate giggling
- Agitated behavior
- Vivid hallucinations
- Unshakable delusions
- Disorganized thinking
- Nonsensical or jumbled speech

NEGATIVE SYMPTOMS

- Those symptoms that reflect the absence of appropriate behavior, such as blank faces, monotonic voices, and motionless bodies
- Flat emotional expressions
- Motionless or rigid postures
- Also shows **MUTISM** (refusal to talk)

PROBLEMS IN THINKING AND LANGUAGE

- Schizophrenia has been called a "thought disorder"
- Problems in thinking, language, memory and attention
- Speech may become jumbled
- Many people with schizophrenia have unshakeable **DELUSIONS** of grandeur, persecution, or reference
  - **DELUSIONS** - false, persistent beliefs that are unsubstantiated by sensory or objective evidence
  - **Delusions of grandeur** - a person may believe that he is a famous historical figure
  - **Delusions of persecution** - may believe that they are sought after
  - **Delusions of reference** - erroneously believe that other people are talking about them or referring to them

PROBLEMS IN PERCEPTION

- **HALLUCINATIONS** - imagery in the absence of external stimulation that the person cannot distinguish from reality
- Auditory hallucinations (hearing voices) are the most common

PROBLEMS IN ACTIONS, EMOTIONS, AND SOCIAL INTERACTION

- Motor activity may become wild or so slowed that the person is said to be in a **STUPOR**
  - **STUPOR** - condition in which the senses, thought, and movement are dulled
- **CATATONIA** - psychotic condition characterized by striking motor impairment
  - Maintaining unusual postures for hours
  - Sudden shifts between stupor and agitation
- **WAXY FLEXIBILITY** - feature of catatonic schizophrenia in which people can be molded into postures that they maintain for quite some time
- Emotional responses may be flat or blunted or completely inappropriate
- Individuals withdraw socially, becoming absorbed in their own thoughts and fantasies

TYPES OF SCHIZOPHRENIA

- **PARANOID** - Characterized by intense delusions and auditory hallucination
- **DISORGANIZED** - Marked by severely disorganized thoughts and behavior
- **CATATONIC** - Dominated by motor disturbances and abnormal movements

EXPLAINING SCHIZOPHRENIA

PSYCHOLOGICAL FACTORS

- Theorists have explained schizophrenia in terms of conditioning and observational learning
- Schizophrenic behavior may be reinforced in socially unrewarding or punitive environments
- In observational learning, patients may mimic behaviors they observe in others
- Schizophrenic behaviors may be unintentionally reinforced because they attract more attention

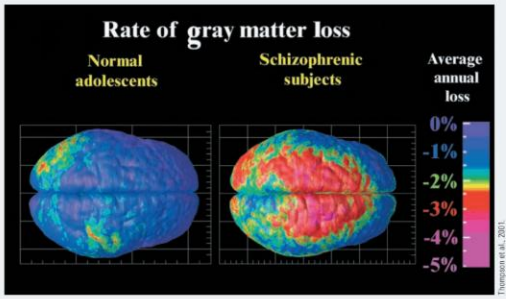
SOCIOCULTURAL FACTORS

- Critics note that many people who are reared in socially punitive settings are apparently immune to the disorder
- Some sociocultural theorists suggest that treatment of schizophrenia requires alleviation of poverty and other social ills rather than changing people whose behavior is deviant
- Poverty, poor parenting, discrimination, and overcrowding might contribute to schizophrenia, particularly in people genetically vulnerable to the disorder
- Lower socioeconomic status may result from schizophrenia rather than cause it, as individuals drift downward due to impaired social and cognitive abilities

BIOLOGICAL VIEWS FACTORS

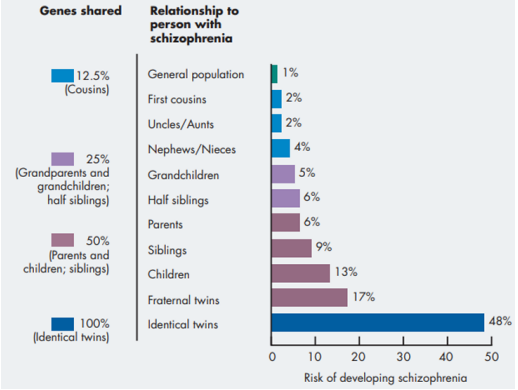
- Brain of schizophrenic people differ from those of normal people
- Brain structure: Reduced gray matter, smaller brain and prefrontal cortex, and lower activity in this region, associated with diminished planning and decision-making abilities

FIG.12.2 AVERAGE RATES OF LOSS OF GRAY MATTER AMONG NORMAL ADOLESCENTS AND ADOLESCENTS DIAGNOSED WITH SCHIZOPHRENIA



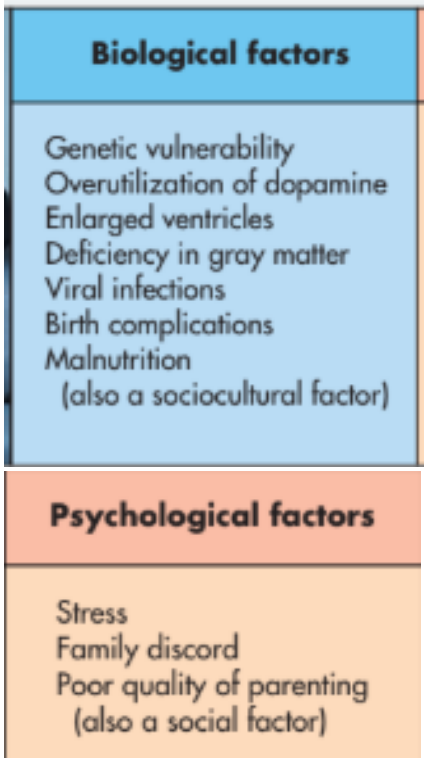
- **GENETICS:**
  - Schizophrenia runs in families
  - Risk increases with familial connection
  - Identical twins show a 48% concordance rate
  - Genetic predisposition is linked to multiple genes
- **PRENATAL and BIRTH COMPLICATIONS**:
  - Maternal flu, malnutrition, and prolonged labor are connected to increased risk
- **DOPAMINE THEORY**:
  - According to dopamine theory, people with schizophrenia "overutilize" dopamine, although they may not produce more of it
  - Excess dopamine activity and higher receptor density contribute to symptoms
  - Research suggests that they have increased concentrations of dopamine at the synapses in the brain, and also larger numbers of dopamine receptors

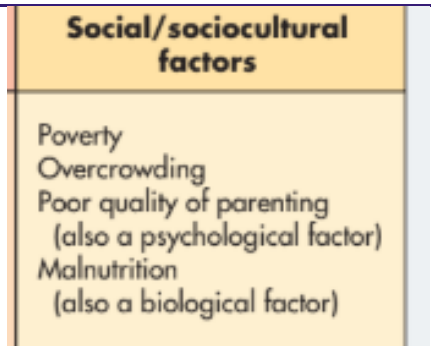
FIG.12.3 RELATIONSHIP TO A PERSON DIAGNOSED WITH SCHIZOPHRENIA AND LIKELIHOOD OF BEING DIAGNOSED WITH SCHIZOPHRENIA ONESELF



BIOPSYCHOSOCIAL MODEL

- People with a genetic vulnerability to the disorder experience increased risk for schizophrenia when they encounter problems
- People without the genetic vulnerability would not develop schizophrenia despite problems





PERSONALITY DISORDERS

- Mental health condition that involves long-lasting, disruptive patterns of thinking, behavior, mood, and relating to others
- Enduring patterns of maladaptive behavior that are sources of distress to the individual or others
- Inflexible and maladaptive

PARANOID PERSONALITY DISORDER

- Tendency to interpret other people's behavior as threatening or demeaning
- Persistent suspiciousness but not involving the disorganization of paranoid schizophrenia

SCHIZOID PERSONALITY DISORDER

- Personality disorder characterized by social withdrawal
- Indifference to relationships and flat emotional response

SCHIZOTYPAL PERSONALITY DISORDER

- Personality disorder characterized by oddities of thought, perception, and behavior but not involving bizarre psychotic behaviors
- Excessive fantasy and suspiciousness, feelings of being unreal
- Symptoms include:
  - Lack of close friends
  - Bizarre behavior
  - Suspicious attitudes towards others
  - Superstitious beliefs
  - Excessive social anxiety that does not improve with familiarity

ANTISOCIAL PERSONALITY DISORDER

- Disregard for the rights of others and lack of empathy
- Diagnosis given to a person who is in frequent conflict with society, yet who is undeterred by punishment and experience little or no guilt and anxiety
- Often show a superficial charm and are at least average in intelligence
- Symptoms include:
  - Hostility toward others
  - Impulsive behavior
  - Disregard for rules
  - Feelings of superiority
  - Manipulating others
  - Not accepting responsibility

BORDERLINE PERSONALITY DISORDER

- Instability in relationships, self-image, mood, and lack of control over impulses
- Uncertain of their values, goals, loyalties, careers, choice of friends, and sexual orientations
- Cannot tolerate being alone
- Can be clinging and demanding but often push away people they depend on

- Viewing others as either all good or all bad, inconsistency
- May engage in self-mutilation (scratching writs, or burning arms)
- Symptoms include:
  - Unstable self-image
  - Intense and unstable relationships
  - Chronic feelings of emptiness
  - Impulsive behavior

AVOIDANT PERSONALITY DISORDER

- Person is unwilling to enter into relationships without assurance of acceptance because of fears of rejection and criticism

ORIGINS OF PERSONALITY DISORDERS

- Cognitive psychologists find that antisocial adolescents tend to interpret other people's behavior as threatening, even when it is not

BIOLOGICAL FACTORS

- Personality traits are to some degree heritable, and many personality disorders seem to be extreme variations of normal personality traits

PSYCHOLOGICAL FACTORS

- Learning theorists suggests that childhood experiences can contribute to maladaptive ways of relating to others in adulthood
- Cognitive psychologists find that antisocial adolescents encode social information in ways that bolster their misdeeds

SOCIOCULTURAL FACTORS

- Some theorists believe we live in fragmented and alienating times that tend to create problems in forming a stable identity and stable relationships

14: METHODS OF THERAPY

WHAT IS PSYCHOTHERAPY

- A systematic interaction between a therapist and a client that applies psychological principles to affect the client's thoughts, feelings, or behavior in an effort to help the client overcome psychological disorders, adjust to problems in living or develop as an individual

SYSTEMATIC INTERACTION

- Therapist's theoretical point of view interacts with the client's problems to determine how the therapist and client relate to each other

PSYCHOLOGICAL PRINCIPLES

- It is based on psychological theory and research in areas such as personality, learning, motivation, and emotion

THOUGHTS, FEELINGS & BEHAVIOR

- Can be aimed at any or all of the aspects of human psychology

PSYCHOLOGICAL DISORDERS, ADJUSTMENT PROBLEMS, AND PERSONAL GROWTH

- Often used to treat people who have psychological disorders

ECLECTIC APPROACH

- Most popular form of therapy
- Combination of cognitive, behavioral, and other forms of therapy

THE HISTORY OF THERAPIES

- People believed that people who behaved in strange ways were possessed by demons, leading to treatments that involve cruel practices—generally, they died early

ASYLUM

- Originated in European monasteries
- First institutions intended primarily for housing people with psychological disorders
- Asylums primarily served as storage facilities rather than treatment centers. Overcrowding, noise, and disease worsened inmates' conditions, and many were subjected to physical abuse like chaining and beatings.
- **BEDLAN** was derived from St. Mary's of Bethlehem, the London asylum that opened in 1547
  - People with psychological disorders were chained, whipped, and allowed to lie in their own waste
- Reform movements began in the **18th century**
  - In Paris, **Philippe Pinel** unchained patients at La Salpêtrière, who most reentered society instead of running amok
- Movements to reform institutions were led by **William Tuke** in England and **Dorothea Dix** in America

MENTAL HOSPITALS

- In United States, mental hospitals gradually replaced asylums
- Function for treatment, not warehousing
- Because of high patient populations and understaffing, many patients received little attention

THE COMMUNITY MENTAL HEALTH MOVEMENT

- In 1960s, efforts have been made to allow people with serious psychological disorders to remain in their communities

- Attempt to maintain new patients as outpatients, and serve patients who have been released
- Many do not receive adequate follow-up care, some become homeless

PSYCHODYNAMIC THERAPIES

- Based on thinking of **Sigmund Freud**, founder of psychodynamic theory
- Assume that psychological problems reflect early childhood experiences and internal conflicts
- Conflicts involve shifting psychic energy among id, ego, and superego, which determine behavior

TRADITIONAL PSYCHOANALYSIS

PSYCHOANALYSIS

- Clinical method devised by Freud
- Aims to provide insight into the conflicts that are presumed to be roots of a person's problems
- **INSIGHT** - knowledge of the experiences that lead to conflicts and maladaptive behavior, recognition of unconscious feelings and conflicts, and conscious evaluation of one's thoughts, feelings, and behavior
- **CATHARSIS** - In psychoanalysis, the expression of repressed feelings and impulses to allow the release of the psychic energy associated with them
- Freud also sought to replace impulsive and defensive behavior with coping behavior
- Freud found that hypnosis allowed his clients to focus on repressed conflicts and talk about them, but some clients denied accuracy once they were out of trance, leading Freud to turn to free association
- **FREE ASSOCIATION** - In psychoanalysis, the uncensored uttering of all thoughts that come to mind
- Ego persists in trying to repress unacceptable impulses and threatening conflicts, resulting to clients showing **resistance** to recalling and discussing
- **RESISTANCE** - tendency to block the free expression of impulses and primitive ideas—a reflection of the defense mechanism of repression

TRANSFERENCE

- Freud believed that clients not only responded to him as an individual but also in ways that reflected their attitudes and feelings toward other people in their lives
- Responding to one person in a way similar to how one responded to another person in childhood

DREAM ANALYSIS

- Freud often asked clients to jot down their dreams upon waking so they could discuss them in therapy
- Freud considered dreams the "royal road to unconscious"
- Unconscious impulses were expressed in dreams as **wish fulfillment**
- **WISH FULFILLMENT** - in dreams, the acting out of ideas and impulses that are repressed when one is conscious

MODERN PSYCHODYNAMIC APPROACHES

- Suggest helpful behavior instead of focusing on insight alone
- There is more focus on the ego as the "executive" of personality and less emphasis on the id—reason why many psychodynamic therapists are called **ego analysts**

HUMANISTIC THERAPIES

- Focus on the quality of the client's subjective, conscious experience
- Focus on what clients are experiencing here and now
- Potential for self-fulfillment
- Focus on present and future
- Focus on conscious thoughts
- Take responsibility for actions

CLIENT-CENTERED THERAPY

- **CARL ROGERS** believed that we are free to make choices and control our destinies, despite burdens of the past
  - Also believed that we have natural tendencies toward health, growth, and fulfillment
- Aims to provide insight into the parts of us that we have disowned so that we can feel whole
- Creates an atmosphere that encourages self-exploration and self-expression
- Self-acceptance frees client to make choices that develop unique potential
- Nondirective, but has the ff. qualities:
  1. **Unconditional positive regard** - respect for clients as human beings with unique values and goals
  2. **Empathy** - recognition of the clients' experiences and feelings
     - Therapists view the world through the client's frame of reference by setting aside their own values and listening closely
  3. **Genuineness** - openness and honesty in responding to the client
- Widely in college and university counseling centers, not just to help students experiencing anxiety and depression, but also to help them make decisions
- **ACTIVE LISTENING** - central to client-centered therapy
  - empathetic listening

GESTALT THERAPY

- Originated by **Fritz Perls**
- Assumes that people disown parts of themselves that might meet with social disapproval or rejection
- **Gestalt** is a German word meaning "unified whole"
- Perls wanted to help clients integrate conflicting parts of their personalities
- Therapist leads clients though planned experiences to heighten their awareness of inner conflict and take responsibility for their behavior

BEHAVIOR THERAPY

- Also called **behavior modification**
- Applies principles of learning to directly promote desired behavioral changes
- Systematic application of the principles of learning to the direct modification of a client's problem behaviors
- Rely on principles of conditioning and observational learning
- Help clients discontinue self-defeating behavior patterns
- Aid clients in acquisition of adaptive behavior patterns
- May help clients gain insight into maladaptive behaviors by helping the person become aware of the circumstances in which behaviors occur

FEAR-REDUCTION METHODS

- Behavior-therapy methods for reducing fears

SYSTEMATIC DESENSITIZATION

- Wolpe's method for reducing fears by associating hierarchy of images of fear-evoking stimuli with deep muscle relaxation
- Gradual process in which the client learns to handle increasingly disturbing stimuli, while anxiety to each one is being counterconditioned

VIRTUAL TECHNOLOGY EXPOSURE THERAPY

- Follows principle of desensitization, but may use more elaborate equipment
- "Idea behind the treatment is to systematically expose the patient to aspects of their experience in a graded fashion so they can confront their fear of the trauma" - Dr. JoAnn Difede

MODELING

- Relies on observational learning
- Clients observe and then imitate people who approach and cope with the objects or situations that the clients fear
- Also increases self-efficacy expectations in coping with feared stimuli

AVERSIVE CONDITIONING

- Procedure that pairs painful or aversive stimuli with unwanted impulses in order to make the impulses less appealing
- Behavior-therapy technique in which undesired responses are inhibited by pairing repugnant or offensive stimuli with them

OPERANT CONDITIONING PROCEDURES

- We tend to repeat behavior that is reinforced. Behavior that is not reinforced tends to be extinguished. Behavior therapists have used these principles of operant conditioning with psychotic patients as well as with clients with milder problems.

THE TOKEN ECONOMY

- Controlled environment in which people are reinforced for desired behaviors with tokens that may be exchanged for privileges

SUCCESSIVE APPROXIMATIONS

- Used to help clients build good habits
- Series of behaviors that gradually become more similar to a target behavior

SOCIAL SKILLS TRAINING

- Decrease social anxiety and build social skills through operant conditioning procedures that employ:
  - Self-monitoring - keeping a record of one's own behavior to identify problems and record successes
  - Coaching
  - Modeling
  - Roleplaying
  - Behavior rehearsal - practicing
  - Feedback - receiving information about the effectiveness of behavior
- Helps people in their interpersonal relations

BIOFEEDBACK TRAINING (BFT)

- Help clients become more aware of, and gain control over, various bodily functions
- Systematic feeding back to an organism of information about a bodily function so that the

organism can gain control of that function

COGNITIVE THERAPIES

- 49% of therapies
- Focuses directly on thoughts and encourages ideas that will help you solve problems rather than blow out of proportion
- Focuses on changing beliefs, attitudes, and automatic types of thinking that create and compound people's problems
- Aim to foster self-insight, but mainly aim to help make people more aware of current cognitions
- Form of therapy that focuses on how clients' cognitions lead to distress and may be modified to relieve distress and promote adaptive behavior
- Teaches people new and more adaptive ways of thinking
- Teach people new, more constructive ways of thinking

AARON BECK'S COGNITIVE THERAPY

- **AARON BECK** - noticed depressed people were similar in the way they view the world
- Focus on arguing clients out of beliefs that are making them miserable and exposing them to situations they avoid because of irrational fear
- Beck notes "cognitive errors" that contribute to clients' miseries:
  1. Clients may **selectively perceive** the world as a harmful place and ignore evidence to the contrary
  2. Clients may **overgeneralize** on the basis of a few examples
  3. Clients may **magnify**, or blow out of proportion, the importance of negative events
  4. Clients may engage in **absolutist thinking**, or looking at the world in black and white rather in shades of gray

ALBERT ELLIS' RATIONAL EMOTIVE BEHAVIOR THERAPY (REBT)

- Beliefs about events, not just the events themselves, shape our responses to them
- Encourages clients to challenge and correct irrational expectations and maladaptive behaviors

COGNITIVE BEHAVIORAL THERAPY

- Approach to therapy that uses cognitive and behavioral techniques that have been validated by research
- Albert Ellis developed his own methods of cognitive therapy—**rational emotive therapy—**and came to see value of behavioral techniques
  - He renamed his method rational emotive behavior therapy

GROUP THERAPIES

- Members might be jointly desensitized to anxiety-evoking stimuli or might practice social skills together
- Treatment of psychological problem in which two or more patients/clients interact with each other on both an emotional and
- Advantages:
  1. Economical - allows therapist to work with several clients at once
  2. Provides more information and life experience for clients to draw on

3. Appropriate behavior received support
4. Affiliating with people with similar problems is reassuring
5. Group members who show improvement provide hope for other members
6. Many individuals seek therapy because of problems in relating to others

- Not for everyone
- **Goals:**
  -

COUPLE THERAPY

- Helps couples enhance their relationship by improving communication skills and helping them manage conflict
- Teaches couples communication skills, ways of handling feelings such as anger, and ways of solving problems

FAMILY THERAPY

- Form of group therapy in which one or more families constitute the group
- **SYSTEMS APPROACH** - family interaction is studied and modified to enhance the growth of individual family members and of the family unit as a whole
- Family therapist attempts to teach family to communicate and encourages growth and autonomy in each family member

SELF-HELP AND SUPPORT GROUPS

- Self-help and support groups, popular in the US and globally, offer people with psychological challenges a space to connect and share solutions
- Groups often focus on specific issues
- **Alcoholic Anonymous (AA)** is the most well-known, using a 12-step program that emphasizes personal accountability, spiritual support, and making amends
  - Its success is linked to long-term participation, reduced depression, and increased confidence in resisting alcohol

DOES PSYCHOTHERAPY WORK?

- In 1952, **Hans Eysenck** claimed that psychotherapy was no more effective than no treatment at all, suggesting many people improved due to spontaneous remission
  - He argued that ⅔ of people
  - with anxiety or depression improved within 2 years, with or without therapy
- Later research contradicted Eysenck, showing therapy's effectiveness through meta-analyses and studies
- Psychodynamic and client-centered therapies help motivated, verbal clients with mild anxiety, depression, and relationship issues
  - These therapies are not effective for severe psychotic disorders like schizophrenia or bipolar disorder
- Cognitive therapy, developed by Ellis and Beck, effectively treats anxiety, depression, OCD, and personality disorders
  - Also helps with major depressive disorder, once thought treatable only by biological methods
- Behavioral and cognitive therapies improve social skills, self-control, and sexual dysfunctions.

- They support couples and families in distress and address health-related behaviors like chronic pain and substance abuse

EVIDENCE-BASED PRACTICES

- Randomized controlled experiments are important standards in psychological research for testing therapy effectiveness
- Researchers often use treatment manuals to ensure consistency in therapy methods
- Therapies proven effective through rigorous testing are called **evidence-based practices**
- Other treatments may be added as more evidence accumulate

TABLE 13.1 EXAMPLES OF EVIDENCE-BASED PRACTICES

Treatment	Condition for Which Treatment Is Effective
Cognitive therapy	Depression Headache
Behavior therapy or behavior modification	Depression Developmental disabilities Enuresis (bed-wetting)
Cognitive behavioral therapy	Panic disorder with and without agoraphobia Generalized anxiety disorder Bulimia nervosa
Exposure treatment	Agoraphobia and specific phobia
Exposure and response prevention	Obsessive–compulsive disorder
Parent training programs	Children with oppositional behavior

- Many therapists argue that evidence-based practices favor cognitive behavioral therapies because they are more readily standardized in treatment manuals and can be followed more accurately by practitioners in experiments
- Observers also note that practices common among various therapies lead to positive outcomes

ETHNICITY AND PSYCHOTHERAPY

- Americans from ethnic minority groups are less likely than European Americans to seek therapy due to:
  - Lack of awareness that therapy would help
  - Lack of information about the availability of professional services or inability to pay for them
  - Distrust of professionals, particularly European American professionals and male professionals
  - Language barriers
  - Reluctance to open up about personal matters to strangers
  - Cultural inclination toward other approaches to problem solving
  - Negative experiences with professionals and authority figures

AFRICAN AMERICANS

- Often reluctant to seek psychological help because of cultural assumptions that people should manage their own problems
- Some are suspicious of therapists—especially European American ones
- May withhold personal information because of society's history of racial discrimination

ASIAN AMERICANS

- Some stigmatize people with psychological disorders, which leads to denying problems and refusing to seek help
- May not understand or believe in Western approaches to psychotherapy
- Many Asians also experience psychological problems as physical symptoms

LATIN AMERICANS

- Measures like the ff. may help bridge the gaps between

psychotherapists and Latin Americans:
  - Interacting with clients in the language requested by them or, referring them to professionals who can do so
  - Using methods that are consistent with the client's values and levels of acculturation
  - Developing therapy methods that incorporate clients' cultural values
    - **Cuento therapy** - form of therapy uses traditional folktales (cuentos) with characters who serve as models for adaptive behavior

NATIVE AMERICANS

- Some psychological disorders they experienced involve disruption of their traditional culture caused by European colonization
- Efforts to prevent psychological disorders may well focus on strengthening Native American cultural identity, pride, and cohesion

BIOLOGICAL THERAPIES

- Apply people's biological structures and processes to the amelioration of psychological disorders

DRUG THERAPY

ANTIANXIETY DRUGS

- Most antianxiety drugs, such as Valium and Ativan, belong to **Benzodiazepines**
- Usually prescribed to outpatients who complain of generalized anxiety or panic attacks
- Depress the activity of CNS, which in turn decreases sympathetic activity, reducing the heart rate, respiration rate, and feelings of nervousness and tension
- It can be easily tolerated
  - When tolerance occurs, dosages must be increased for the drug to remain effective
- Most common side effect is **sedation** (feeling tired or drowsy)
- Withdrawal from these drugs may lead to **REBOUND ANXIETY**
  - **REBOUND ANXIETY** - anxiety that can occur when one discontinues use of a tranquilizer
- Can induce physical dependence, as evidenced by withdrawal symptoms such as tremors, sweating, insomnia, and rapid heartbeat

ANTIPSYCHOTIC DRUGS

- Used to treat Schizophrenia
- Also called major tranquilizers
- In most cases, it reduces agitation, delusions, and hallucinations
- Ex. Phenothiazines, Clozapine
- Blocks dopamine receptors in the brain

ANTIDEPRESSANTS

- Used to treat major depression
- Also helpful for people with eating disorders, panic disorders, obsessive-compulsive disorder, and social anxiety disorder
- Increase levels of neurotransmitters, which affect both depression and appetite
- **SELECTIVE SEROTONIN-REUPTAKE INHIBITORS (SSRIS)** - (ex. Prozac, Zoloft) block the reuptake of serotonin by presynaptic neurons
- **SEROTONIN-NOREPINEPHRINE REUPTAKE INHIBITORS (SNRIs)** - (ex. Effexor) inhibit reuptake of both serotonin and norepinephrine

MOOD STABILIZERS

- Ancient Greeks and Romans were among first to use metal lithium as mood stabilizer
- A salt in Lithium (lithium carbonate) flattens cycles of manic behavior and depression in most people
- Lithium can also be used to strengthen effects of antidepressant medication
- People with bipolar disorder may have to use lithium indefinitely
- Other examples of mood stabilizers: Divalproex sodium, Lamotrigine, Carbamazepine, and Oxcarbazepine

ELECTROCONVULSIVE THERAPY

- Treatment of disorders by passing an electric current (that causes a convulsion) through the head
- Introduced by Italian psychiatrist **Ugo Cerletti** in 1939
- Cerletti noted that some slaughterhouses used electric shock to render animals unconscious
- Cerletti believed that convulsions brought by electric shack were incompatible with schizophrenia and other major psychological disorders
- Now used mainly for people with major depression who do not respond to antidepressants
- Research suggests that post-ECT problems in learning and memory may be lessened by use of **Ketamine**

PSYCHOSURGERY

- Known today as **Prefrontal lobotomy**
- Used with people with severe disorders
- Pick-like instrument severs the nerve pathways that link the prefrontal lobes of the brain to the thalamus
- Intended to sever thought from emotion and enable severely disturbed patients to regain control
- Pioneered by **Antonio Egas Moniz** and was brought to the US in the 1930s
- Side effects include hyperactivity and distractibility, impaired learning ability, overeating, apathy and withdrawal, epileptic-type seizures, reduced creativity, and, occasionally, death

DOES BIOLOGICAL PSYCHOTHERAPY WORK?

- Antipsychotic drugs reduce the need for restraints and hospitalization in schizophrenia patients
  - These drugs have enabled many former patients to live independently, work, and maintain family lives
- Cognitive therapy offers coping skills that lower the risk of depression recurrence after treatment ends
- A meta-analysis found combined psychotherapy and drug treatment more effective than therapy alone, though the added benefit of antidepressants was unclear
- Cognitive behavioral therapy is now recommended for schizophrenia, helping reduce symptoms by addressing thought patterns
- Antianxiety drugs are useful short-term but often misused to avoid confronting life problems

- Electroconvulsive therapy (ECT), despite controversy, helps those unresponsive to antidepressants and may serve as a last resort
- Drug therapy and ECT can be effective for treatment-resistant disorders
- For anxiety and mild depression, psychotherapy is generally preferred over medication
- Psychotherapy teaches problem-solving and cognitive change—skills drugs can't provide

15: SOCIAL PSYCHOLOGY

- Field of psychology that studies the nature and causes of behavior and mental processes in social situations

ATTITUDES

- Not just feelings or emotions
- Enduring behavioral and cognitive tendencies that are expressed by evaluating particular people, places, or things with favor or disfavor
- Can foster love or hate
- Lead to social conflict or to the resolution of conflicts
- Can change, but not easily
- Concern your feelings and beliefs about things
- Developed along with early learning experiences

ATTITUDE FORMATION

- Not fully mechanical

GENETICS

- Personality characteristics are moderately associated with genetic heritage and social/political attitudes

LEARNING

- Attitudes are formed early through experiences, conditioning (association of positive and negative stimuli), and observational learning.
- Attitudes formed through direct experience may be easier to recall, but we also acquire attitudes by observing, listening, or reading works of other people

COGNITIVE APPRAISAL

- Attitudes are formed or changed based on new information, with initial attitudes acting as cognitive anchors
- People are motivated to understand the environment so that they can make predictions and exercise some control over it
- People also sometimes form or change attitudes on the basis of new information
- Initial attitudes act as **cognitive anchors**

PREDICTING BEHAVIOR FROM ATTITUDES

- Definition of attitude implies that our behavior is consistent with our attitudes

A-B PROBLEM

- Links between attitudes (A) and behaviors (B) tend to be weak to moderate
- The issue of how wwii we can predict behavior on the basis of attitudes

FACTORS THAT AFFECT THE LIKELIHOOD TO PREDICT BEHAVIOR FROM ATTITUDES:

- **SPECIFICITY** - Specific attitudes better predict specific behaviors
- **STRENGTH OF ATTITUDES** - Stronger attitudes are more likely to determine behavior
- **VESTED INTEREST** - Greater vested interest leads to a higher likelihood of acting on attitudes
- **ACCESSIBILITY** - Accessible attitudes (brought to mind) are more likely to influence behavior

CHANGING ATTITUDES THROUGH PERSUASION

ELABORATION LIKELIHOOD MODEL

- Describes ways in which people respond to persuasive messages

- View that persuasive messages are evaluated (elaborated) on the basis of central and peripheral cues

TWO ROUTES TO PERSUADING OTHERS TO CHANGE ATTITUDES:

- **CENTRAL ROUTE** - Thoughtful consideration of arguments and evidence
- **PERIPHERAL ROUTE** - Associates objects with positive or negative cues
  - Peripheral factors include: messenger, context of the message, and the audience

THE PERSUASIVE MESSAGE

- Repeated exposure to people and things enhances their appeal
- **TWO-SIDED ARGUMENTS** - Recounting and refuting opposing arguments
- **FEAR APPEAL** - Persuasive communication arousing fear
  - Influences behavior on the basis of arousing fear instead of rational analysis of the issues
  - Most effective when the audience believes the risks are serious and they can change their behavior to avert risks
- Audiences tend to believe arguments that appear to run counter to the vested interests of the communicator

THE PERSUASIVE COMMUNICATOR

- Characterized by expertise, trustworthiness, attractiveness, or similarity to their audiences
- People find it painful when they are confronted with information that counters their own views
- **SELECTIVE AVOIDANCE** - diverting one's attention from information that is inconsistent with one's attitudes
- **SELECTIVE EXPOSURE** - deliberately seeking and attending to information that is consistent with one's attitudes

CONTEXT OF THE MESSAGE

- Environment and approach can increase the likelihood of persuasion

THE PERSUADED AUDIENCE

- May be that people with high self-esteem and low social anxiety are more likely to resist social pressure
- People who comply with unreasonable requests are more apt to report thoughts such as:
  - "I was worried about what the other person would think of me if I said no."
  - "It is better to help other people than to be self-centered."
  - "If I had said no, the other person might have been hurt or insulted."
- People who refuse unreasonable requests reported thoughts like these:
  - "It doesn't really matter what the other person thinks of me."
  - "I am perfectly free to say no."
  - "The request was unreasonable."

CHANGING ATTITUDES AND BEHAVIOR BY MEANS OF COGNITIVE DISSONANCE

COGNITIVE-DISSONANCE THEORY

- View that we are motivated to make our cognitions or beliefs consistent with each other and with our behavior
- People are thinking creatures who seek consistency in their behaviors and their attitudes
- **ATTITUDE-DISCREPANT BEHAVIOR** - behavior that ran

counter to what they actually thought
    - Behavior inconsistent with an attitude that may have the effects of modifying an attitude
- Learning theorists might predict that the more we are reinforced for doing something the more we should like it
- Cognitive-dissonance theorists rightly predicted that because the ideas of "I was paid very little" and "I told someone that this assignment was interesting" are dissonant, people will tend to engage in **effort justification**
- **EFFORT JUSTIFICATION** - Explaining behavior to make unpleasant undertaking seem worth it
    - Tendency to seek justification for strenuous efforts
- Discomfort of cognitive dissonance motivates people to explain their behavior to themselves in such a way that unpleasant undertakings seem worth it

PREJUDICE AND DISCRIMINATION

PREJUDICE

- Attitude toward a group that leads people to evaluate members of that group negatively—even though they thought they have never met them
- On a cognitive level, it is linked to expectations that members of the target group will behave poorly
- On an emotional level, it is associated with negative feelings
- In behavioral terms, it is on

DISCRIMINATION

- Behavior resulting from prejudice

STEREOTYPING

- Erroneous assumptions that all members of a group share the same traits
- May lead people to view members of groups in a biased fashion
- Can be positive or negative

SOURCES OF PREJUDICE

- People who are susceptible to feelings of prejudice tend to be low in agreeableness and low in openness to experience

FACTORS THAT INDICATE HOW PREJUDICED PEOPLE SELECT THEIR TARGETS

- **Dissimilarity** - we are apt to like people who share our attitudes
- **Social Conflict** - there is often social and economic conflict between people of different races and religions
- **Social learning** - children acquire some attitudes from others, especially their parents
- **Information processing** - prejudices act as cognitive filters through which we view the social world
- **Social categorization** - we tend to divide our social world into "us" and "them"
    - People usually view those who belong to their own groups more favorably than those who do not

COMBATING PREJUDICE

- Encourage intergroup contact and cooperation
- Present examples of admired individuals within stigmatized groups
    - Verbal reminders should work as well
- Attack discriminatory behavior - It is sometimes easier to change

people\s behavior than to alter their feelings
    - Cognitive-dissonance theory suggests that when we change people\s, their feelings may follow along
- Hold discussion forms - many campuses conduct workshops and discussion groups on gender, race, and diversity
- Examine your own beliefs

ATTRACTION AND LOVE

ATTRACTION

- In social psychology, an attitude of liking or disliking (negative attraction)
- Factors contributing to attraction are physical appearance, similarity, and reciprocity

PHYSICAL APPEARANCE

- Key factor in attraction and in the consideration of romantic partners
- Many standards for beauty appear to be cross-cultural

GENDER DIFFERENCES IN SELECTION OF A ROMANTIC PARTNER

- Physical appearance may be a major factor in the selection of a romantic partner
- Men tend to place relatively greater emphasis on physical allure
- Women tend to place greater emphasis on traits such as professional status, consideration, dependability, kindness, and fondness for children

ATTRACTION-SIMILARITY HYPOTHESIS

- States that people tend to develop romantic relationships with people who are similar to themselves in physical attractiveness and other traits

PROPINQUITY

- Marriages are made in the neighborhood and not in heaven

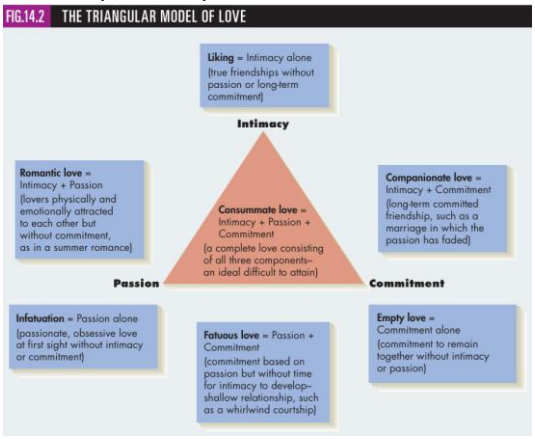
RECIPROCITY

- We tend to be more open, warm, and helpful when we are interacting with people who seem to like us

LOVE

- A strong positive emotion
- **ROMANTIC LOVE** - dramatic form of heated, passionate love
    - Intense, positive emotion that involves sexual attraction, feelings of caring, and the belief that one is in love
    - In Sternberg's scheme, is characterized by passion and intimacy
- **TRIANGULAR MODEL OF LOVE** - does not refer to two men wooing the same woman
    - Robert Sternberg's view that love involves three components: intimacy, passion, commitment
- **Intimacy** - couple's closeness, to their mutual concern and sharing of feelings and resources
    - Close acquaintance and familiarity
    - Characteristic of a relationship in which partners share their innermost feelings
    - Involves closeness and caring
    - Championing interests of loved one even if it entails sacrificing one's own
- **Passion** - Romance and sexual feelings; most crucial in short-term relationships

  - Involves fascination (preoccupation with loved one), sexual craving, and the desire for exclusiveness
  - Strong romantic and sexual feelings
- **Commitment** - deciding to enhance and maintain the relationship
- Intimacy and commitment are more important in enduring relationships
- **CONSUMMATE LOVE** - ideal form of love within Sternberg's model
  - Combination of intimacy, passion, and commitment
- **LIKING** = Intimacy alone
- **COMPANIONATE** = Intimacy + Commitment
- **EMPTY LOVE** = Commitment alone
- **FATUOUS LOVE** = Passion + Commitment
- **INFATUATION** = Passion alone
- **ROMANTIC LOVE** = Intimacy + Passion
- People are cognitively biased toward evaluating their partners positively



SOCIAL PERCEPTION

- Concerns ways in which we perceive other people
- Subfield of psychology that studies the ways in which we form and modify impressions of others

PRIMACY AND RECENCY EFFECTS

PRIMACY EFFECT

- Tendency to evaluate others based on first impressions

RECENCY EFFECT

- Tendency to evaluate others based on the most recent impression

ATTRIBUTION THEORY

- **ATTRIBUTION** - belief concerning why people behave in a certain way
- **ATTRIBUTION THEORY** - processes by which people draw conclusions about the factors that influence one another's behavior

DISPOSITIONAL ATTRIBUTION

- Ascribe a person's behavior to internal factors, such as personality traits and free will
- Attributing behavior to internal factors (personality)

SITUATIONAL ATTRIBUTION

- Attribute a person's actions to external factors, such as circumstances, social influence, or socialization
- Attributing behavior to external factors (circumstances)

FUNDAMENTAL ATTRIBUTION ERROR

- Bias in the attribution process
- Overemphasizing dispositional factors in others' behavior
- We tend to infer traits from behavior
- But in collectivist cultures that stress interdependence
  - People are more likely to attribute another person's behavior to that person's social roles and obligations

ACTOR-OBSERVER EFFECT

- Tendency to attribute our own behavior to situational factors but to attribute the behavior of others to dispositional factors
- When we see other people doing things that we do not like, we tend to judge them as willful actors
- when we find ourselves doing things that we ordinarily disapprove of, we tend to see ourselves as victims of circumstances

SELF-SERVING BIAS

- Attributing our successes to internal factors, but failures to external factors
- We are likely to ascribe our successes to internal, dispositional factors but our failures to external, situational influences
- Another interesting attribution bias is a gender difference in attributions for friendly behavior.
  - Men are more likely than women to interpret a woman's smile or friendliness toward a man as flirting

SOCIAL INFLUENCE

- Area of social psychology that studies the ways in which people alter the thoughts, feelings, and behavior of others

OBEDIENCE TO AUTHORITY

THE MILGRAM STUDIES

- Demonstrates the extent to which people obey immoral requests from authority figures
- **Signs of stress**:
  - Trembling
  - Stuttering
  - Biting lips
  - Groaning
  - Sweating
  - Digging fingernails into their flesh

ON DECEPTION AND TRUTH

- Milgram debriefed his participants after the experiment was complete
- He explained purpose and methods of his research in detail
- Emphasized the fact that they had not actually harmed anyone
- Ethics of Milgram studies have been debated by psychologists for decades

WHY DID PEOPLE IN THE MILGRAM STUDIES OBEY THE EXPERIMENTERS

- Many people obey the commands of others even when they are required to perform immoral tasks
- **Factors contributing to obedience according to Milgram**:
  - **Socialization** - we are socialized from early childhood to obey authority figures
  - **Lack of social comparison** - In Milgram's experimental settings, experimenters displayed command of the situation, but "teachers" (participants) did not have the opportunity to compare their ideas and feelings with those of other people in the same situation.
  - **Perception of legitimate authority**
  - **Foot-in-the-door technique** - method for inducing compliance in which a small request is followed by a larger request
  - **Inaccessibility of values** - People are more likely to act in accordance with their attitudes when their attitudes are readily available, or accessible.
  - **Buffers between perpetrator and the victim**

CONFORMITY

- We are said to conform when we change our behavior to adhere to social norms
- **Social norms** are widely accepted expectations concerning social behavior
- **Explicit social norms** - often made into rules and laws
- **Implicit social norms** - unspoken social norms
- Changing behavior to adhere to social norms
- Tendency to conform to social norms is often good, because most have evolved because they promote comfort and survival
- **Group pressure** also promotes maladaptive behavior

ASCH STUDY

- Demonstrating conformity to erroneous group judgements
- Three out of four agreed with the majority's wrong answer at least once

FACTORS THAT INFLUENCE CONFORMITY

- Belonging to a **collectivist** rather than an individualistic society
- **Desire to be liked** by other members of the group
- Low self-esteem
- Social shyness
- Lack of familiarity with the task
- Group size
- Social support

AGGRESSION

- Behavior intended to cause harm or pain

BIOLOGY, CHEMISTRY, AND AGGRESSION

- Evolutionary psychologists believed that "genetic whisperings" promote aggression to some degree
- In response to certain stimuli, many nonhumans show instinctive aggressive reactions
- Electrical stimulation of part of the hypothalamus triggers stereotypical aggressive behaviors in many lower animals
  - However, in humans, whose brains are more complex, other brain structures apparently moderate possible aggressive instincts.
- Testosterone appears to affect the tendencies to dominate and control other people.
  - Testosterone levels also vary with the occasion

PSYCHOLOGICAL ASPECTS OF AGGRESSION

- **Empathy** encourages helping behavior; lack of empathy increases aggression
- Freud believed that aggression is a natural and instinctive reaction to the **frustrations** of daily life
- Cognitive psychologists assert that our behavior is influenced by our values, by how we interpret situations, and by choice.
  - We are more likely to respond aggressively to a provocation when we magnify the importance of the "insult" or otherwise stir up feelings of anger
- **Learning** through reinforcement
- From social cognitive perspective, aggressive skills are mainly

acquired by observation of other people
    - believe that consciousness and choice play key roles in aggressive behavior among humans
    - Social cognitive theorists believe that we are not likely to act aggressively unless we believe that aggression is appropriate under the circumstances and likely to be reinforced.
- Situational factors:
    - Milgram studies showed us that majority of "normal people will act aggressively when they are pushed by other people to do so

GROUP BEHAVIOR

- Groups help us satisfy our needs for affection, attention, and belonging

SOCIAL FACILITATION

- Process by which a person's performance is increased when other members of a group engage in similar behavior
- Increased performance when others are present
- May be influenced by evaluation apprehension and arousal
- **EVALUATION APPREHENSION** - Concern that others are evaluating our behavior
- **SOCIAL LOFING** - fail to make a significant effort—or because they believe they will not be found out and held accountable
- **DIFFUSION OF RESPONSIBILITY** - spreading or sharing of responsibility for a decision or behavior within a group
    - Each person may feel less obligation to help because others are present, especially if others are perceived as being capable of doing the job
    - Group members may also reduce their efforts if an apparently capable member makes no contribution but "rides free" on the efforts of others

GROUP DECISION MAKING

- Organizations use groups such as committees or juries to make decisions in the belief that group decisions are more accurate or valid than individual decisions
- **SOCIAL DECISION SCHEMES** that govern much of group decision making
    - **Majority-wins scheme** - group arrives at the decision that was initially supported by the majority
        - Appears to guide decision-making most often when there is no single objectively correct decision
    - **Truth-wins scheme** – as more information is provided and opinions are discussed, group comes to recognize that one approach is objectively correct
    - **Two-thirds majority scheme** - juries tend to convict defendants when ⅔ of the jury initially favors conviction
    - **First-shift rule** - group tends to adopt decision that reflects the first shift in opinion expressed by any group member

POLARIZATION AND THE "RISKY SHIFT"

- **POLARIZATION** - Taking of an extreme position

GROUP THINK

- Originated by **Irving Janis**
- Problem that may arise in group decision making when group members are swayed more by group cohesiveness and a dynamic leader than by the realities of the situation
- Has been connected with historic fiascos such as the botched Bay of Pigs invasion of Cuba, the Watergate scandal in which members of the Nixon administration believed they could cover up a theft, and NASA\s decision to launch the **Challenger** space shuttle despite engineers\ warnings about the dangers created by unusually cold weather
- Characteristics of group think that contribute to flawed decisions:
  - Feelings of invulnerability
  - Group's belief in its rightness
  - Discrediting information contrary to the group's decision
  - Pressures on group members to conform
  - Stereotyping members of the out-group

MOB BEHAVIOR AND DEINDIVIDUATION

- Frenchman **Gustave Le Bon** branded mobs and crowds as irrational, resembling a "beast with many heads"
- **DEINDIVIDUATION** – state in which they are willing to follow a norm that emerges in a specific situation, even if it means ignoring their own values
- Reduced self-awareness and concern for social evaluation
- Factors that lead to deindividuation:
  - Anonymity
  - Diffusion of responsibility
  - Arousal due to noise and crowding
  - Focus on emerging group norms rather than on one\s own values

ALTRUISM AND THE BYSTANDER EFFECT

ALTRUISM

- Selfless concern for the welfare of others

BYSTANDER EFFECT

- Tendency to avoid helping other people in emergencies when other people are also present and apparently capable of helping
- Reduced likelihood of helping when others are present

FACTORS INVOLVED IN HELPING BEHAVIOR

- Observers are more likely to help when they are in a good mood
- People who are empathic are more likely to help people in need
- Bystanders may not help unless they believe that an emergency exists
- Observers must assume the responsibility to act
- Must know what to do
- Observers are more likely to help people they know
- Observers are more likely to help people who are similar to themselves
  - Being able to identify with the person in need promotes helping behavior

Summary of Chapter 10: The Voyage Through the Life Span

Prenatal Development

- **Zygote** – a fertilized ovum (egg cell).
- **Germinal Stage** – conception to implantation.
- **Embryonic Stage** – implantation to 8th week; major organ systems form, head and organs develop before extremities.
- **Amniotic Sac** – a sac within the uterus that contains the embryo or fetus; fluid protects and regulates temperature.
- **Placenta** – a membrane that permits exchange of nutrients and waste between mother and child (bloodstreams don't mix).
- **Umbilical Cord** – tube connecting embryo to placenta, conducting nutrients and waste.
- **Fetal Stage** – 3rd month until birth; organs mature, fetus moves, gains weight, and prepares for independent life

Childhood: Physical Development

- Infants double birth weight in 5 months, triple by 1 year.
- Height increases ~10 inches in first year.
- Growth continues steadily until adolescence.

Reflexes

- **Rooting** – turning of infant's head toward a touch (e.g., mother's nipple).
- Other reflexes: sucking, withdrawal, Moro (startle), grasp/palmar, Babinski (toes fan).
- Reflexes are inborn, automatic, and essential for survival.

Motor Development

- Progression from lifting head → rolling → sitting → crawling → walking → running.
- Sequence is invariant (genetically programmed), though timing varies.
- Influenced by both maturation (brain development) and environment (nutrition, stimulation).

Perceptual Development

- Newborns can track light, discriminate colors by 3 months, and develop depth perception by 6–8 months.
- Prefer human faces.
- Hearing is well developed; infants prefer mother's voice.
- Can discriminate odors; maternal-infant odor recognition aids bonding.

Childhood: Cognitive Development

Piaget's Theory

- **Assimilation** – inclusion of a new event into an existing schema.
- **Accommodation** – modification of schemas to integrate new information.

- **Schema** – a mental structure for organizing knowledge.

Stages of Cognitive Development

1. **Sensorimotor Stage (0–2 yrs)**
   - Learning through senses and motor activity.
   - **Object Permanence** – recognition that objects removed from sight still exist.
2. **Preoperational Stage (2–7 yrs)**
   - Use of words/symbols; illogical, egocentric thinking.
   - **Egocentrism** – assumption others see the world as oneself.
   - **Animism** – attributing life to objects.
   - **Artificialism** – belief natural events are human-made.
   - **Objective Responsibility** – judging wrongness by damage done, not intent.
   - **Law of Conservation** – recognition that mass/weight/volume remain the same despite changes in shape.
3. **Concrete Operational Stage (7–12 yrs)**
   - Logical thought about tangible objects.
   - **Decentration** – focusing on more than one dimension.
   - **Subjective Moral Judgment** – judging based on motives as well as damage.
   - **Reversibility** – recognition that processes can be undone.
   - Less egocentric; can take others' perspectives.
4. **Formal Operational Stage (12+ yrs)**
   - Abstract, logical, hypothetical thinking.
   - Deductive reasoning, classification, and hypothetical problem-solving.

Vygotsky's Sociocultural Theory

- **Zone of Proximal Development (ZPD)** – range of tasks a child can do with guidance from a more skilled person.
- **Scaffolding** – temporary support provided by adults/teachers until the child can function independently.
- Emphasizes social and cultural influences on learning.

Kohlberg's Theory of Moral Development

- Moral reasoning develops in stages, always in sequence.

Levels & Stages

1. **Preconventional Level** (up to ~9 yrs)
   - **Stage 1: Obedience & Punishment** – right/wrong judged by consequences.
   - **Stage 2: Individualism & Exchange** – morality based on self-interest and reciprocity.
2. **Conventional Level** (adolescents & many adults)
   - **Stage 3: Interpersonal Relationships** – guided by social approval.

    - **Stage 4: Maintaining Social Order** – morality tied to obeying laws/rules.
3. **Postconventional Level** (some adolescents/adults)
    - **Stage 5: Social Contract** – laws are social contracts, can be changed if unjust.
    - **Stage 6: Universal Ethical Principles** – morality based on justice, equality, dignity, even against laws.

👫 Childhood: Social & Emotional Development

Erikson's Stages (early years)

- **Trust vs. Mistrust** (infancy)
- **Autonomy vs. Shame & Doubt** (toddlers)
- **Initiative vs. Guilt** (preschool)
- **Industry vs. Inferiority** (school age)

Attachment (Ainsworth & Bowlby)

- **Attachment** – enduring emotional tie between infant and caregiver.
- **Strange Situation** – method to study attachment.
    - **Secure Attachment** – mild protest, comforted on reunion.
    - **Avoidant Attachment** – little distress, ignores caregiver.
    - **Ambivalent/Resistant Attachment** – intense distress, mixed response on reunion.
- **Contact Comfort** – hypothesized primary drive to seek physical comfort through contact.
- **Critical Period** – specific time when attachment/imprinting occurs.
- **Imprinting** – instinctive attachment response in animals.

Parenting Styles (Baumrind)

1. **Authoritative** – strict but warm, use reason, supportive.
2. **Authoritarian** – rigid, demand obedience, less warmth.
3. **Permissive** – few rules, indulgent, poor supervision.
4. **Uninvolved** – little warmth, few demands, detached.

🧑 Adolescence

- **Puberty** – period when sexual reproduction becomes possible.
- **Secondary Sex Characteristics** – traits distinguishing males/females not directly tied to reproduction.
- **Formal Operational Stage** – abstract, hypothetical reasoning.
- **Adolescent Egocentrism** – self-focused thinking.
    - **Imaginary Audience** – belief others are as concerned with us as we are.
    - **Personal Fable** – belief our feelings/ideas are unique and invulnerable.
- **Ego Identity vs. Role Diffusion** (Erikson) – firm sense of self vs. lack of clarity.

🧑 Adulthood

Early Adulthood

- Peak physical strength, sensory sharpness, sexual functioning.
- **Intimacy vs. Isolation** (Erikson) – forming close relationships vs. loneliness.

Middle Adulthood

- Gradual physical decline.
- **Menopause** – cessation of menstruation.
- **Generativity vs. Stagnation** (Erikson) – contributing to society vs. feeling unproductive.
- **Sandwich Generation** – caring for both children and aging parents.

Late Adulthood

- Physical decline (bones, senses, immune system).
- **Crystallized Intelligence** – accumulated knowledge/skills (increases with age).
- **Fluid Intelligence** – mental flexibility, problem-solving (declines with age).
- **Alzheimer's Disease** – progressive brain disease with memory and cognitive loss.
- **Ego Integrity vs. Despair** (Erikson) – acceptance of life vs. regret.
- **Selective Optimization with Compensation** – focusing on meaningful activities despite decline.

Summary of Chapter 8: Thinking, Language, and Intelligence

🧠 Thinking

- **Thinking** – paying attention to information, representing it mentally, reasoning, and making judgments/decisions.
    - Conscious and planned (not like dreaming/daydreaming).
- **Concepts** – mental categories grouping objects, events, or qualities with common properties.
- **Cognitions** – mental activity of understanding, processing, and communicating information.
- **Prototypes** – best examples of a category (e.g., a dog as a prototype of animals).
- **Over-extension** – when children include too many instances in a category (e.g., calling all 4-legged animals "dogs").

🧩 Problem Solving

- **Problem Solving** – process of finding solutions to obstacles or challenges.

Methods:

1. **Understanding the Problem** – mental representation must match reality.
2. **Algorithms** – step-by-step procedures that guarantee solutions (e.g., formulas, systematic random search, anagrams).
3. **Heuristics** – shortcuts or "rules of thumb" (faster but not guaranteed).
    - **Means-End Analysis** – reduce gap between current state and goal.
    - **Analogy Heuristic** – apply past solutions to new problems.

Factors Affecting Problem Solving:

- **Expertise** – experts solve faster using memory, mental images, and parallel processing.
- **Mental Sets** – tendency to use old methods even when not useful.
- **Insight** – sudden realization of a solution.
- **Incubation** – stepping back can lead to insight later.
- **Functional Fixedness** – difficulty using objects in new ways.

⚖️ Judgment & Decision Making

- People rely on **heuristics**:
    - **Representative Heuristic** – judge by similarity to a category.
    - **Availability Heuristic** – judge by how easily examples come to mind.
    - **Anchoring & Adjustment** – rely on initial estimate (anchor) and adjust slightly.
- **Framing Effect** – wording/context influences decisions.
- **Overconfidence** – overestimating accuracy of judgments.

- **Motivated Reasoning** – decisions based on emotion, not evidence.

Language

- **Language** – communication of thoughts/feelings using symbols and grammar.
  - Properties: **Semanticity** (words as symbols), **Infinite Creativity**, **Displacement** (talk about past/future).
- **Nonhuman Communication** – animals communicate but lack full language properties.
- **Do Apes Use Language?** – apes can learn signs but debate exists on whether it's true language.

Language & Culture

- **Linguistic-Relativity Hypothesis (Whorf)** – language shapes perception of the world.
- Criticism: thought can exist without language; vocabulary reflects importance, not limits.

Language Development

- **Prelinguistic Vocalizations** – crying, cooing, babbling.
- **Holophrases** – single words with complex meaning.
- **Telegraphic Speech** – two-word sentences (e.g., "Mommy go").
- **Overregularization** – applying grammar rules incorrectly (e.g., "mouses" instead of "mice").
- Vocabulary grows rapidly; by age 6, ~10,000 words.

Theories:

- **Learning Theory** – language learned through imitation and reinforcement.
- **Nativist Approach (Chomsky)** – inborn **Language Acquisition Device (LAD)** + universal grammar.
- **Psycholinguistic Theory** – interaction of environment and innate ability.
- **Motherese (Infant-Directed Speech)** – simplified, high-pitched speech helps learning.

Intelligence

- **Intelligence** – general mental capability to reason, plan, solve problems, think abstractly, learn quickly, and from experience.

Theories:

1. **Spearman's g & s** – general intelligence (g) + specific abilities (s).
2. **Thurstone's Primary Mental Abilities** – 8 abilities (e.g., memory, word fluency, reasoning).
3. **Gardner's Multiple Intelligences** – linguistic, logical-mathematical, musical, bodily-kinesthetic, spatial, interpersonal, intrapersonal, naturalist, existential.
4. **Sternberg's Triarchic Theory** – analytical (academic), creative, practical ("street smarts").
5. **Emotional Intelligence** – recognizing, regulating, and understanding emotions.

Creativity

- **Creativity** – ability to generate novel/useful solutions.

- **Convergent Thinking** – narrowing to one best answer.
- **Divergent Thinking** – generating many possible solutions.

Measuring Intelligence

- **Stanford-Binet Intelligence Scale (SBIS)** – measures IQ (Mental Age ÷ Chronological Age × 100).
- **Wechsler Scales** – verbal & performance IQs; deviation IQ (compared to age group).
- **Group Tests** – used in schools/armies but less accurate.
- **Reliability & Validity** – consistent and predictive of school performance.

Differences in Intelligence

- **Socioeconomic/Ethnic** – differences often due to environment, not race.
- **Gender** – girls stronger in verbal, boys in spatial; overall differences small.
- **Genetics & Environment** – both influence IQ (heritability ~40–60%).
- **Flynn Effect** – IQ scores rising over generations due to environment/education.